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BIF401 Final term preparation Highlighted notes from chapter 4 to 6

Important MCQs, Short and Long Questions

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Chapter 4 - RNA Secondary Structure Prediction

Module 001: DNA TO RNA SEQUENCES

MOTIVATION

In early days RNA was a considered as a structure which was involve between DNA and protein, means takes information from DNA and converts that information into protein synthesis.

Short question: Types of Now we know that it has multiple types like mRNA, tRNA, miRNA and siRNA. And they perform most of the work in gene expression and proteins.

MCQ: Not All RNA molecules are same, they differ in nucleotide sequences and functions also.

✓ Short question: RNA virsus defiation:

Many viruses assemble their genomes **from RNAs**. They are therefore called **RNA viruses**. Examples include

MCQ: Human Immunodeficiency Virus and Hepatitis C Virus.

Difference Between between RNA and DNA:

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There is little difference between RNA and DNA:

- Thymine is replaced by Uracil in RNA molecule.
- RNA molecule is single stand.
- RNA contain ribose sugar.

S.Q: Why RNA have shorter life spam

Because RNA **has two (OH) groups** that's why it has **shot life spam** because of **both (OH) repulsion**.

Module 002: TYPES OF RNA &THEIR FUNCTIONS

There are two categories of RNA:

- Coding RNA
- Non-Coding RNA

MCQ: Coding RNA perform their coded function in **protein** synthesis.

Mcq: And Non-coding RNA helps in **translation process.**

Q: Write the types of RNA

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Mcq: How many types of RNA

TYPES OF RNA

There are many types of RNA according to their funtions like:

- Messenger RNA (mRNA)
- Transfer RNA (tRNA)
- Ribosomal RNA (rRNA)
- Micro RNAs (miRNA)
- Small Interfering RNA (siRNA)

Short question: Write a note on Mrna.

Mcq: % age of mrna in cell? Only 5-10%

MESSENGER RNA

Only 5-10% of this RNA type is present in cell. Which has variable sequence, variable size and it carries the genetic information form DNA to Ribosomes where proteins to be assembled. Messenger **RNA 5' end is capped with (7-Methyl Guanosine Triphosphate)** which helps the

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Ribosomes to identify the mRNA. And 3' end of the mRNA is **poly A tail** (around 30-200 adenylate residues) which help shield against 3' exonucleases)

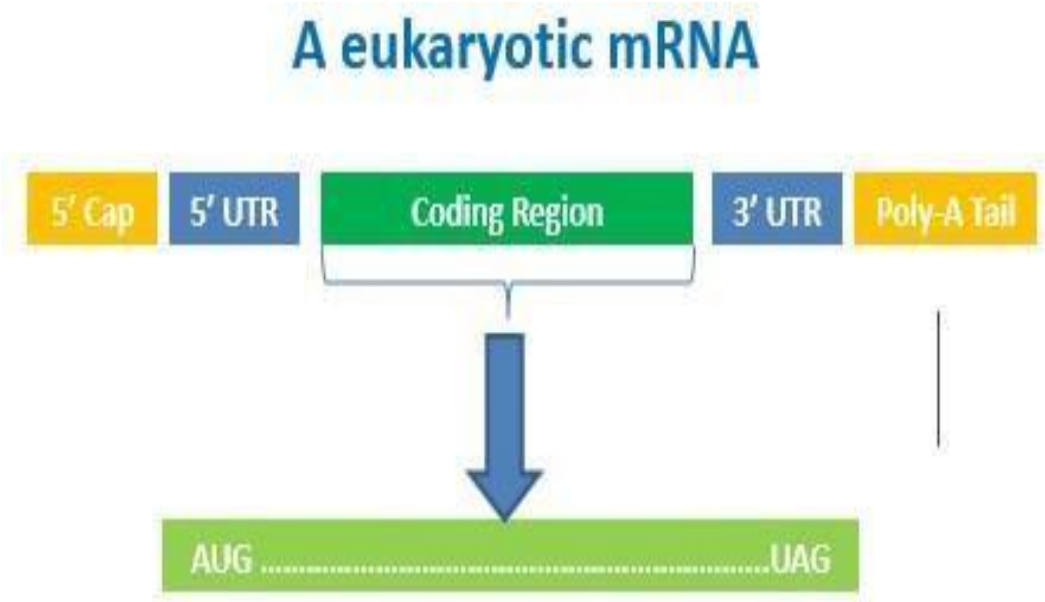


Figure 2 RNA sequence is complementary to the DNA sequence and is translated as codons of three nucleotides

S.Q **How RNA differe?**

As RNA has differ in nucleotides sequences therefore differ in functions.

Module 003: SIGNIFICANCES OF RNA STRUCTURE

S.Q: What is the : SIGNIFICANCES OF RNA STRUCTURE

RNA can form **3D structures** such structural properties helps the RNA molecule to perform different functions.

S.Q: Explain or write the Composition of RNA

As RNA is composed of sugars, phosphate and nucleotides and these nucleotides have ability to form hydrogen bonds.

- A' can make hydrogen bonds with 'U'
- 'G' makes hydrogen bonds with 'C'
- 'G' can also make hydrogen bonds with 'U' (Wobble Pair)

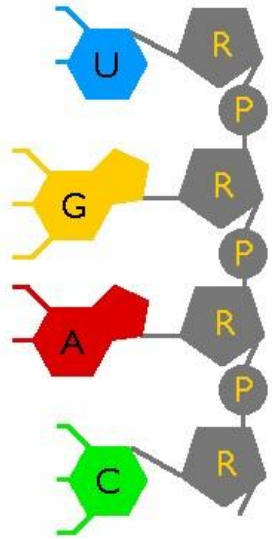


Figure 3 In RNA ribose is used in place of deoxyribose 3 In RNA uracil is used in place of thymine

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Long and S. Q: Write the Functions of RNA short and Long question:

Due to this ability of bonding RNA forms many structures and due to variety of structures RNA performs many functions in cell like:

- DNA information transfer
- Regulatory roles
- Catalytic roles
- Defense & immune response
- Structure-based special roles

Module 004: RNA FOLDING AND ENERGY FOLDING

S.Q: Define Gibs Free Energy ? how DNA folds?

RNA molecules form many structures for stability and different functions. **"Gibs Free Energy" (LANGRIDGE and KOLLMAN 1987) is the free energy available for RNA molecule for reactions and RNA structure formation takes place at this**

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lower energy. Incase if RNA has two structures we can select the one with lowest energy state.

	Y:	A	C	G	U
X:	A	.	.	.	-2.1
	C	.	.	-3.3	.
	G	.	-2.4	.	-1.4
	U	-2.1	.	-2.1	.

We can calculate the overall energy of RNA structures by summing up energies given out during the process of folding. For knowing the positive and negative values of calculations of stabilizing and destabilizing energies we may factor in ways in which RNA can be destabilized.

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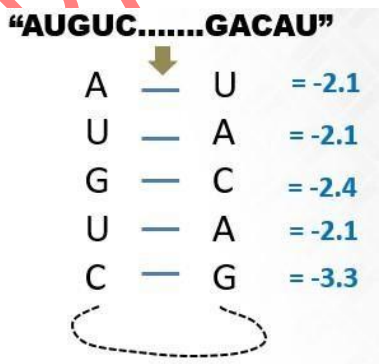
Module 005: CALCULATING ENERGIES OF FOLDING- AN EXAMPLE

S.I.Q: Composition of RNA?

RNA is composed of four nucleotides (A, U, C and G) and these nucleotides are attached with ribose sugar in backbone. And these nucleotides have hydrogen bonding between them. G always bond with C and Always bonds with U through hydrogen bonding and energy is released.

That's why RNA molecule become more stable.

S.Q : Why RNA molecule become more stable?



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5 nucleotides formed H-Bonds. This bond formation released energy (-12.0 kcal/mol) RNA molecule took up a 2' structure. Hence became more stable.

Module 006: TYPES OF RNA SECONDARY STRUCTURES-I

All the **complimentary bases of RNA combine together to form RNA secondary structures. A simple nucleotide sequences of RNA is called as Primary structure** and denoted by 1' while when these nucleotides fold together and form a complex structure that is called secondary structure and denoted by 2'.

Short & Long question: How many types of secondary structure of RNA?

The preferred structure of RNA is 2' which has many structural patterns like **Helices, Loops, Bulges and Junctions**

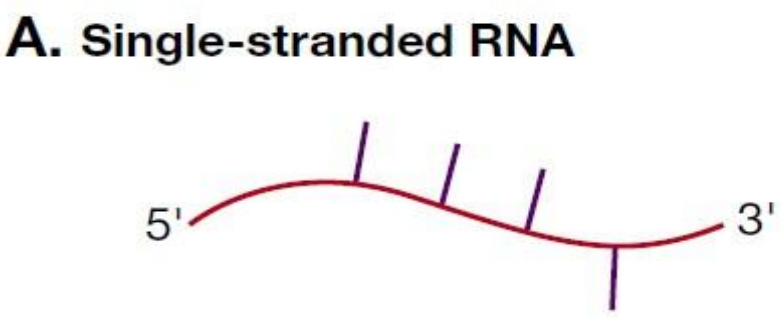
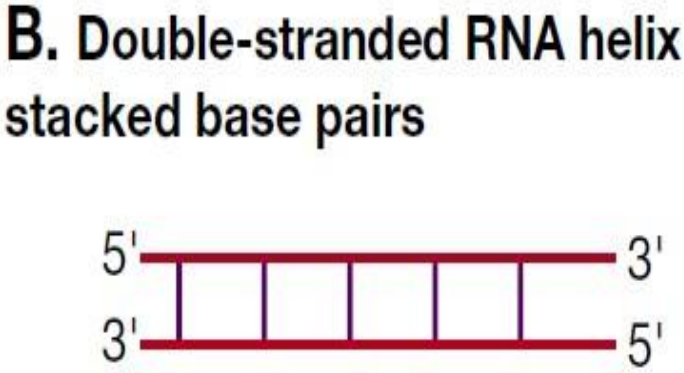


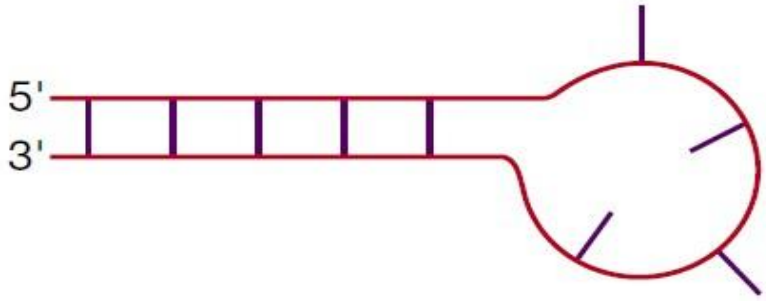
Figure 8 RNA sequence extends from its 5' end to 3' end. Upon folding, 3' end may fold on to the 5' end

The first 2' RNA structure is called helix. Unlike the DNA helix, the RNA helix is formed when the RNA folds onto itself.



The second 2' structure is the hairpin loop

C. Stem and loop or hairpin loop.



The loop of the hairpin must at least **four bases long** to avoid **steric hindrance** with base-pairing in the stem part of the structure.

Note that hairpins reverse the **chemical direction** of the RNA molecule.

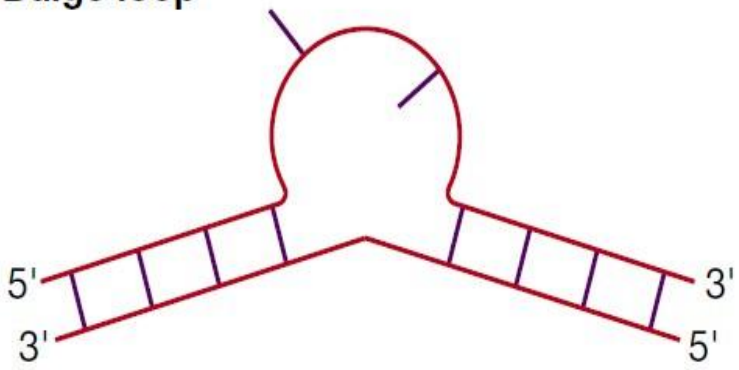
Module 007: TYPES OF RNA SECONDARY STRUCTURES-II

RNA 1' structure fold the (5'-3') ends and make RNA 2' structure just like helix and hairpin structure.

Mcq: The third type of 2' structure **is bulge loop**.

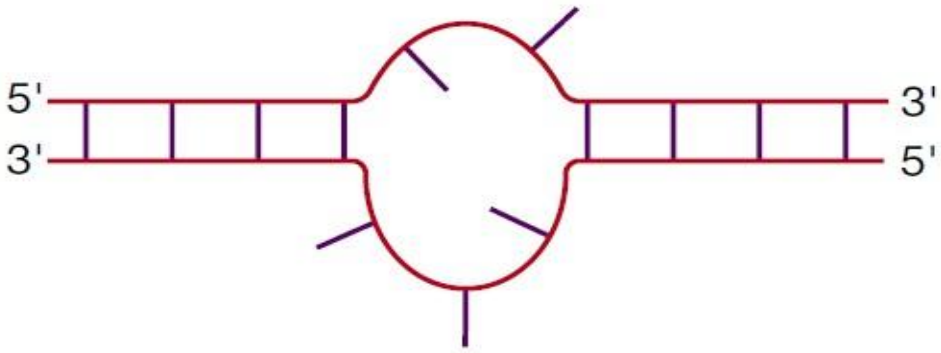
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D. Bulge loop



Mcq: Bulges, are formed when a double-stranded region cannot form base pairs perfectly. Bulges can be asymmetric **with varying number of base pairs** on one side of the loop. Bulge loops are commonly found in helical segments of cellular RNAs and used to measure the helical twist of RNA in solution. (Tang and Draper 1990)The forth type of 2' RNA structure is interior loop.

E. Interior loop



Mcq: Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.**(Turner, Sugimoto et al. 1988)**

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Module 008: TYPES OF RNA SECONDARY STRUCTURES-III

S. L : Q: Define or explain 2' RNA structure junction

Another 2' RNA structure is the **Junction or Intersection.**

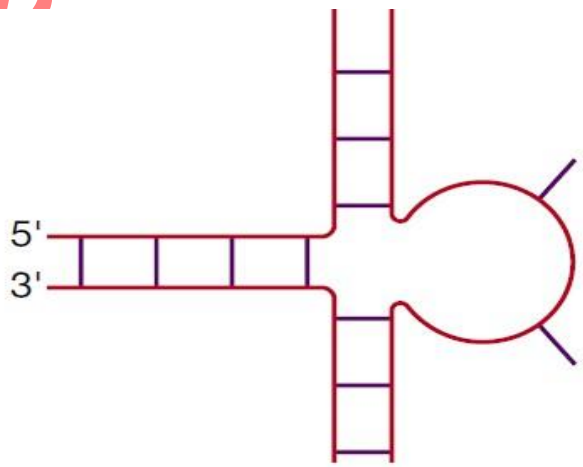


Figure 9 2' RNA structure called junction Z

S I

Junctions include two or more double-stranded regions converging to form a closed structure. The unpaired bases appear as a bulge.**(Zuker and Sankoff 1984)**

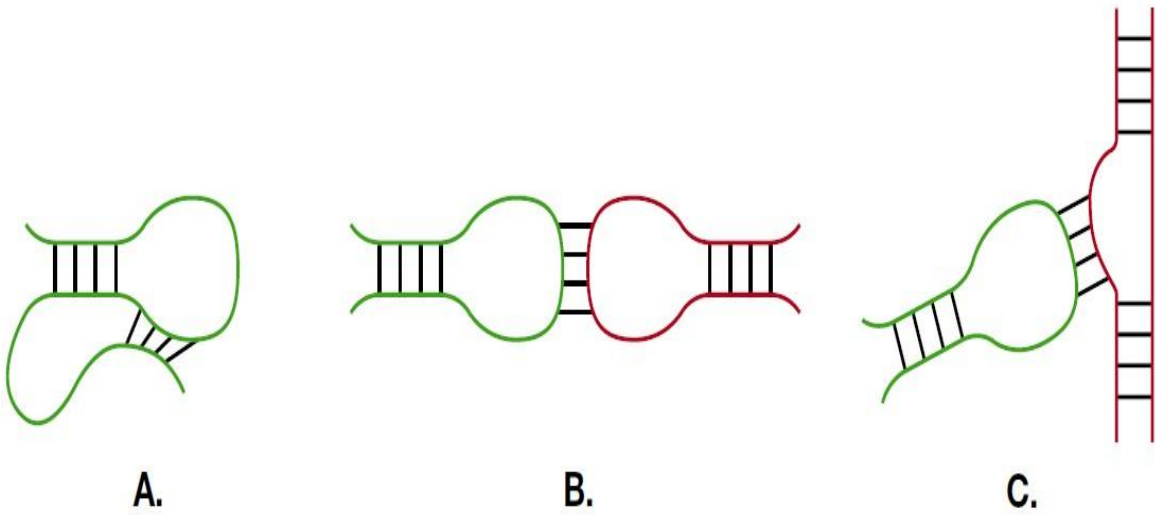


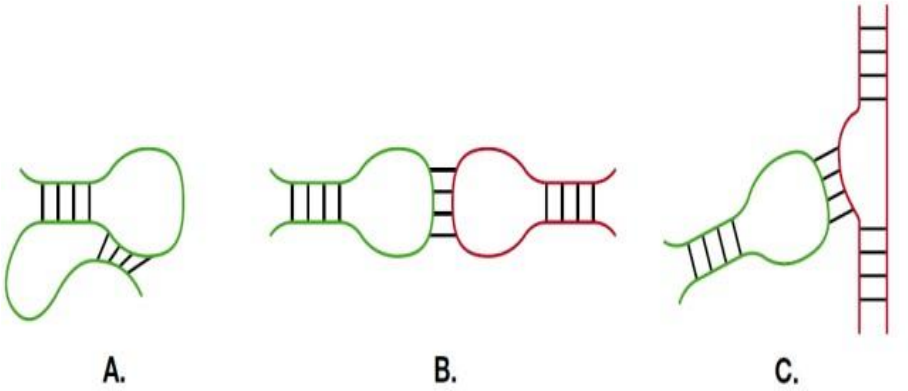
Figure 10 Unpaired bases in two 2' structures form hydrogen bonds with each other

RNA tertiary structures are formed when RNA unpaired base bond in 2' region bond.

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Module 009: RNA TERTIARY STRUCTURES

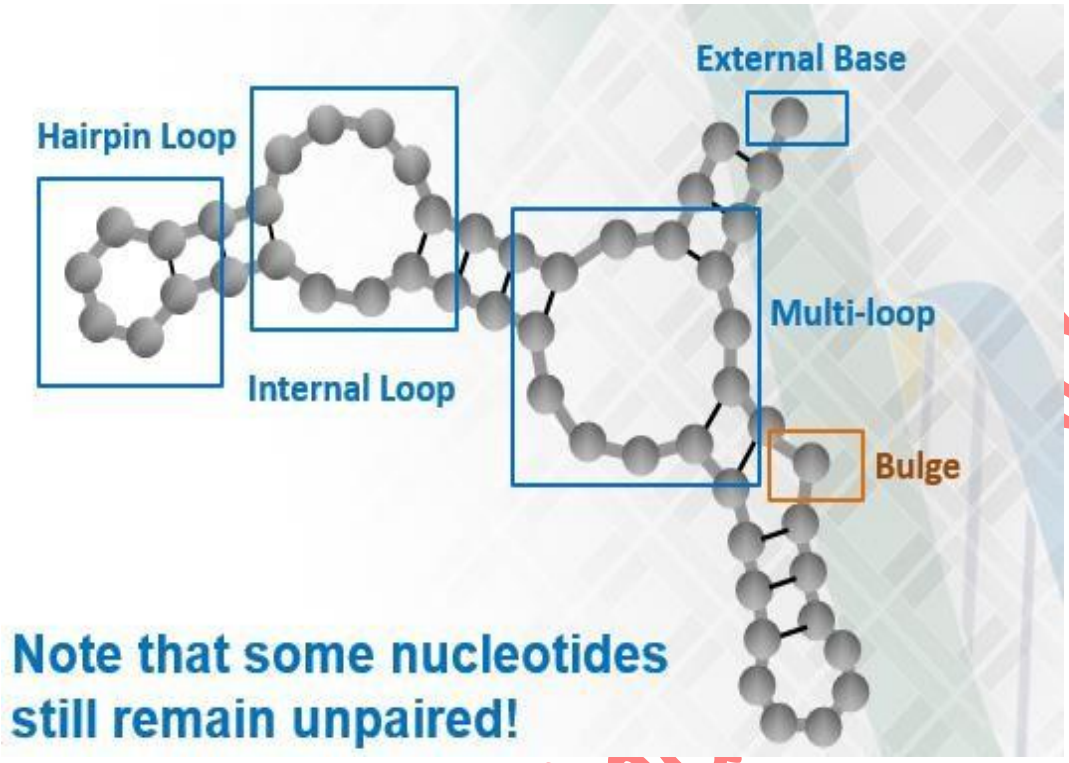
2’ RNA structures is formed due to folding of nucleotide with in RNA molecule but after folding some nucleotides remain open **for interaction.** And they form hydrogen bonds together.



s.q: Define structure tertiary 3’ structure

These unpaired nucleotides of 2’ structure interact with other unpaired nucleotides and form a third **structure called tertiary 3’** structure. For example 4 nucleotides in hairpin loop structure does.

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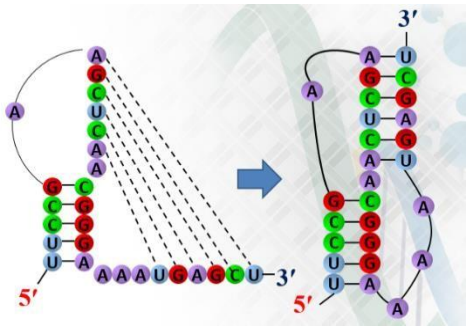
The above figure:

1. Indicate how these 2' structures come together
2. Indicate the difference between internal loop and multi loop
3. Indicate the yet **unpaired bases**

Question : Define pseudoknots)

The unpaired bases in 3' structure remain paired by abnormal folding called **(pseudoknots)** but instead of pairing they remain available or pairing.

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pseudoknots

Module 010: CIRCULAR REPRESENTATION OF STRUCTURES

Tertiary or 3’ structure of RNA may form pseudoknots to detect the pseudoknots in RNA structure we need “circular plot” which is a graphical approach.

Intersecting arcs in circular plot are the pseudoknot.

Question,: short and long :Name the Experiment METHODS TO DETERMINE RNA STRUCTURES? How

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many METHODS TO DETERMINE RNA STRUCTURES?

Module 011: EXPERIMENTAL METHODS TO DETERMINE RNA STRUCTURES

RNA has 1', 2' and 3' structures. 1' has simple nucleotide sequence and 2' has nucleotides folding and 3' has knots.

Question: What is X-ray crystallography

For measuring the RNA structure we use X-ray crystallography (Smyth and Martin 2000), which works according to the principle of diffraction. Crystallized RNA diffracts X-rays which helps estimate atomic positions

Mcq : All isotopes that contain an odd number of protons and/or of neutrons (see Isotope) have an **intrinsic magnetic moment and angular momentum,** in other words a nonzero spin, while all nuclides with even numbers of both have a total spin of zero. The most commonly studied nuclei are 1H and 13C, al

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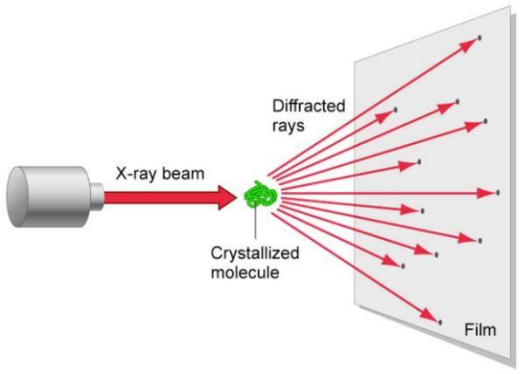


Figure 13 X-ray Crystallography https://260h.pbworks.com/w/page/30814223/X%20Ray%20Crystallography

Question: Explain the Atomic Force microscopy, define, uses?

Another method to measure the RNA structure is called as **Atomic Force microscopy** in this technique a laser connected to a Si_3N_4 piezoelectric probe scans an RNA sample. It **works well in air and liquid environment.**

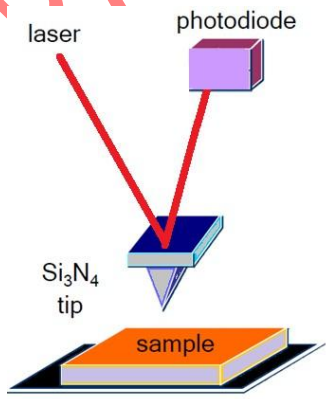
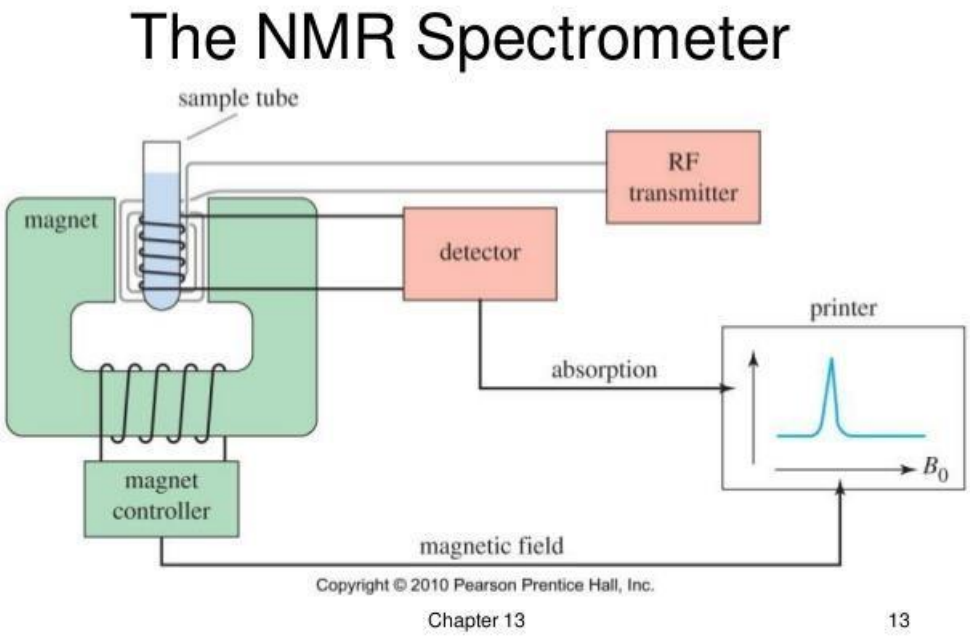


Figure 14 Atomic microscopy

Question: what is Nuclear Magnetic Resonance Imaging, uses definition, explanation?

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The third method for measuring the RNA structure is **Nuclear Magnetic Resonance Imaging in this method Hydrogen atoms in RNA resonate upon placement in a high magnetic field. It Works well without crystallizing RNA**



http://www.slideshare.net/Oatsmith/13-nuclear-magnetic-resonance-spectroscopy-wade-7th

. **STORAGE OF STRUCTURES**

Reported structures are stored in online databases. Example includes RNA Bricks and RMDB etc. Bioinformaticians can refer to these databases for RNA structure studies

MCQS: RNA Bricks is a database of…… **RNA 3D structure motifs** and their contacts, both with themselves and with proteins

Stanford University's RNA Mapping Database is an archive that contains results of diverse structural mapping experiments performed on ribonucleic acids.

N

Module 012: Strategies for RNA Structure Prediction

RNA structure 2' and 3' can be measured experimentally, but RNA molecule readily degrade due to their short shelf life.

Give 1' RNA structure creates the 2' structure because the simple nucleotides folds and form 2' structure.

Mcqs: And on the base of folding we can predict the **stability of the RNA molecule**.

For example.

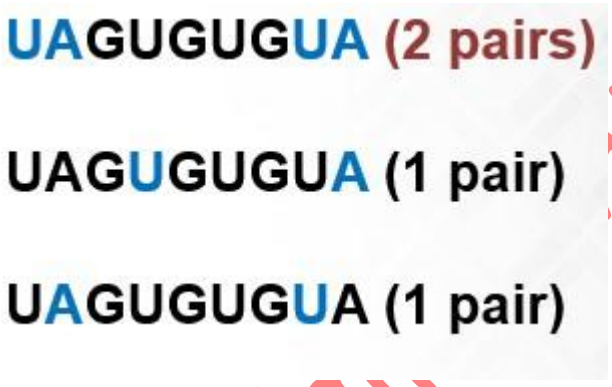


Figure 15 pairs represent the stability of the RNA molecule

Mcqs:

Maximizing the number of nucleotides can **increase the structure** and we have to select the structure according to the stability.

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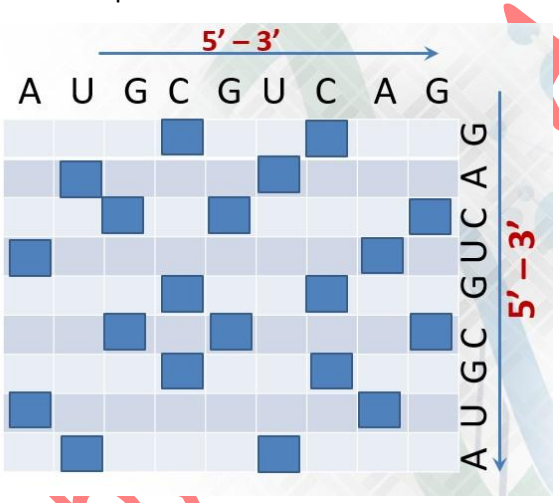
Module 013: Dot Plots for RNA 2' Structure Prediction

Structure measurement through experiments is slow and costly and there is maximum chances of more than one structure existence.

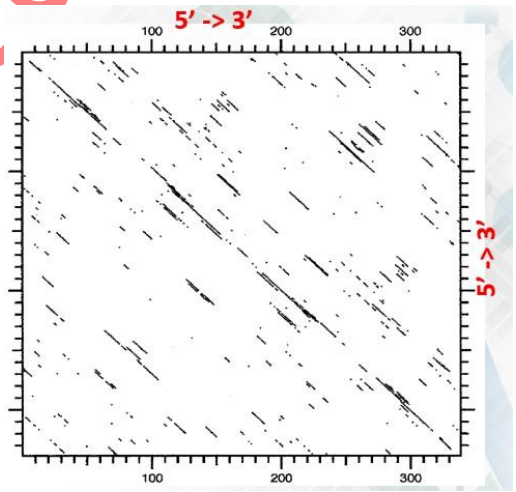
S,& L Question: Describe the dot plot method for rna structure prediction

s.q Use of dot plot method?

The dot plot method for RNA structure prediction is easy. Draw a square and partition by drawing gridlines. Put RNA sequence on top and left sides of the square. Put a "DOT" on complementary nucleotides For example:



Connect regions of paired nucleotides to form 2' structures in following image.



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In longest RNA nucleotides the gaps between complementary nucleotides become bulges and loops of the structure.

Module 014: ENERGY BASES METHODS

Mcqs: Experimental prediction of RNA structure is **slow and costly** that's why a few 2' RNA structures are reported experimentally.

While prediction we get many possible 2' structures of RNA and for optimal structure selection we calculate their overall **stability**.

	Y:	A	C	G	U
X:	A	.	.	.	-2.1
	C	.	.	-3.3	.
	G	.	-2.4	.	-1.4
	U	-2.1	.	-2.1	.

Figure 18 energy table

Long Q: What is STABILIZING ENERGY and DESTABILIZING ENERGY of RNA Structure predictiuon?

- **STABILIZING ENERGY**

Energy table helps us to find the optimal prediction of structure because energy is released when complementary nucleotides make bonds.

- **DESTABILIZING ENERGY**

Remaining unpaired nucleotides destabilized the RNA structure in form of hairpin or bulge structure.

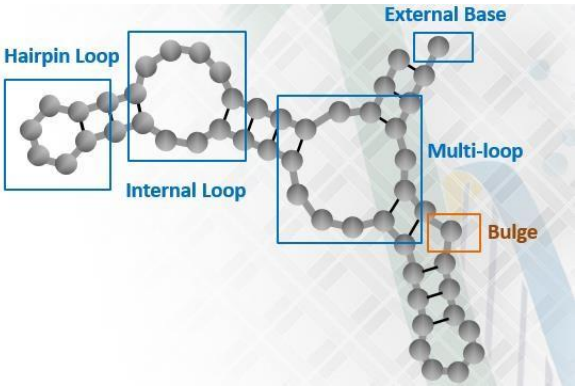


Figure 19 Hairpin+IntLoop+ExtLoop+Bulge+Hairpin

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- **SUM OF ENERGIES**

Sum of stabilizing and destabilizing energies can help determine the quality of a 2' RNA structure. 2' structure with longest coupled sequences vs. one with lowest energy

Module 015: Zuker's Algorithm

L. S. Q: Define or describe Zuker's Algorithm?

Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm.

Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and –ve energies.

	Y:	A	C	G	U
X:	A	.	.	.	-2.1
	C	.	.	-3.3	.
	G	.	-2.4	.	-1.4
	U	-2.1	.	-2.1	.

Figure 20 stacking energies

Figure 21 destabilizing energies

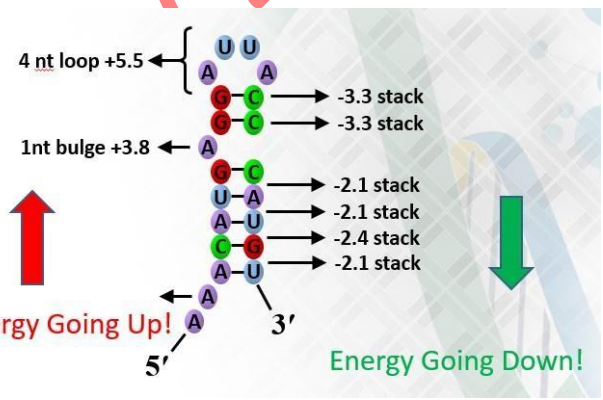


Figure 22 working principle of Zuker's Algorithm (2003)

It Compute energies of all possible 2' structures. Generate combinations of all computed 2' structures. Select the one with lowest energy.

Module 016: Zuker's Algorithm EXAMPLE

Zuker's Algorithm involves computing stabilizing and destabilizing energies of a 2' structure. All possible 2' structures are generated. The best 2' structure is selected!

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Module 018: Martinez Algorithm

S, Q: Difference between Zuker's Algorithm and Martinez Algorithm

Zuker's Algorithm involves computing **stabilizing and destabilizing energies of 2' structure**. And it also computes the overall energy by summing up the positive and negative energies. Martinez Algorithm is improvement on it.

Making combination of all possible structures is time consuming**, Martinez Algorithm favors those 2' structure which are energetically more feasible**.

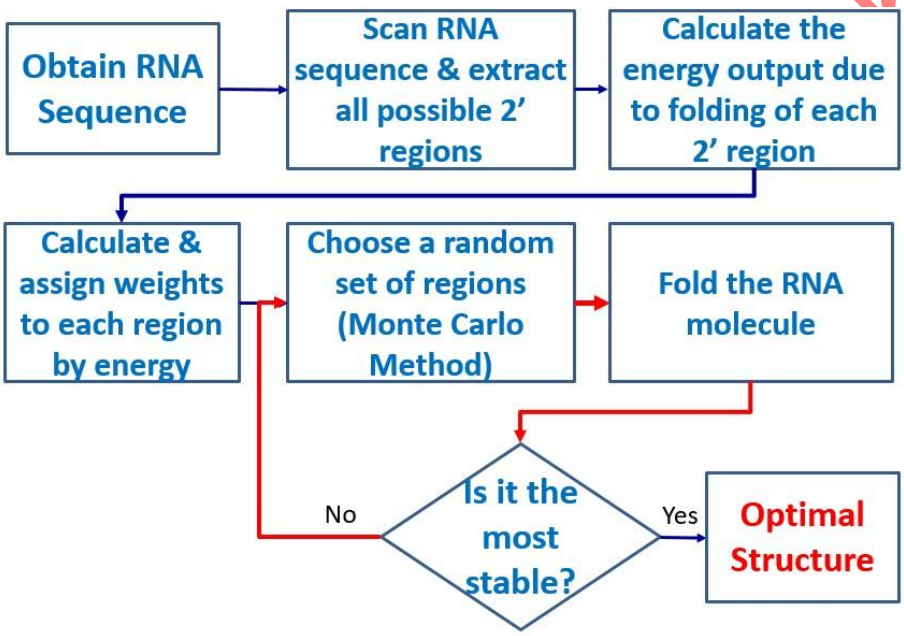


Figure 25 Martinez Algorithm flow chart

In Martinez algorithm all the 2' structures are weighed by its stability and **optimal one** is sorted out. Monte Carlo methods (or Monte Carlo experiments) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other mathematical methods.

And Monto Carlo method do not provide a definitive solution.

Module 019: Dynamic Programming Approaches

RNA sequence **contains 4 type of nucleotides G/C, G/U and A/U** and it may contain hundreds of nucleotides it means there is possibility of many combinations.

In 2' RNA structure there may be large number of nucleotide sequence with large number of combinations hence it is hard to find the optimal one and for this prediction we us Dynamic Programming (DP) which breaks the larger problems into smaller one.

S & L .Q: Describe the PRINCIPLE OF DYNAMIC PROGRAMMING

S.Q: why we use dynamic programming? Uses of dynamic programming?

For optimal structure combination selection we use the Dynamic Programming (DP) and we select the sequence of RNA nucleotides and list all the possible complementary positions for nucleotides in the given complete sequence.

Dynamic Programming then recombines such combinations in a process called "**Traceback**" to ensure that the highest coupled 2' structure is reported

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-Jacobson Algorithm –

Module 020: Nussinov An Overview

S.L.Q: Define and describe the useses of Nussinov?

Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal **RNA 2' structures, Proposed in 1980**. Computes 2' structures with most nucleotide coupling.

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HOW IT WORKS

Steps:

- Create a matrix with RNA sequences on top and right
- Set diagonal & lower tri-diagonal to zero
- Start filling each empty position in matrix by choosing the maximum of 4 scores

	J	1	2	3	4	5	6	7	8	9	
I		G	G	C	A	A	A	U	G	C	
1	G	0									
2	G	0	0								
3	C		0	0							A
4	A			0	0						
5	A				0	0					G
6	A					0	0				
7	U						0	0			
8	G							0	0		
9	C								0	0	

Figure 27 A.Note the I and J labels, B. Initialize tri-diagonal and lower tri-diagonals to zero.

The score $S(i, j)$ is the maximum of the following four possibilities

-Jacobson Algorithm –

Module 021: Nussinov

The Flowchart NJ algorithm is actually a dynamic programming (DP) approach to predict the 2' RNA structure. A scoring matrix is initialized to record scores in NJ Algorithm .For filling scoring matrix, the maximum score from 4 matrix positions is chosen.

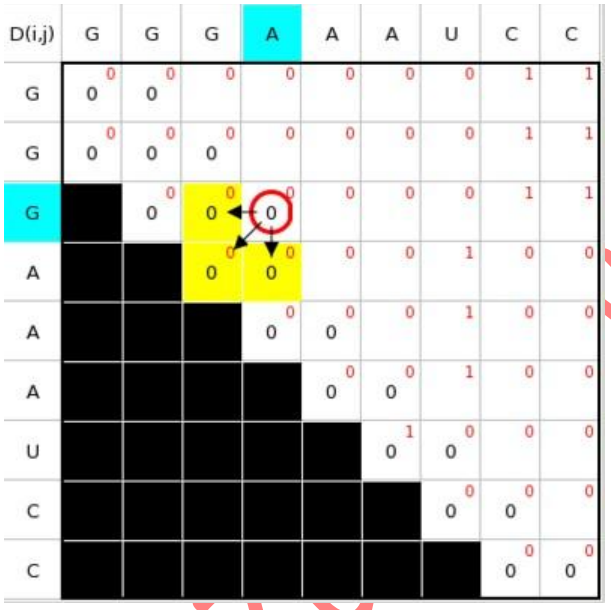


Figure 0.3 for maximum score 4 positions are used in scoring

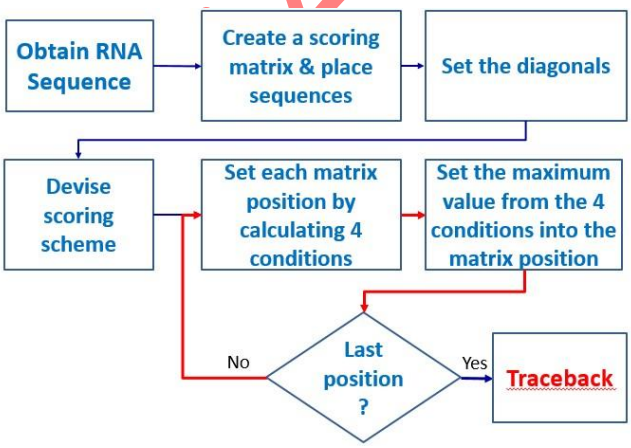


Figure 28 flow chart of NJ Algorithm.

MCQ: Traceback is used to **report the coupling of structures in sequences**.

-Jacobson Algorithm –

Module 022: Nussinov EXAMPLE

S.Q: The **main points to be focused in N-J Algorithm are:**

- Scoring Matrix
- Matrix Initialization
- Scoring method
- The 4 different positions to be considered for calculating matrix

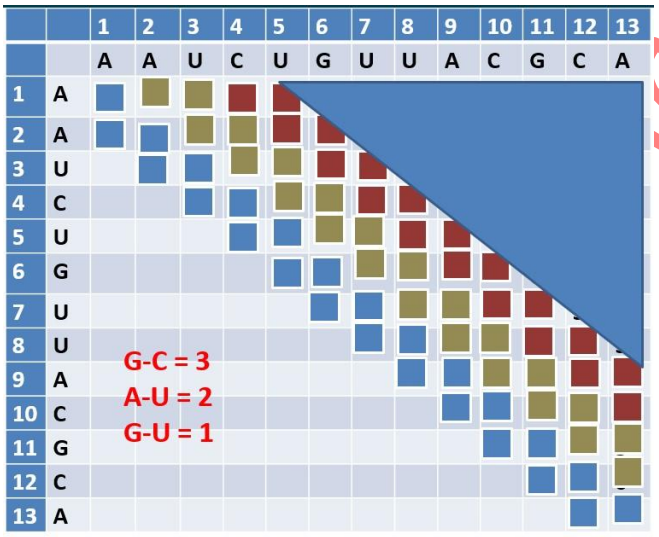


Figure 29 N-J Algorithm scoring

Mcq: The matrix is filled by **four different** positions. **Left, Bottom, Diagonal, and Left/Bottom elements.** In this way all complementary nucleotides coupling is catered.

Prepared by

-Jacobson Algorithm –

Prepared by Zainab Arshad

Module 023: Score Calculations & Traceback

From four positions the score is calculated and from each position we calculate the score contribution. And maximum score is sorted out.

		1	2	3	4	5	6	7	8	9	10	11	12	13
		A	A	U	C	U	G	U	U	A	C	G	C	A
1	A	0	0	2	2	4	5	7	7	9	9	12	12	12
2	A	0	0	2	2	2	5	5	5	7	7	10	10	12
3	U		0	0	0	0	3	3	3	5	5	8	8	10
4	C			0	0	0	3	3	3	5	5	8	8	10
5	U				0	0	1	1	1	3	5	6	8	10
6	G					0	0	1	1	3	5	6	8	8
7	U						0	0	0	2	2	5	5	7
8	U							0	0	2	2	5	5	5
9	A	G-C = 3							0	0	0	3	3	3
10	C	A-U = 2								0	0	3	3	3
11	G	G-U = 1									0	0	3	3
12	C											0	0	0
13	A	4th condition of max{S(i,k)+S(k+1,j)}.											0	0

Figure 30 scoring and traceback in N-J Algorithm.

There can be many traceback.

S,Q: Use of trace back?

MCQ: Each traceback is used to make the **RNA secondary structure**. And traceback with highest number of nucleotide coupling is selected.

Prepared by Zia ul Haq

Module 024: Comparison of Algorithms

RNA has three different structures 1', 2' and 3'. For these structures predictions there are many algorithms.

S.Q: But in all algorithm **there are two main strategies:**

1. **Nucleotides stacking**
2. **Energy minimization**

Answer: **Comparasioin between the Nussinov An Overview and Energy minimization**

Nussinov An Overview **MCQ: Zuker's Algorithm** involves **energy minimization**. It is updated version and incorporate the phylogenetic information. It is improved. Overcomes the pseudo knots assumes them and accommodate them. And this algorithm helps to predict the structures of RNA based on nucleotides.

- **NUCLEOTIDES STACKING ALGORITHM.**

NJ's Algorithm comes under this category. It involves the maximizing the nucleotides pairing. Traceback helps to find best 2' structure.

Mcq: It predicts the **75% accurate 2' structure**. Because there may be more than two equal scores as it is calculated from four different positions. To get best results we need to combine the stacking and energy minimization methods together.

For further improvements in results we take help from:

- Sequences
- Comparison
- Nucleotide
- Covariance analysis

Module 025: WEB RESOURCES-I

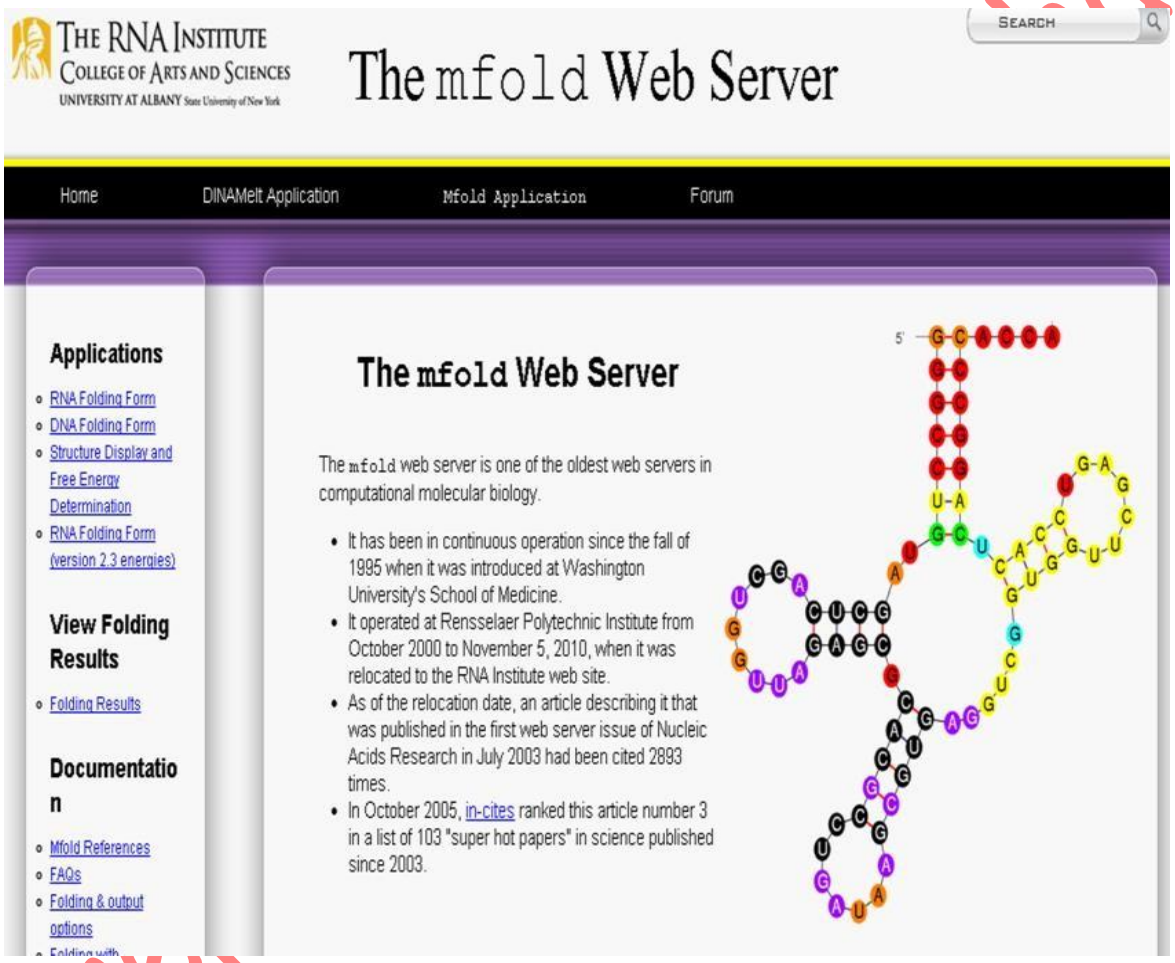
Prepared by Zainab Arshad

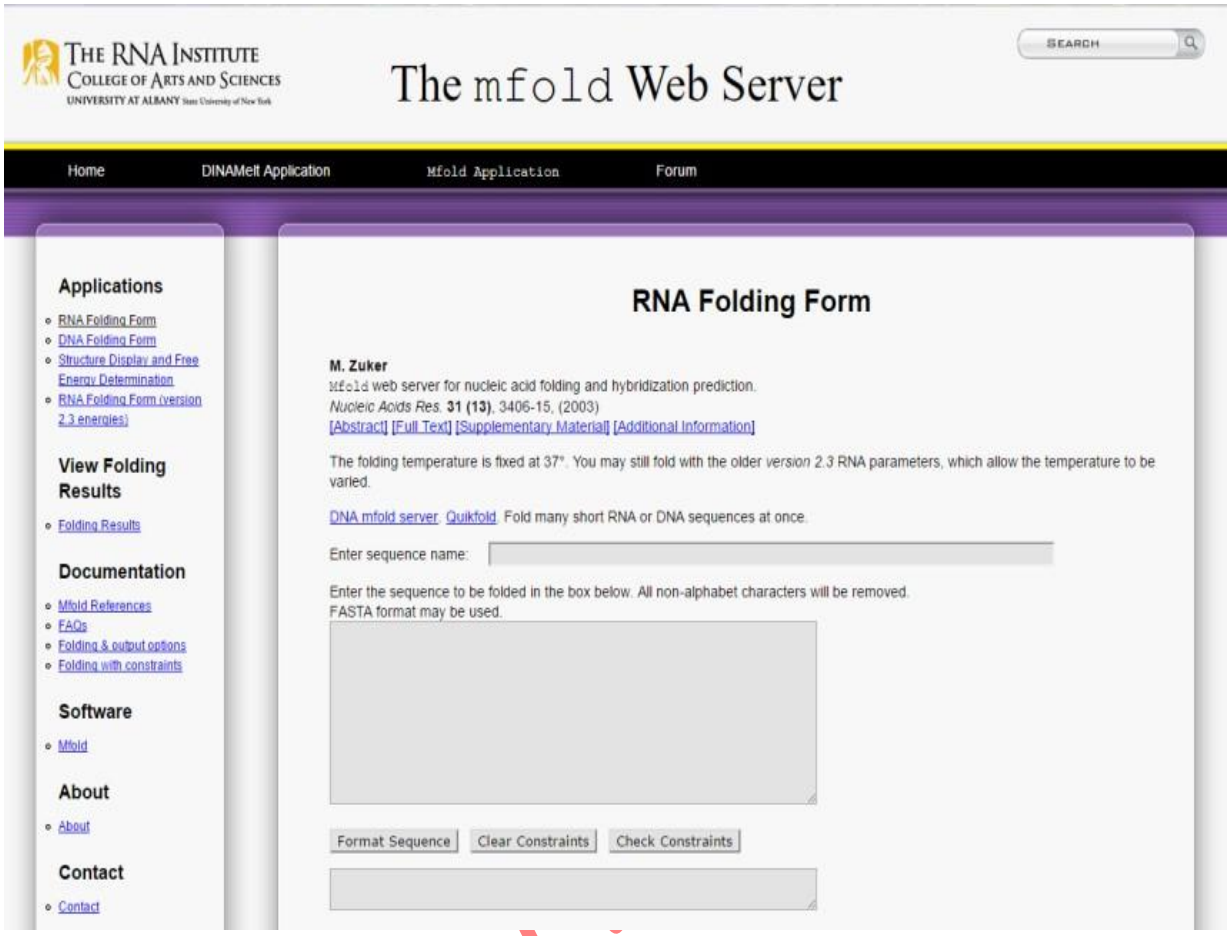
For prediction of 1' and 2' structure of RNA we use different algorithm like Zuker's, Martinez and N-J. Online tools also.

S.Q: Define and uses:

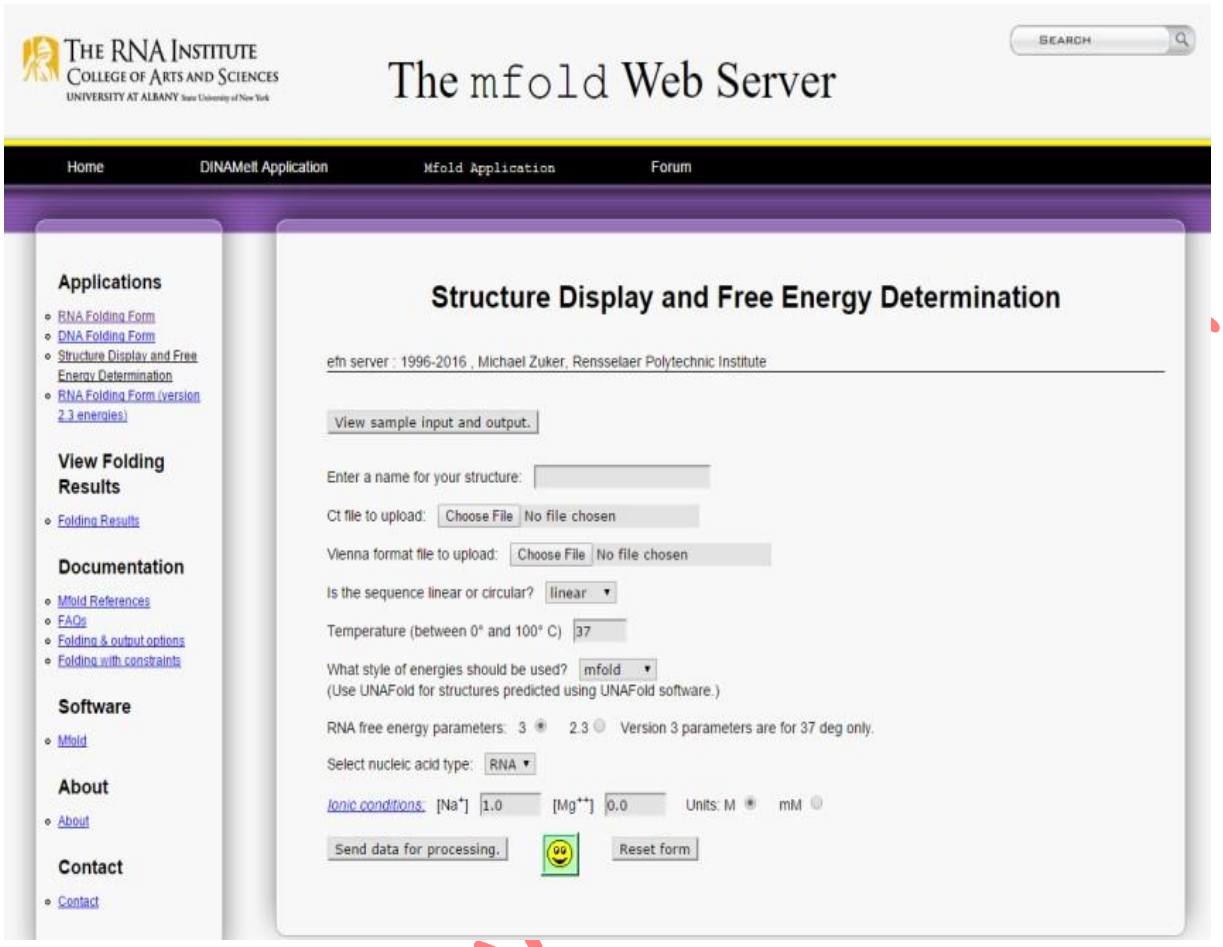
MCQS: The mfold web server is one of the oldest web servers in computational molecular biology. Mfold is upgraded version of Zuker's algorithm.

MCQS: MFOLD is computationally expensive and can give results for **1' and 2'** structures that have sequences less than **8000 nucleotides.**




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RNA Folding Form
DNA Folding Form
Structure Display and Free Energy Determination
RNA Folding Form (version 2.3 energies)
View Folding Results
Folding Results
Documentation
Mfold References
FAQs
Folding & output options
Folding with constraints
Software
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Contact
RNA Folding Form
M. Zuker
Mfold web server for nucleic acid folding and hybridization prediction.
Nucleic Acids Res. 31 (13), 3406-15, (2003)
[Abstract] [Full Text] [Supplementary Material] [Additional Information]
The folding temperature is fixed at 37°. You may still fold with the older version 2.3 RNA parameters, which allow the temperature to be varied.
DNA mfold server. Quikfold. Fold many short RNA or DNA sequences at once.
Enter sequence name:
Enter the sequence to be folded in the box below. All non-alphabet characters will be removed.
FASTA format may be used.
Format Sequence
Clear Constraints
Check Constraints

Prepared by

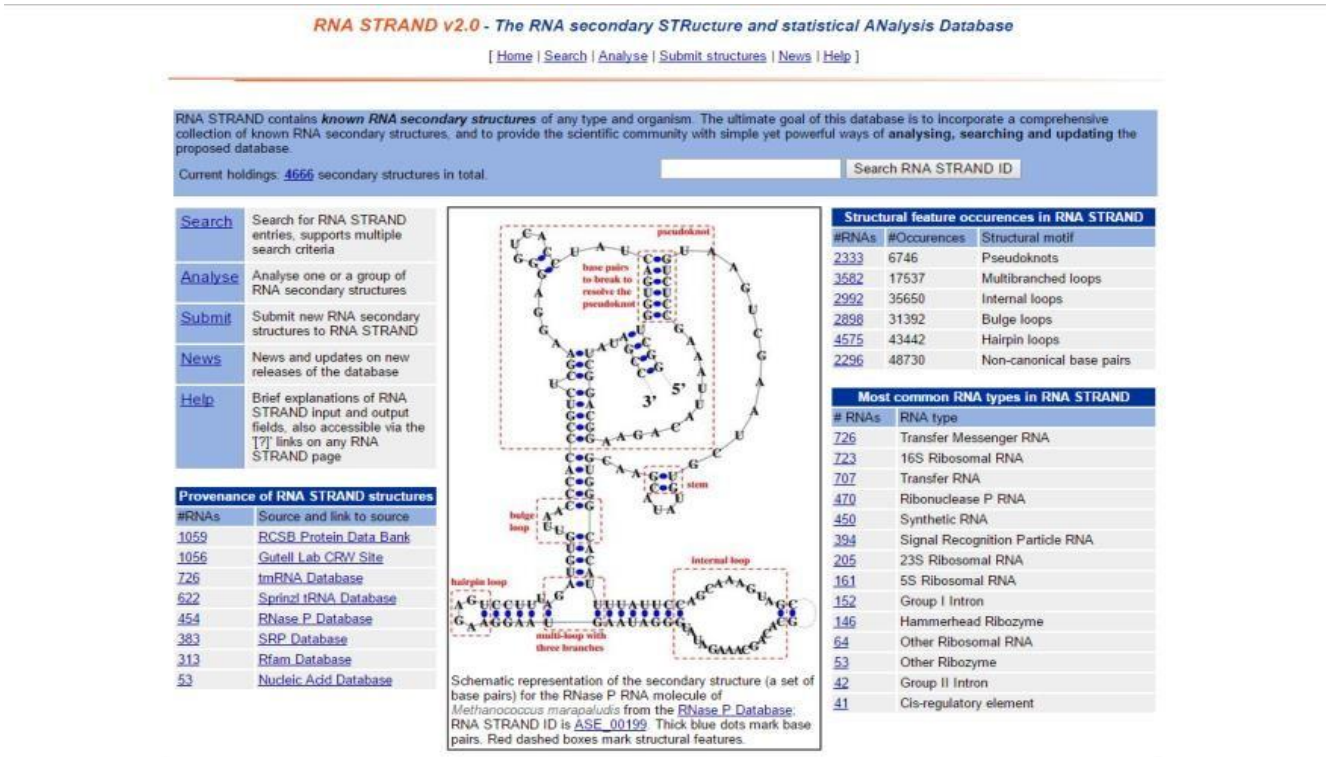


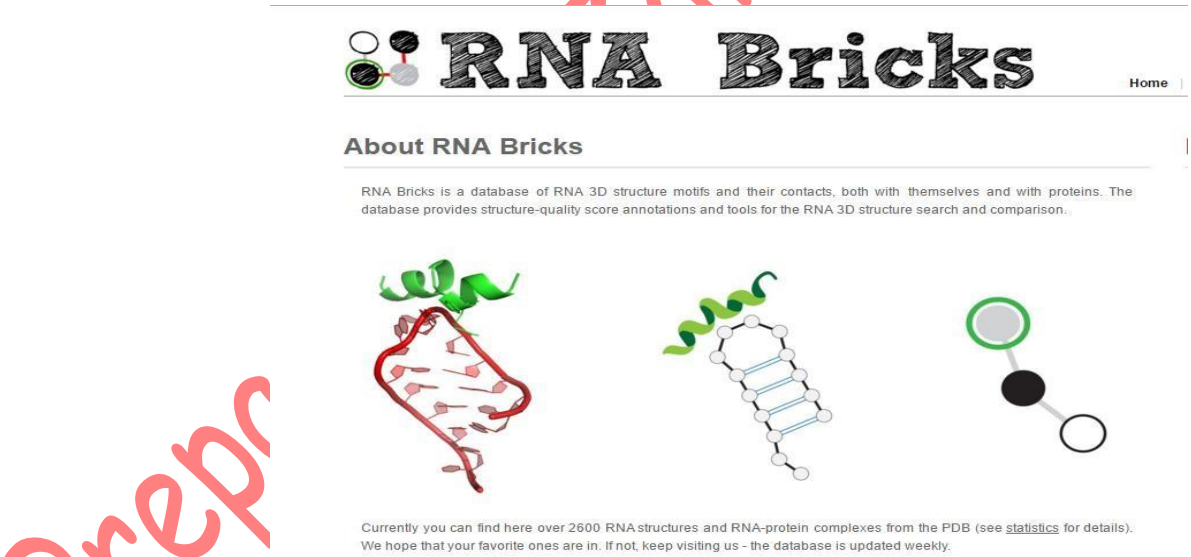
Mcqs: MFOLD helps fold an **RNA nucleotide sequence** into its possible 2' structures. MFOLD gives out several structures along with their energetic stability

Module 026: WEB RESOURCES-II

RNA nucleotides folds to form **2' structure** from simple portion of 1' nucleotides. For example CUUCGG occurs a wide variety in RNA and it mostly forms the stable hairpin loop. So we can make the list of all likely 2' structures arising from 1'.

Prepared by





RNA 1' folds and makes **RNA 2' structure** and this online database is established for 2' RNA structure and it act as dictionary for 2' RNA structure.

Chapter 5 - Protein Sequences

Module 001: FREFINE OM DNA/RNA SEQUENCES TO PROTEIN

MSQ: & S.Q : define central dogma.

We are aware that DNA has four **nucleotides bases (A, C, T & G). RNA contains (A, C, U & G).** And **protein contains 20 different amino acids**. **DNA to RNA then Protein is called as central dogma. This includes translation, transcription and protein modifications.**

Mcq & Defination: A set of three nucleotides called codon, codes the information for specific amino acids in protein synthesis.

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Figure 0.2 codon (set of three nucleotide) codes for specific amino acid.

s.q What is the Function of codon?

Cordons select the amino acids and ribosomes make the protein by polymerization process and these nucleotides coil together to form 3D structure.

Prepared by ... shad

Module 002: CODING OF AMINO ACIDS

Nucleotides (A, G, C, and T) make set of three called codons for amino acid selection in protein synthesis. **More than one codon can code for same amino acids as there are 20 amino acids involved in protein synthesis.**

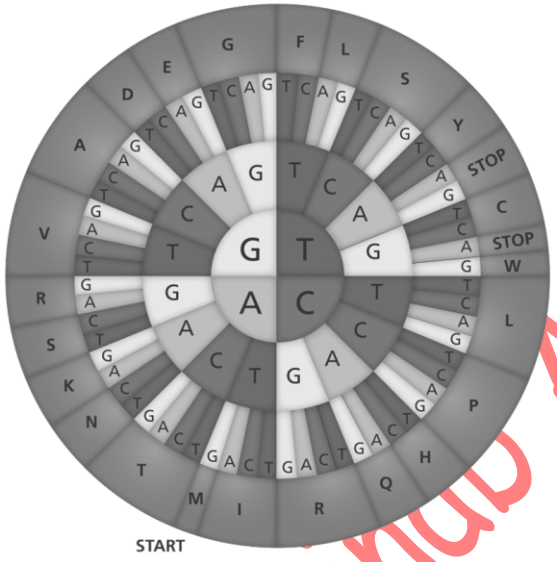


Figure 3 coding of amino acids

Prepared by Zainab Arshad

What is OPEN READING FRAMES?

Module 003: OPEN READING FRAMES

Codons codes information for amino acid and there are three stop codons and one start codon. For the valid open reading frame it must have longest sequence.

In molecular genetics, **an open reading frame (ORF) is the part of a reading frame that has the potential to code for a protein or peptide. An ORF is a continuous stretch of codons that do not contain a stop codon (usually UAA, UAG or UGA).**

https://en.wikipedia.org/wiki/Open_reading_frame

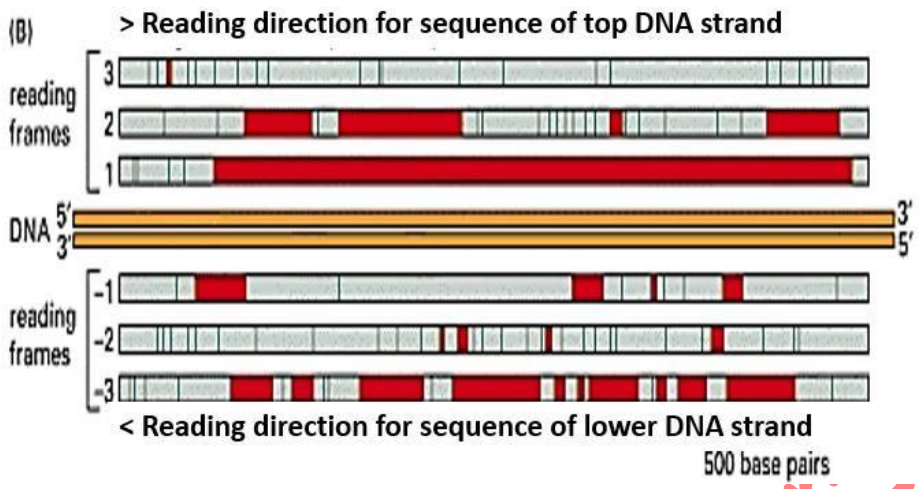


Figure 5 ORF 1 is valid, as it is the longest

There is online tool from which we can find ORF in any sequence.

MCqs: Six ORF exist in any DNA sequence and longest one is marked and first stop codon will be marked end of the protein.

Module 004: ORF Extraction – A Flowchart

Codons of 3 nucleotides code for each Amino Acid. There are **1 start and 3 stop codons**. Selection of ORF **is based on its length if it the longest one from others** than it would be suitable for protein synthesis reaction.

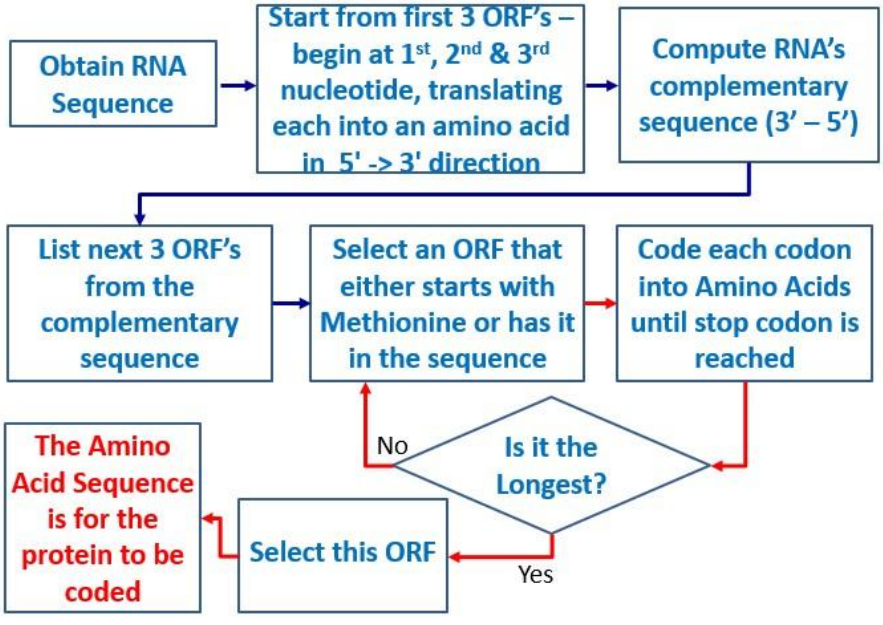


Figure 7 ORF extraction flowchart

Both reverse and forward RNA sequences are considered which may have many ORF and selection is based upon longest protein sequences having.

Module 005: SEQUENCING PROTEINS

S.Q: why we use Adam degradation method in protein sequencing?

Given the DNA/RNA sequence**, ORFs can be extracted and protein sequence** can be determined. **But there are chances that protein may be unknown, that's why we use Adam degradation method in protein sequencing**.

MCQ: **Edman degradation, developed by Pehr Edman, is a method of sequencing amino acids in a peptide.**

In this method, the amino-terminal residue is labeled and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues.it starting from the N-terminal and removing one amino acid at a time

Cyclic degradation of peptides by Phenyl-iso-thio-cyanate (PhNCS). PhNCS attaches to the free amino group at N-terminal residue. 1 amino acid is removed as a PhNCS derivative.

S.Q &: DRAWBACKS Edman degradation,

- It is restricted to chain of 60 residues.
- It is very time consuming process 40-50 amino acids per day.

Modern techniques for this are Tandem mass spectrometry.

Module 006: Application of Mass Spectrometry in Protein Sequencing

S.Q: WHAT ARE THE ADVANTGAGE Edam Degradation methods: Edam Degradation methods helps us to sequence the protein which is unknown. But it is restricted to 60 amino acids only.

Protein can be charged with electrons or protons and if moving charges are placed in between the magnetic field they get deflected. And their deflection is proportional to their momentum.

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \text{(Lorentz force law)}$$

$$\mathbf{F} = m\mathbf{a} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \quad \text{(Newton's second law of motion)}$$

Figure 10 Moving charged particles in a magnetic field

Where:

Prepared by Zainab Arshad

- **F** is the force applied to the ion
- **m** is the mass of the particle,
- **a** is the acceleration
- **Q** is the electric charge,
- **E** is the electric field
- **v × B** is the cross product of the ion's velocity and the magnetic flux density.

COMPUNENTS

- Sample Injection
- Ionization Source
- Mass Analyzer
- Ion Detector
- Spectra search using computational tools

CONCLUSION

Charged proteins can be set into motion within a **magnetic field**. Their deflections accurately correspond to their **molecular mass**. Deflections can be measured (hence protein's mass)

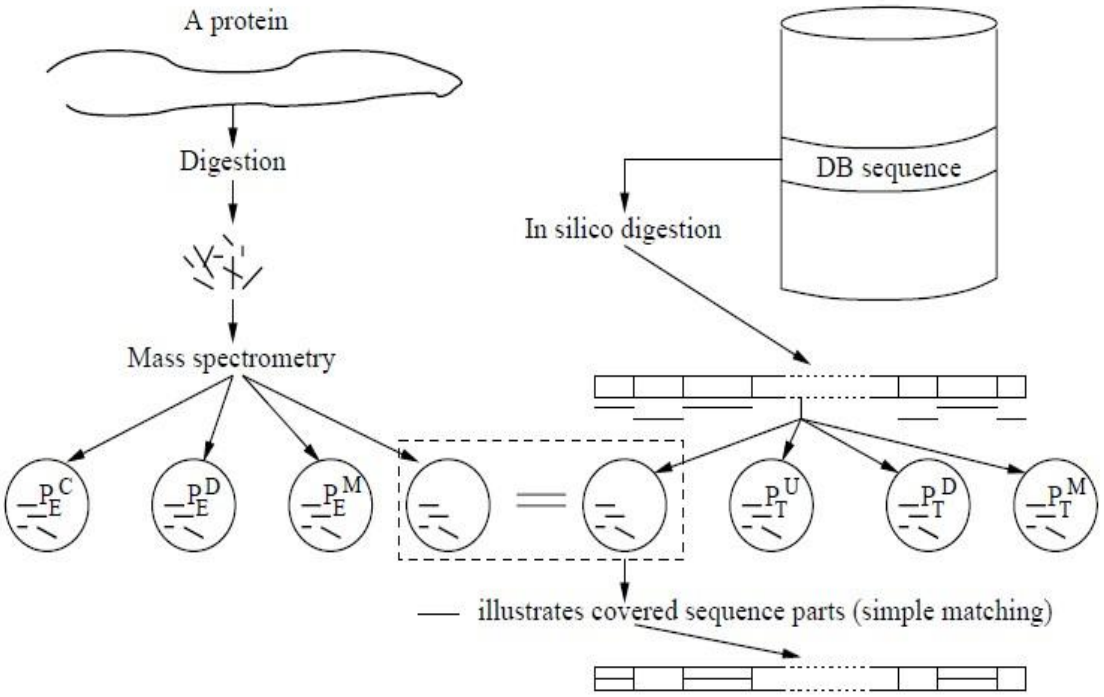
Module 007: Techniques for MS Proteomics

Question: Describe the techniques for MS Proteomics:

MCQ: MS proteomics works on the **principle** of protein ionization which is placed in very high **magnetic field**. Each protein deflects to its **proportion** which is equal to its molecular weight in this way **molecular mass** is measured. The **protein mass** of unknown protein is compared with the **masses** of proteins in database and matching one is selected.

MCQ: Example for protein sequences database is **uniProt, swissprot** etc.

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Proteins are measured and sequenced if are unknown than matched with existing database if matched than are shortlisted.

Module 008: Types of MS-based proteomics

Explain:

Proteins can be sequenced by **Edam's degradation and Mass spectrometry**. MS based proteomics helps us to sequence the larger and bigger proteins more quickly.

Following steps are involved in MS:

- Separation
- Ionization
- Mass analysis
- Detection

S.Q & MCQs: Two methodologies are involved

1. Bottom up proteomics
2. Top down proteomics

Bottom up proteomics measures the peptide masses produced after protein enzymatic digestion. And Top down proteomics measures the intact proteins followed by peptides after fragmentation.

Prepared by Zaina

S.Q & L.Q: BOTTOM UP PROTEOMICS

In this methodology the protein complex is treated with site specific enzymes which cleave them into **amino acid residue and resultant peptides** are measured for their masses. One peptide is selected at one time for processing and when all are processed than protein search engine is used for matches.

S.Q & L.Q: TOP DOWN PROTEOMICS

In this methodology proteins are ionized and measured for their masses and one protein is mass selected at a time for **fragmentation.** And resultant peptide fragments are measured for mass.

We can say that **bottom up proteomics** deals with peptides while **top down proteomics** can handle the whole protein.

Module 009: BOTTOM UP PROTEOMICS

S.Q & L.Q: There are two types of proteomics protocol that are usually employed.

1. **Bottom up proteomics**
2. **Top down proteomics**

PROTOCOL

MCQs:

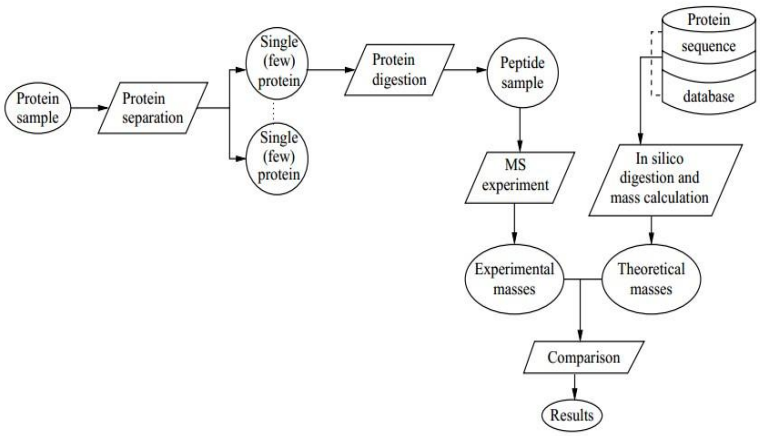
1. Sample containing the mixture of protein from cells and tissues is obtained.
2. **MCqs**. Enzymes such as **trypsin** is use to **cleave** the proteins.
3. Enzyme **cleaves the amino acids** at specific sites of amino acid.
4. **Several peptides** are formed when protein is cleaved.
5. Number of peptide depends upon the number of **sites** where **enzymes cleaved** the protein. For example trypsin cleaves the protein at **lysine (k)**
6. Mass of each peptide is measured.
7. **One** peptide is selected at a time.
8. Different enzyme is use to cleave the protein at different site.
9. This process keep going until the **possible number of peptides** are formed or searched.
10. Peptides are searched in data base and matched.

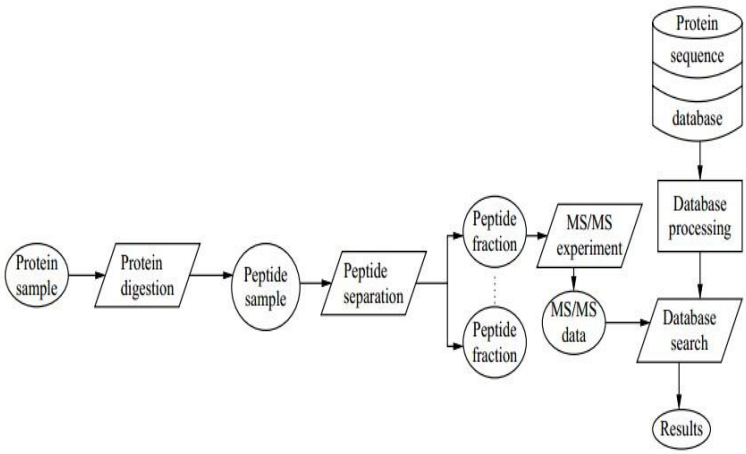
Module 010: Two Approaches for Bottom up Proteomics (BUP)

There are two approaches for **bottom up proteomics**.

Prepared by Zainab Arshad

1. **Peptide Mass Fingerprinting.**
2. **Shotgun Proteomics**





Shotgun Proteomics digest the whole protein and mix first and compared with database. And peptide mass **finger printing** involves in protein separation followed by single protein's peptide analysis.

L.Q & S.Q: Module 011: TOP DOWN PROTEOMICS

Bottom up **proteomics identifies** the proteins by cleaving them into segments at specific sites and was not suitable to measure the **direct protein masse**.

PROTOCOL

1. Sample **containing** the protein mixture from cells and tissues is obtained
2. The entire protein is mixed and analyzed for masses.

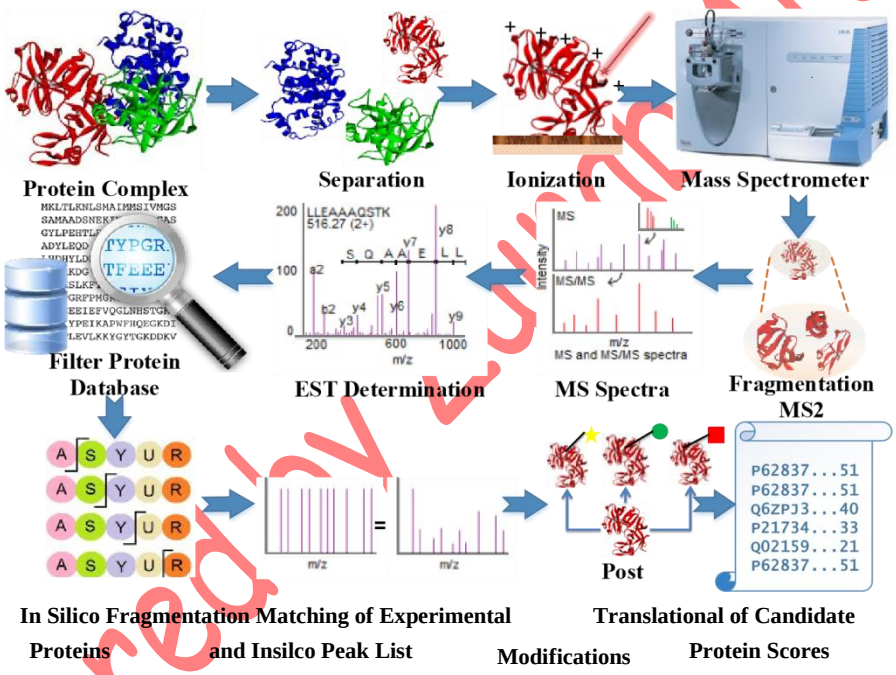
3. The list of masses is obtained.
4. **TDP** Measures all post translational masses of protein.
5. After **MS1 one protein** is selected at a time and fragmented to obtain its peptides.
6. The process is repeated many times.

Comparison is done from protein database **uniProt and swissProt**.

TDP also measure the masses of intact proteins and masses of **post transcriptional** changes.

Module 012: PROTEIN SEQUENCE IDENTIFICATION:

MCQs: Mass spectrometry helps us to measure the **molecular weight** of **proteins and peptides**, but several proteins can have same masses to identify them we follow the flow chart of following techniques.



The flowcharts discussed above can help us arrive at the sequence of the protein in question. Scoring schemes are required to quantitatively represent the quality of results

Module 013: PROTEIN IONIZATION TECHNIQUES

Protein ionization is used in **Mass spectrometry** based on proteomics protocols. Ionization involves loading of proton in protein or removal of protein. Ionizations can increase or decrease the **mass** of protein or peptide.

Prepared by Zuhair Ahmad

SALIENT IONIZATION

Is the technique which includes Matrix Assisted Laser Desorption Ionization MALDI) & Electro Spray Ionization (ESI) for example:

$[M+H]^+, [M], [M-H]^-$

monoisotopic or average masses

MALDI

In this technique one proton is added to **protein or peptide** and the molecular weight is increases by one and **Mass spectrometry** reports the molecule at **+1.**

ESI

ESI adds many protons to protein or peptides and molecular weight is increased by the number of protons added. But it is **difficult in ESI** to find the molecule with **+1.**

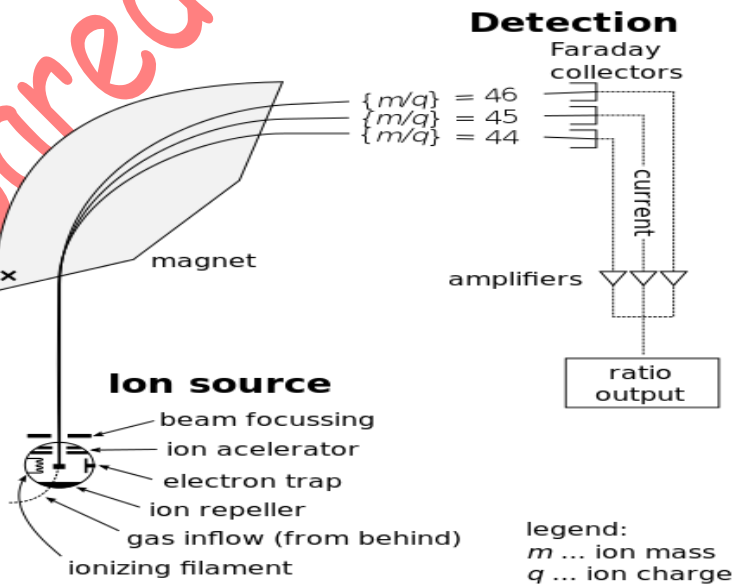
EXAMPLE

MS data from MALDI ionization is easier to handle as the product ions masses are mostly at **"1+mass".**

ESI is difficult to use as it does not easily give away **the +1 charged ion**

Module 014: MS1 & INTACT PROTEIN MASS :

When we ionize the protein, it can be **deflected by a magnetic field** in proportion to its mass and the **mass of protein** can be measured by spectrometry.



Prepared by Zainab Arshad

Mass/charge helps us to calculate the **mass of protein; "Mass Select**" can help to select specific MS1 for further analysis.

MS1 results the **intact masses of the peptides**.

Module 015: SCORING INTACT PROTEIN MASS

MS1 helps us to obtain the intact masses of precursor molecules which depend upon the **proteomics and protocol** applied. Protein masses reported by MS1 are matched with protein database, but before match the masses are converted **into +1 of all molecules.**

- **SCORING**

MCQs: We can score each protein in the way that it gets maximum score and **low quality matches should get low scores.**

MCQs: After **filtering the multiple** charges we get the only the peaks having charge 1

. And after this filter we compare it with protein data base.

- **SCORING SCHEME**

MCQ: All experimental masses are compared with **theoretical masses** of database and mass is selected on the base of **closeness.**

Module 016: PROTEIN FRAGMENTATION TECHNIQUES:

We compare the experimental mass with **theoretical data base mass** of protein and on base of closeness we rank or score it.

If several proteins have same score than selection is done by using another technique **protein fragmentation**. We fragment the protein or peptide and **ionize** it, it helps us to **measure** the **fragment masses** as the same ways as their precursor.

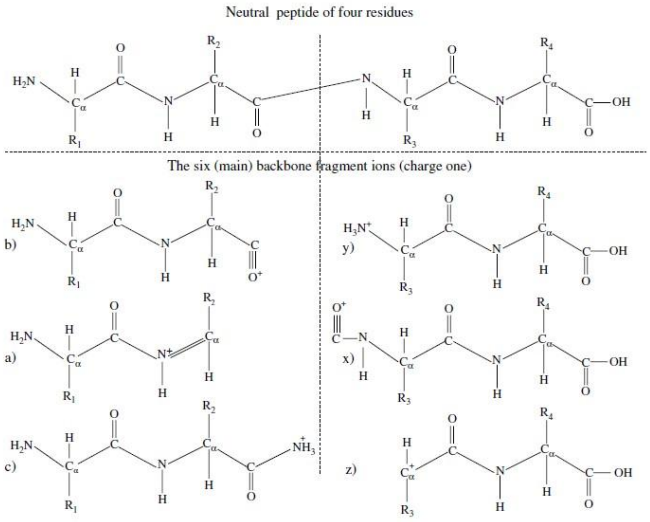
There are **different techniques** for protein fragmentation.

- **Electron Capture Dissociation (ECD)**
- **Electron Transfer Dissociation (ETD)**
- **Collision Induced Dissociated (CID)**

Prepared by Zainab Arshad

Each fragmentation technique gives result of specific type of fragments.

ECD gives out ‘C’ and ‘Z’ ions. CID gives out ‘B’ and ‘Y’ ions, etc.

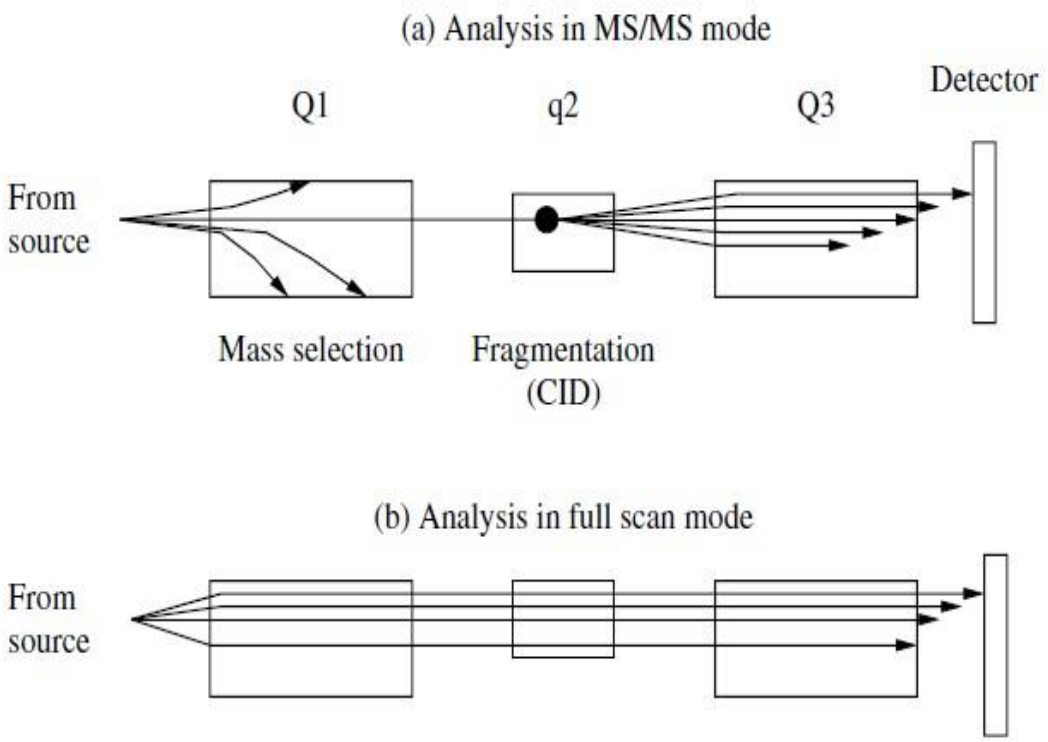


If we can measure the mass of fragments using **MS,** Calculate the **theoretical mass** of the **fragments**. Then, we can award score on the basis of the **similarity** of experimental and theoretical mass.

Module 017: TANDEM MS

Intact masses can measure the intact proteins or peptides. And this can be followed up by their fragmentation in MS chamber.

Prepared by Zuha rshad



Tandem MS can be extended to the fragments of **the intact fragment**. All you need is the **MS instrument capability to,**
(i) Select fragment's mass range.
(ii) Fragment the precursor fragment.

Tandem MS helps us to measure masses of fragments. By this scoring and protein identifications so easy.

Module 018: MEASURING EXPERIMENTAL FRAGMENT'S MASS:

In MS1, the molecular weight of intact sample molecule is measure and then intact molecule is fragmented in two afterward, these two fragments are measured by MS or MS2

FRAGMENTATION TECHNIQUES AND MOLECULAR WEIGHT

MCQS: Fragmentation techniques **include ECD; CID** etc. intact molecule fragmentation splits the molecule into **two parts.**

FRAGMENT MASS

Mass of fragment is produced by **MS2 deepening upon the technique** because each technique splits the protein or peptide at different location.

Prepared by Zaf

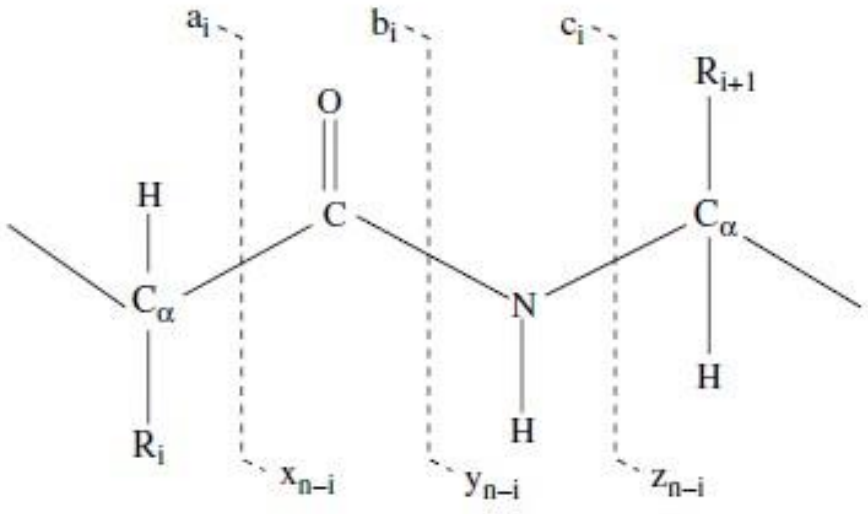


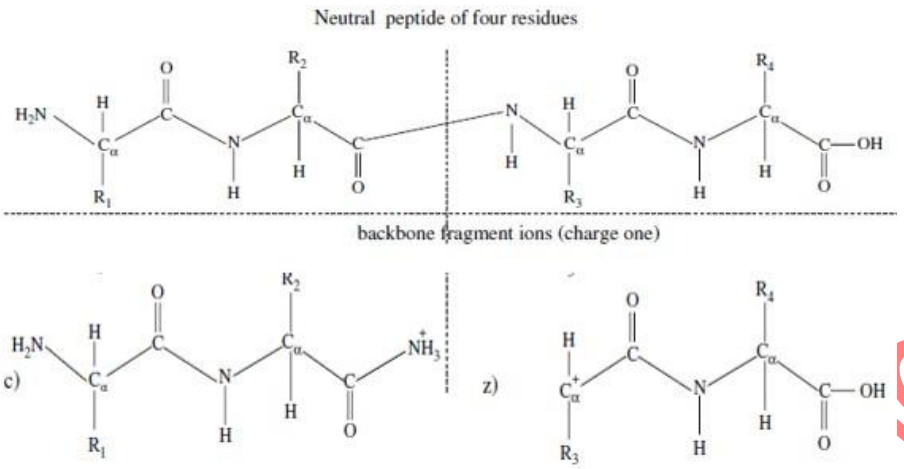
Figure 0.89 Masses after Fragmentation by ECD, CID & ETD

Experimental mass reported from MS2 is matched with theoretical peptides of candidate proteins (from DB). Score is awarded on the basis of the closeness between experimental and theoretical masses.

Module 019: Calculating Theoretical Fragment's Mass

After measuring the mass **of intact molecule from MS2** we compare that mass with theoretical mass of databased proteins.

Prepared by Zainab



$$M_c = R_F + \langle N \rangle + \mathrm{N} + 3\mathrm{H} \qquad M_z = R_F + \langle C \rangle - \mathrm{NH}$$

Figure 0.91 Masses after Fragmentation by ECD

$$M_b = R_F + \langle N \rangle \qquad M_y = R_F + \langle C \rangle + 2\mathrm{H}$$
$$M_a = R_F + \langle N \rangle - \mathrm{CO} \qquad M_x = R_F + \langle C \rangle + \mathrm{CO}$$
$$M_c = R_F + \langle N \rangle + \mathrm{N} + 3\mathrm{H} \qquad M_z = R_F + \langle C \rangle - \mathrm{NH}$$

$$M_b + M_y = R_P + \langle N \rangle + \langle C \rangle + 2\mathrm{H} \; (= M_P + 2\mathrm{H})$$
$$M_a + M_x = R_P + \langle N \rangle - \mathrm{CO} + \langle C \rangle + \mathrm{CO} \; (= M_P)$$
$$M_c + M_z = R_P + \langle \mathrm{N} \rangle + \mathrm{N} + 3\mathrm{H} + \langle C \rangle - \mathrm{N} - \mathrm{H} \; (= M_P + 2\mathrm{H})$$

Figure 22 Masses after Fragmentation by ECD

Prepared

Module 020: PEPTIDE SEQUENCE TAG

s.q:what are peptide sequence tag?

MCQS: Peptide sequence tag are the sequence of **peptide which are produced after MS2**. We can obtain the sequence of peptide through variation in fragmentation site.

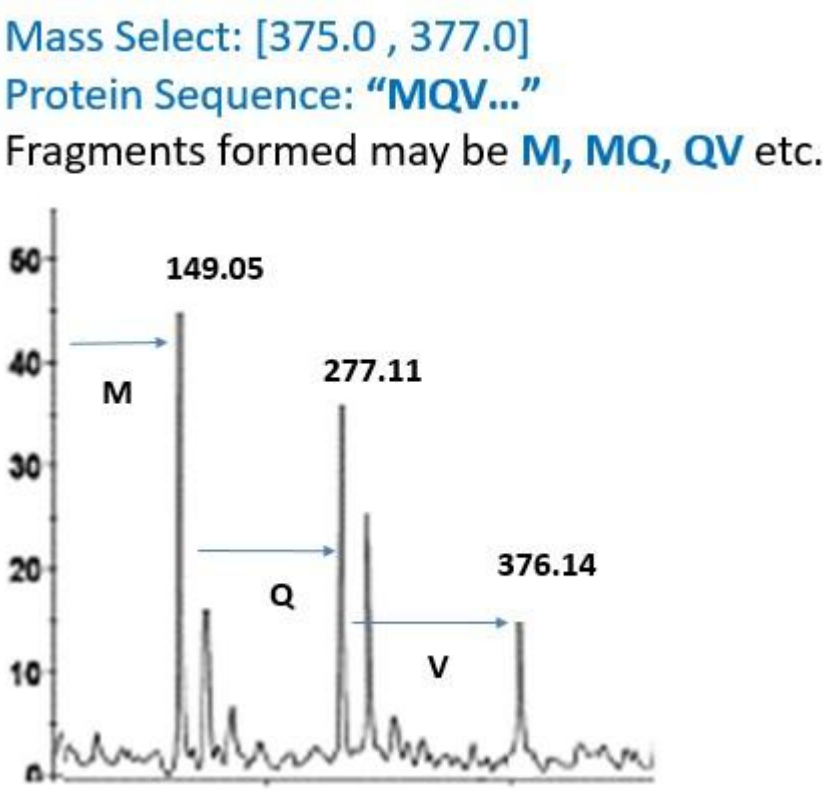


Figure 23 variation sites

MCQS: Precursor proteins or peptides fragmentation leads to formation **of multiple ions** of the same fragment type. However, fragments have variation in their molecular weights due to **variation in site of fragmentation**

Fragmentation at consecutive sites leads to a mass difference equal to that of a single amino acid. Such consecutive peaks can reveal partial peptide sequence tags

Module 021: Extracting Peptide Sequence Tags

PSTs are formed due to sequential cleavage of precursor protein/peptide's backbone.

Prepared by

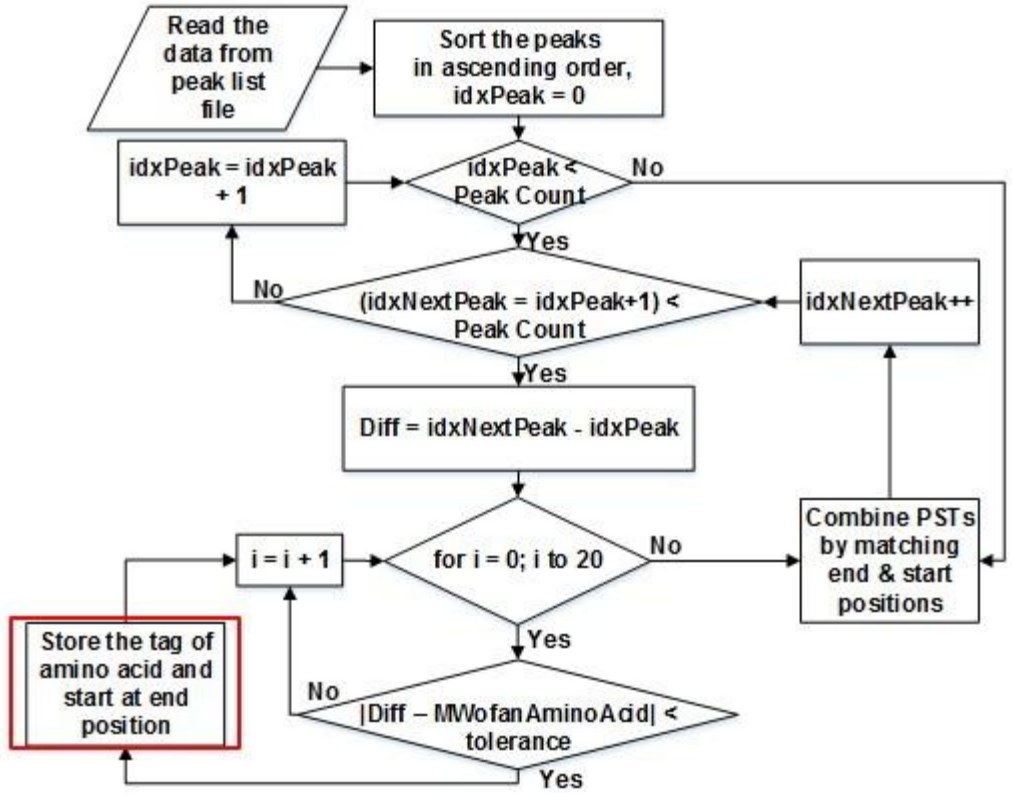


Figure 24 peptide sequencing tagging

Peptide sequence tags can be extracted from **peak list iteratively**. A high quality mass spectrum will produce large number of **PSTs**. The bigger the peptide sequence tags, the better!

Module 022: Using Peptide Sequence Tags in Protein Search

S.Q: What is use of Peptide Sequence Tags in Protein Search

PSTs provide clues of the **precursor protein/peptides sequence**. Consider that we extract the following PSTs: M, MQ, QV etc. Search protein sequence database (e.g. Uniprot, Swissprot)

Sample sequence in protein DB

>>sp|Q6GZ4X|0X1R_FRG3G Putative transcription factor 0X1R OS=Random virus 3 (isolate Goorha) GN=FV3-0X1R PE=4 SV=1

MAFSAEDVLKEYDRRRRMEALLLSLYYPNDRKLLDYKEWSPPRVQVECPKAPVEWNNPPSEKGLIVGHFSGI
KYKGEKAQASEVDVNKMCCWVSKFKDAMRRYQGIQTCKIPGQVLSDLDAKIKAYNLTVEGVEGFVRYSRVT

Prepared by Z.A

KQHVAAFLKELRHSKQYENVNLIHYILTDKRVDIQHLEKDLVKDFKALVESAHRMRQGHMQNVKYILYQLLK
KHGHGPDGPDILTVKTGSMQLYDDSFRKIYTDLGWKFTPL

For all the proteins in the database, we find out which PSTs exist in which proteins. The protein reporting the most PSTs is more probable to be the precursor protein.

If many PSTs report the same number or protein report the longer PSTs than through scoring we find the greater number. After **extracting the PSTs** we search the entire database for protein who report it.

Module 023: Scoring Peptide Sequence Tags

According to scoring scheme if a candidate protein matches 'n' PSTs, then its score can be given by:

$$PST_{Score} = \sum_{i=0}^{n} Length\ (PST_i)^2$$

Additionally, if we include RMSE to the scoring system, then it can highlight better PST matches. And RMSE is the root mean square error.

RMSE for a sequence tag 'i' of length 'n'?

$$RMSE_i = \sum_{i=0}^{n} \sqrt{(M_{Hop} - MAA)^2}$$

So, the updated relationship is:

$$PST_{Score} = \sum_{i=0}^{n} \left(\frac{Length(PSTi)^2}{RMSE_i}\right)$$

Prepared by Zeeshan Arshad

Module 024: In silico Fragment Comparison

MS1 reports the intact mass of molecules (proteins or peptides) in the sample. Intact mass can be compared with every protein's mass in database to identify the molecule in the sample.

Mcqs: Incase multiple candidate proteins are reported, MS2 can be performed. **MS2 helps measure fragment peptide masses. MS2 data can be used to extract peptide sequence tags**

Mcqs: If the protein identification is still not conformed than each experimentally reported MS2 fragment is compared with the in **silico spectra of proteins** from database.

Mcqs: Fragmentation techniques determine product ions e.g. ECD -> c/z and CID -> b/y ions etc. With known fragment types, we can compute the MW of all possible protein fragments

MWs of each fragment individually

For obtaining all possible theoretical fragments in a protein, we need to compute the **MWs of each fragment individually**

Consider a random protein sequence from DB:



Figure 26 random protein sequence

Matching experimental fragments with in silico fragments is the final resort in protein search and identification.

Module 025: In silico Fragment Comparison and scoring

Experimental MS2 can be compared with the in silico spectra of protein from database.

- **MCQS:** Count the matches between in **silico and in-vitro peaks**.
- **MCQS:** Give an equivalent score to **candidate protein**.
- Weigh each of the aforementioned match by the mass error
- Accumulate the score

With "all possible" fragments in in silico spectrum, and "reported" fragments in experimental spectrum, we can match and rank.

Scoring scheme should also consider the errors in peak matching

Prepared by Z Ahmad Arshad

Prepared by Zainab Arshad

Module 026: Protein Sequence Database Search Algorithm

MCQS: **MS1 and MS2 provide us with a host of data towards enabling us in identifying unknown proteins**. A step by step approach combining MW, PSTs and insilico spectral matching is required.

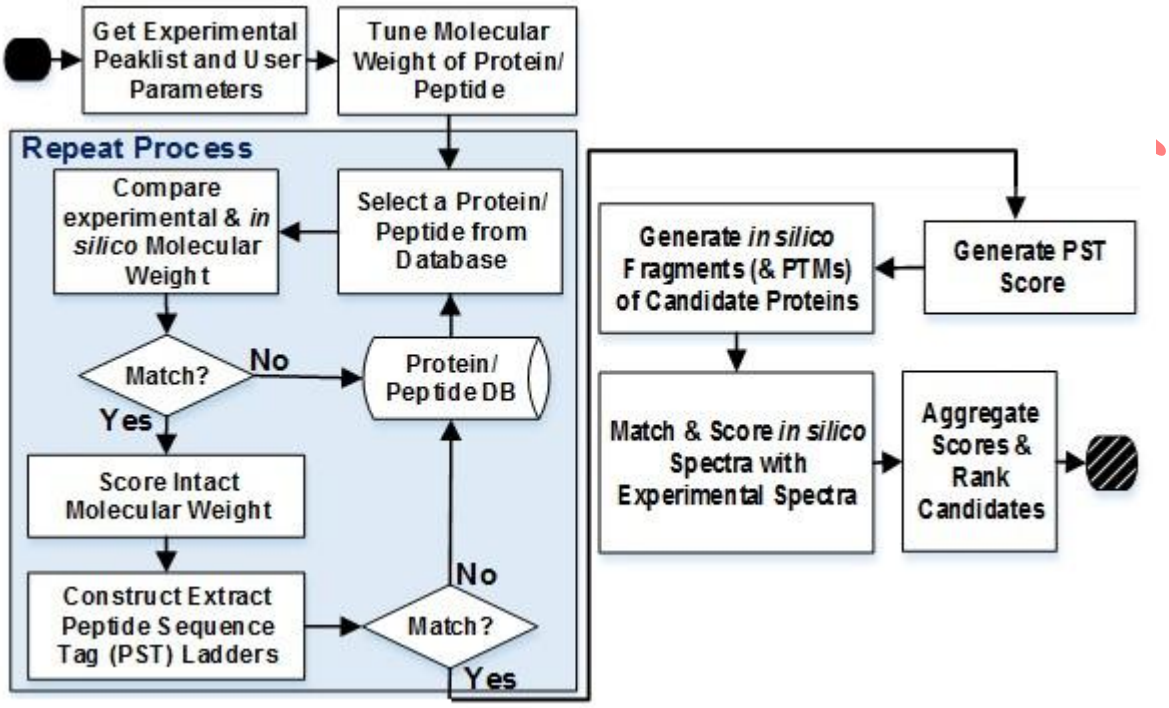


Figure 27 protein sequencing flowchart

MCQS: Integrating MW, PST and insilico comparison algorithms in a workflow can help create a composite protein search engine. A **composite scoring system** is also required for this search engine.

Module 027: Integrative Scoring Schemes

MCQS: **Three individual scores can be obtained:**

- **MW Match score**
- **PST Match score**
- **In vitro & In silico Match score**

For overall cumulative score computation:

- We simply sum the scores up (a linear function).

Prepared by

$$Score_{MW}^2 + Score_{PST}^2 + Score_{Exp <> Thr}^2 = Score^2$$

- Weigh each scoring component up by respective RMSE before summing them up

$$Score^2 = \frac{Score_{MW}}{E_{MW}} + \sum_{i=0}^{m} \frac{Score_{PST}}{RMSE_{PST}} + \sum_{i=0}^{n} \frac{Score_{Exp <> Thr}}{E_{EXP <> Thr}}$$

- Complex non-linear functions integrate the scoring components in Mascot etc.
- Highly proprietary for commercial proteomics software are used.

Composite scoring schemes are needed to combine scores coming in from multiple criteria. The ability of a scoring scheme to better isolate true positives from false positives is important.

Module 028: LARGE SCALE PROTEOMICS

Peptide mixtures in bottom up proteomics are very complex. Tryptic peptides may reach up to an order of 300,000–400,000.In whole proteome samples, protein count may be over 10,000. Experiments have shown that it is difficult or even impossible to analyze all these peptides in a single analysis, as the mass spectrometer is essentially overwhelmed.

Over half a million peptides reported in a typical LSP experiment are redundant.

If we could find a unique peptide for a protein, that would make sequence coverage suffer and we have to strike a compromise between sequences coverage and sample coverage.

TECHNIQUE

One way forward would be to transfer peptides to the MS chamber in a step-by-step manner. However, this imposes a precondition that a peptide is not selected earlier as well (i.e. more than once)

STEP BY STEP TECHNIQUE

1. The instrument alternates between MS and MS/MS modes.
2. Three most intense peaks are chosen for MS/MS analysis.
3. After the initial MS scan, an MS/MS spectrum from peptide A is obtained by selectively fragmenting this mass only.

Prepared by Zainab Arshad

4. Next, a spectrum for peptide B is produced, followed by a recording of the MS/MS spectrum for peptide C.
5. After these three fragmentation spectra have been obtained, a new MS scan is started.

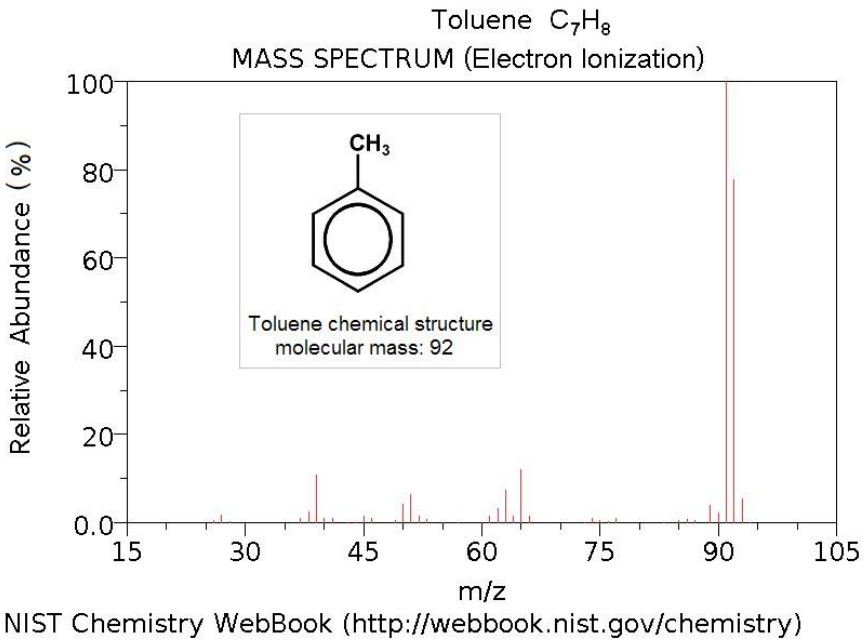
From this scan, three more peptides A B C are selected for fragmentation and the cycle starts over again.

The number of MS1/2 scans can be limited by carefully selecting the peptide peaks. Once the intense peptides are identified, next batch of peptides is chosen for MS2.

Prepared by Shahzaib Arshad

Module 029: PROTEOMICS DATA FILE FORMATS

Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides. Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured.



Followings are the formats for proteomics data:

Agilent	.D/.YEP
Bruker	.BAF
ABI/Sciex	.WIFF
	.t2d
Thermo Xcalibur	.RAW
Micromass	
PerkinElmer	
Waters	
Shimadzu	.QGD

Figure 28 Formats used for proteomics data

OPEN FORMTAS:

mzXML (tools.proteomecenter.org/mzXMLViewer.php)

MGF (proteomicsresource.washington.edu/mascot/help/data_file_help.html)

Multiple MS data formats exist. Proprietary formats exist which come implemented as software with hardware. Also, open software standards exist for interoperability etc.

Module 030: RAW FILE FORMAT

Mass spectrometer outputs data with ionic mass/charge ratios & respective ion intensities. RAW file is a format in which an instrument outputs data in binary form.

Prepared by Zainab Arshad

Vendor	Instrument	Data Type	Converter & Set Up
Thermo	Ion Trap / Orbitrap	raw	MSFileReader
AB SCIEX	Triple TOF	wiff	AB SCIEX Data Converter
	QSTAR / QTRAP	wiff	Analyst + mzWiff
	TOF TOF	wiff	AB Extractor
Agilent	Ion Trap	yep	CompassXtract
	Q-TOF	d	MassHunter
Bruker	QTOF / Ion Trap / TOF TOF	baf / yep / lift	CompassXtract
Shimadzu	AXIMA-CFR	run	Kratos Converter
Varian	FTICR / Ion Trap	sms/xms	VarianMS
Waters	Q-TOF	raw	MassLynx

Figure 29 Raw file formats

Free Software Name	Input RAW Data Type	Output Data Types
ProteoWizard	Thermo, Agilent, Bruker, Waters, AB Sciex	mzML, TraML, mzIdentML, mzXML
OpenMS	All vendor formats	mzML, TraML, mzIdentML, mzData, mzQuantML
Trans Proteomic Pipeline	Thermo, ABI Sciex, Agilent, Waters.[14]	mzML, mzXML, pepXML, protXML (Proteowizard)
PEAKS	Thermo, ABI Sciex, Agilent, Waters, Bruker, Shimadzu, Varian	MSF, tandem, mzML, omx, dat, FASTA

Figure 30 list of tools for raw data processing

Multiple RAW file formats are prevailing in the industry. Each vendor has its own unique RAW file format. You can convert proprietary formats into open formats

Module 031: MGF FILE FORMAT

MGF – Mascot Generic Format. MGF is a simple human-readable format for MS/MS data developed by Matrix Science. Mascot Search Engine available at this link online.
http://www.matrixscience.com/search_form_select.html

Name	Description	Header	Local	Choices/Range	Notes
ACCESSION	Database entries to be searched	✔		List of double quoted, comma separated values	
CHARGE	Peptide charge	✔	✔	1-	M-H- on PMF form
				Mr	
				1+	MH+ on PMF form
				N- to N+ where N is an integer and combinations	Not PMF
CLE	Enzyme	✔		Trypsin etc., as defined in enzymes file	No default, so must be specified
COM	Search title	✔			Applies to the whole search

http://www.matrixscience.com/help/data_file_help.html

CUTOUT	Precursor removal	✔		Pair of comma separated integers	MIS only
COMP	Amino acid composition		✔		
DB	Database	✔		As defined in mascot.dat	
DECOY	Perform decoy search	✔		0 (false)	Default
				1 (true)	
ERRORTOLERANT	Error tolerant	✔		0 (false)	Default
				1 (true)	Not PMF
ETAG	Error tolerant sequence tag		✔		A single query can have multiple ETAGs





PRECURSOR	Precursor m/z	✔		>100	
QUANTITATION	Quantitation method	✔		as defined in quantitation.xml	MIS only
RAWFILE	Raw file identifier		✔	string	MIS only

Prepared by Zainab Arshad

Prepared by Zainab Arshad

Module 032: OPEN MS DATA FORMAT

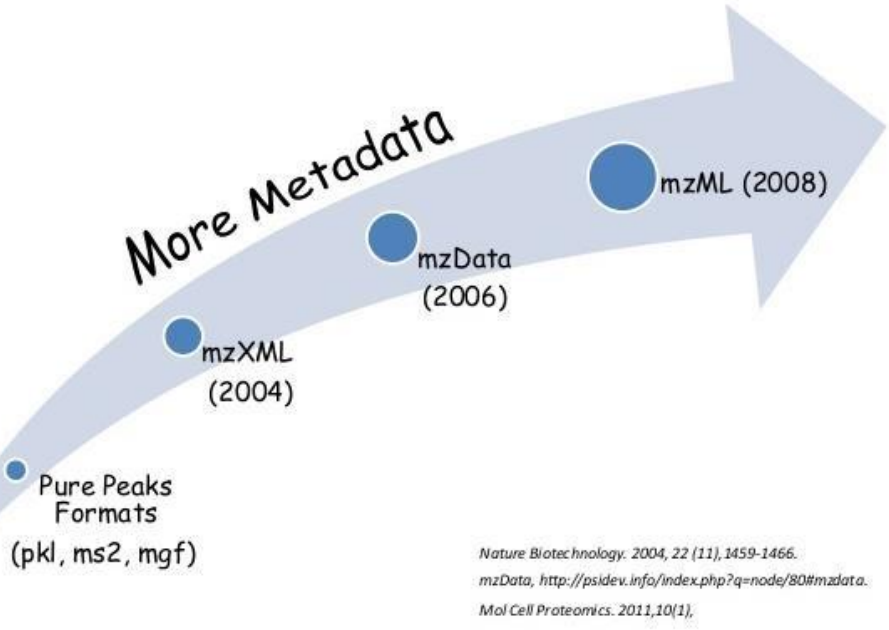
Mass spectrometer outputs data with mass/charge ratios & respective ion intensities. RAW file formats are specific to each instrument and each vendor has its own unique file format. Once an instrument is upgraded, data output from the instrument is also changed. Hence the underlying RAW file format needs to be upgraded as well.

NEED

Proprietary RAW formats are binary formats which are difficult to read and parse. If you have the software from the maker of the MS then you can read the RAW data file as well.

SOLUTION

- mzData was developed by HUPO-PSI
- mzXML was developed at the Institute for Systems Biology
- To combine them, a joint venture produced mzML



Nature Biotechnology. 2004, 22 (11), 1459-1466.
mzData, http://psidev.info/index.php?q=node/80#mzdata.
Mol Cell Proteomics. 2011,10(1),

Pr nab Arshad

Tool	Formats
ProteoWizard	mzML, TraML, mzIdentML, mzXML, vendor formats
OpenMS	mzML, TraML, mzIdentML, mzData, mzQuantML, et al.
Trans-Proteomic Pipeline (TPP)	mzML, mzXML, pepXML, protXML (ProteoWizard)
compomics-utilities	MSF, tandem, mzML, omx, dat, FASTA
jmzReader	mzML, mzXML, mzData, PRIDE XML, dta, MGF, ms2, pkl
jTraML	TraML
multiplierz	Vendor formats
PEFF Viewer	PEFF
PRIDE Converter 2	mzTab, PRIDE XML (jmzReader)
Mascot & Distiller	MGF, mzML, mzXML, mzIdentML, vendor formats
SpectraST	msp, splib, blib, ASF, mzML, mzXML, pepXML, etc.
ProHits	PSI-MI (TPP formats)

Figure 31 Formats used for open use

Prepared by Za

MCQS: Several software exist for converting **RAW file formats into open software formats**. Each open format has its own unique advantages. mzXML and MGF formats are most frequently used

Module 033: Online Proteomics Tools - MASCOT

Matrix Science developed an online Bottom up Proteomics Search Engine. "MASCOT". Mascot can search peptide mass fingerprinting and shotgun proteomics dataset

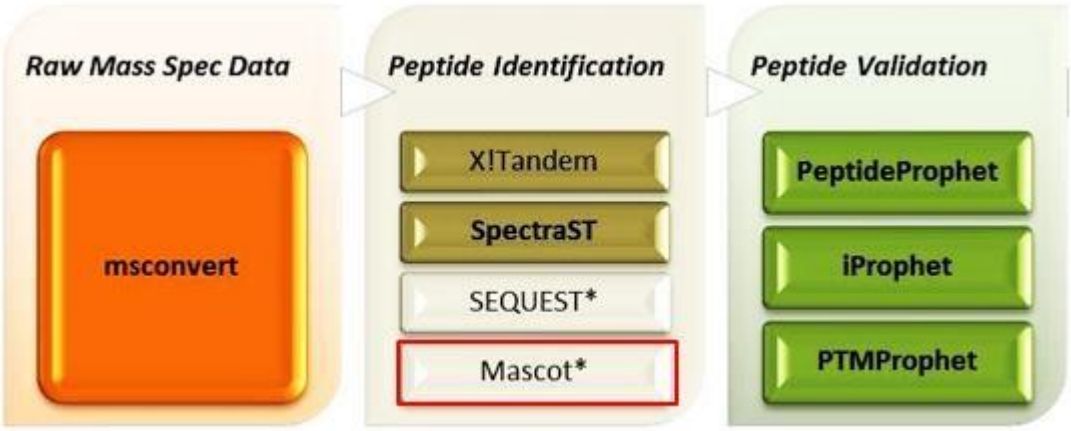


Figure 32 mzML http://tools.proteomecenter.org/software.php

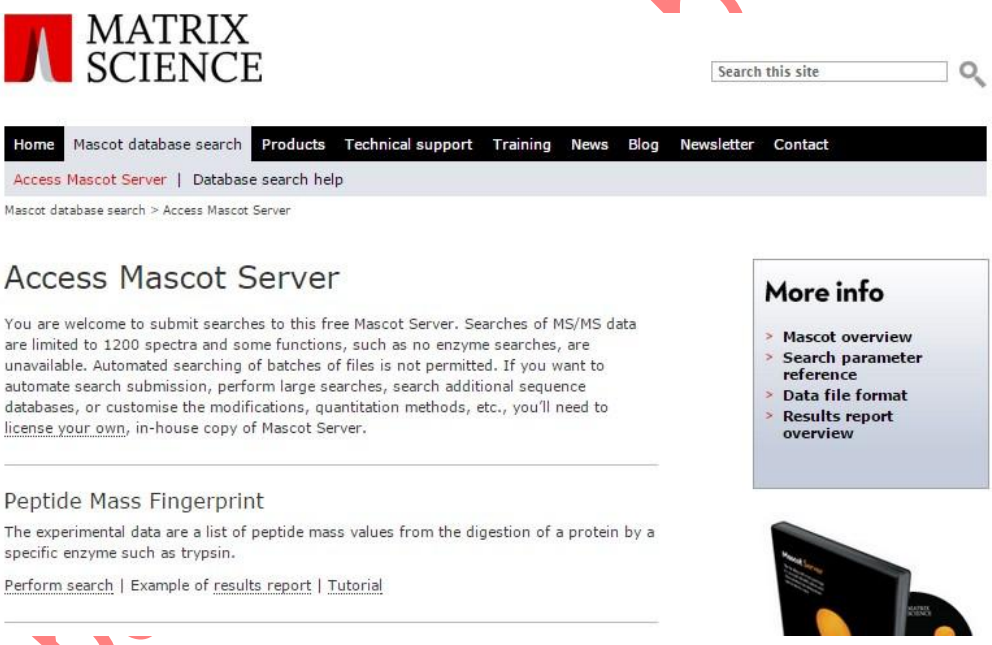


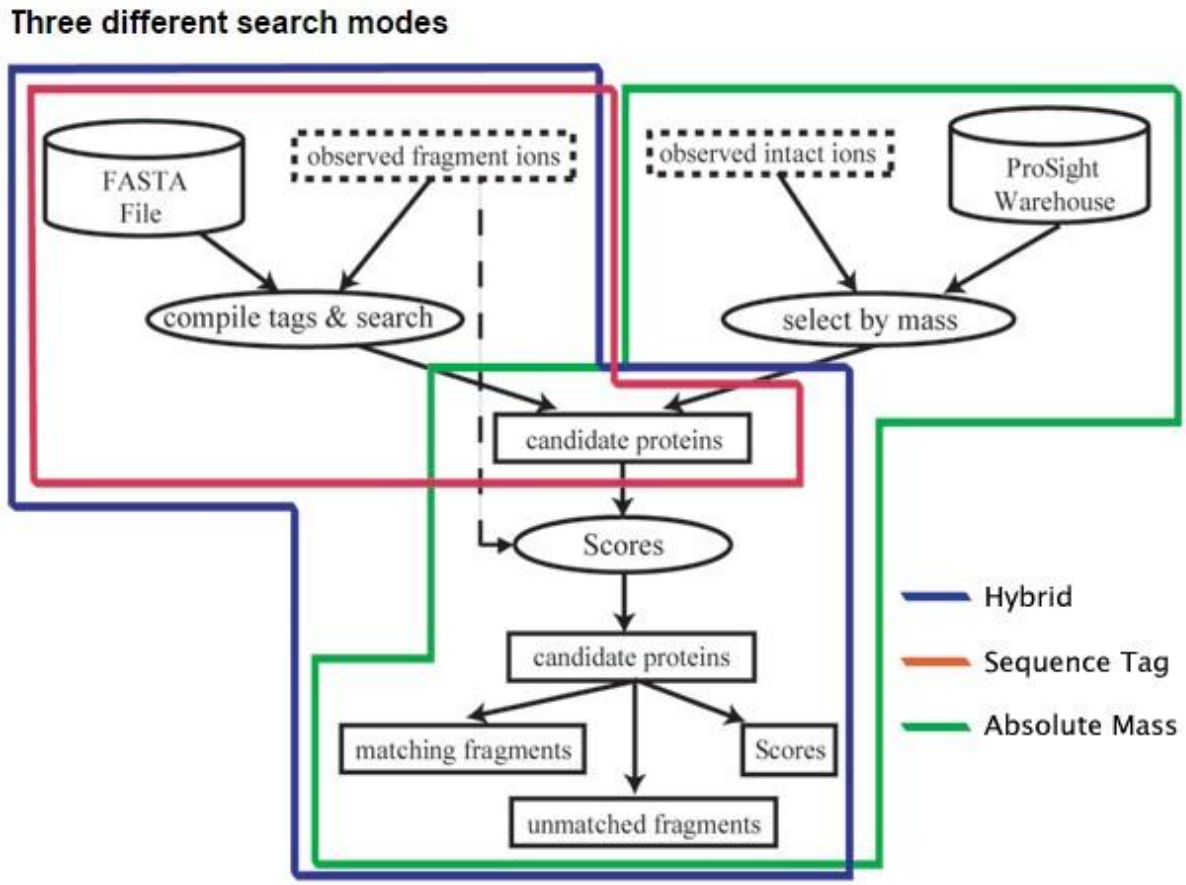
Figure 33 http://www.matrixscience.com/search_form_select.html

MCQS: Mascot is the most widely used online search tool for **proteomics data.** However, it lacks a batch processing mode. Also, it does not cater for **top-down proteomics data**.

Module 034: Online Proteomics Tools – ProSight PTM

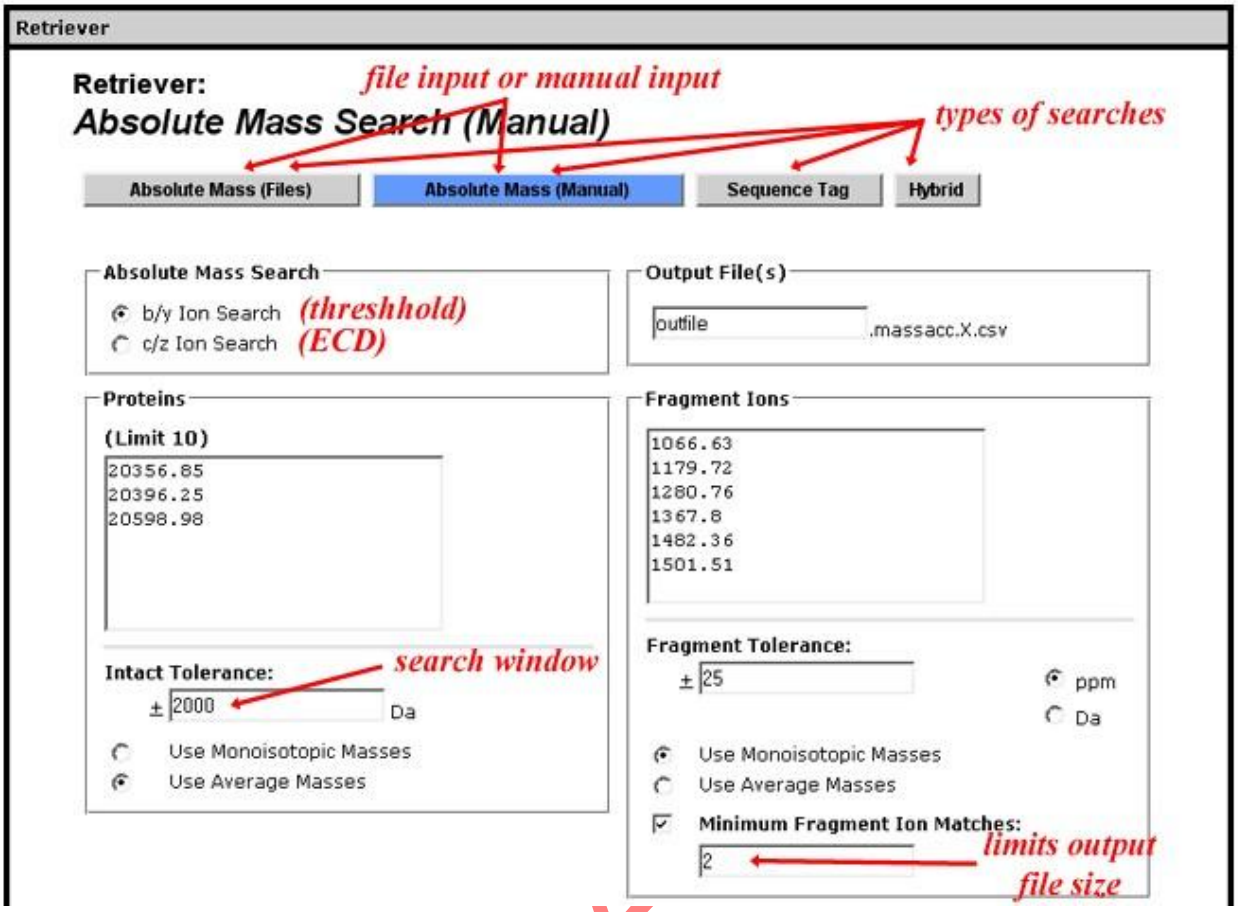
Kelleher et al have developed an online Top down Proteomics Search Engine. "Prosight PTM".

ProsightPTM searches top down proteomics data and reports the precursor protein



https://prosightptm.northwestern.edu/about_retriever.html

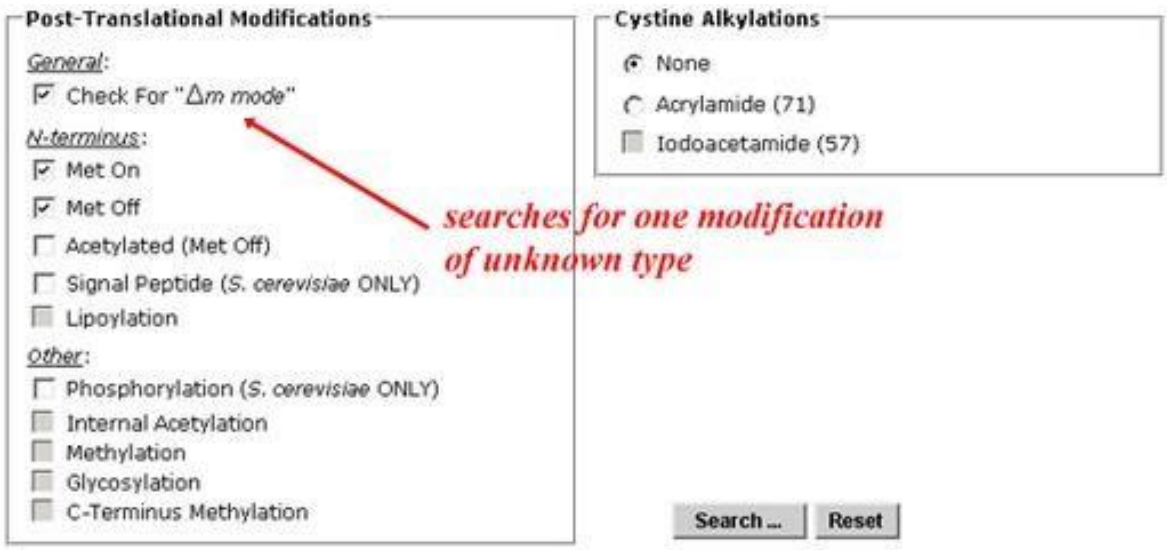
Prepared by



https://prosightptm.northwestern.edu/about_retriever.html

Database

	Organism	Strain	Annotated By	# Protein Forms	Size (MB)
○	*M. pneumoniae*	M129	Herrmann	(1758)	0.9
○	*E. coli*	K12	SWISS-Prot	(22,621)	8
○	*E. coli*	K12	Wisconsin	(22,621)	8
○	*B. subtilis*	168	Kunst/Danchin	(10,301)	3.8
○	*S. oneidensis*	---	SWISS-Prot	(12,504)	4.5
○	*M. jannaschii*	DSM2661	Olsen	(4638)	1.8



MCQS: ProSight PTM is the state of the art in top down proteomics search. Using Prosight PTM, posttranslational modifications can be accurately identified.

Prepared by Zainab A.

Module 035: Example Case Study - I

For case study we follow some steps:

Step 1 – Monoisotopic Peak Detection

Natural elements occur in multiple isotopes. Isotopes differ in their masses.The abundance of each isotopic variant is unique.

Element		Abundance (%)	Mass
Hydrogen	^{1}H	99.99	1.007 83
	^{2}H	0.01	2.014 10
Carbon	^{12}C	98.91	12.000 0
	^{13}C	1.09	13.003 4
Nitrogen	^{14}N	99.6	14.003 1
	^{15}N	0.4	15.000 1
Oxygen	^{16}O	99.76	15.994 9
	^{17}O	0.04	16.999 1
	^{18}O	0.20	17.999 2
Phosphorus	^{31}P	100	30.973 8
Sulfur	^{32}S	95.02	31.972 1
	^{33}S	0.76	32.971 5
	^{34}S	4.22	33.967 6

Figure 34 Isotopic variants of natural elements

Prepared

TYPES OF MASSES

- Nominal Mass
- Monoisotopic Mass
- Average Mass

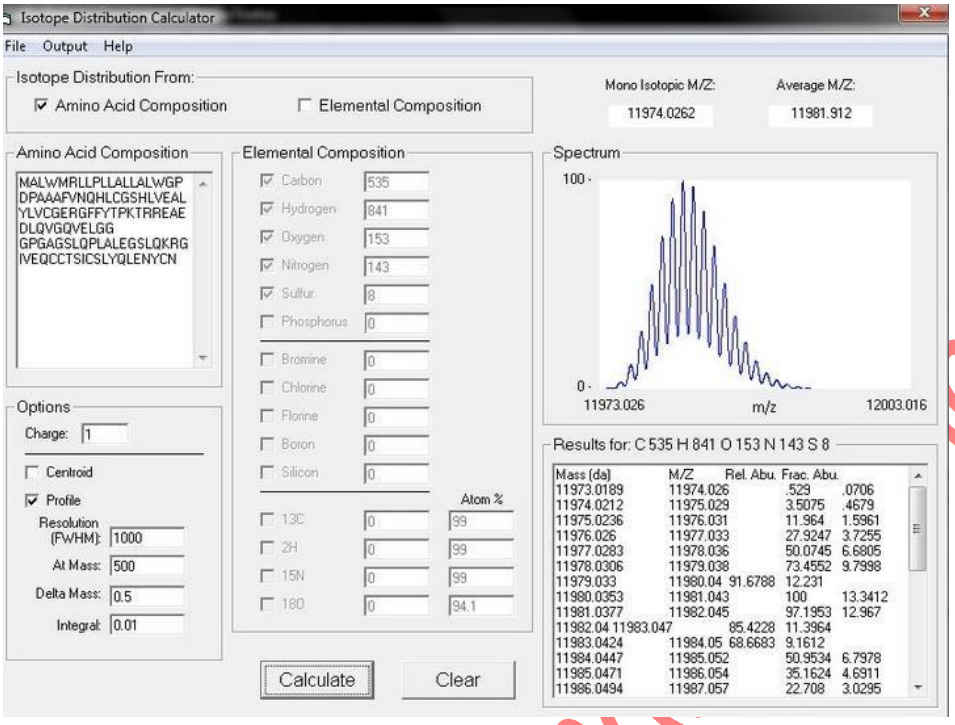


Figure 35 Detecting Monoisotopic Peaks

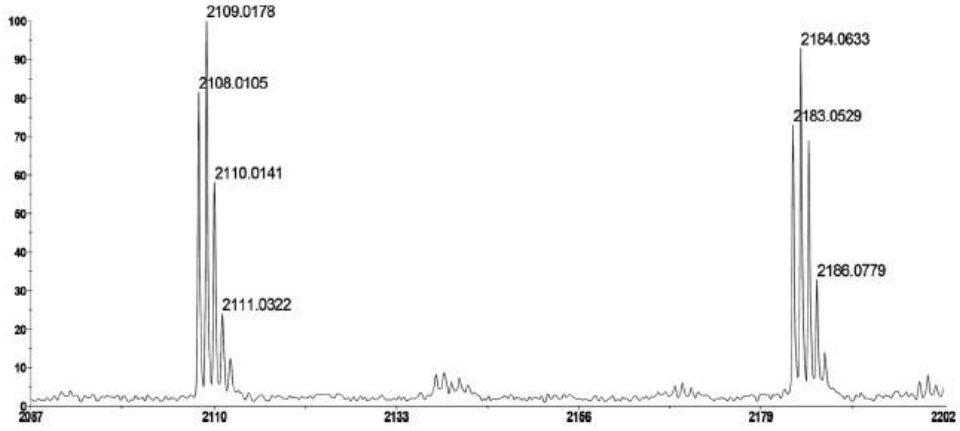


Figure 36 Detecting Monoisotopic Peaks

MS1 data reports the isotopic distribution of intact molecule's mass. Monoisotopic mass value has to be selected from this mass distribution. This value is the highest mass value in the distribution

–

Module 036: Example Case Study II

The first step in protein identification and characterization using mass spectrometry involves intact protein/peptide mass measurement. Next, we fragment the protein. A protein or peptide backbone may be fragmented anywhere along the peptide backbone.

This results in formation of two fragments i.e. N-term fragment and C-Term fragment.

For possible fragments let's take an example protein with 100 residues. Such a molecule's backbone can be fragmented at 100 different locations. The total number of possible fragments is then 200

S.Q: TANDEM MS

The mass of 200 fragments can then be measured by using an MS again. The necessary condition for this measurement is that all 200 fragments are ionized.

To ensure that all fragments of precursor molecule are also charged, we can use Electrospray ionization (ESI).ESI induces multiple charges on the intact molecule

Role of Electrospray Ionization

Since ESI induces multiple charges on the precursor molecule, there is a good chance that upon precursor's fragmentation, each fragment will have a portion of the charge. ESI allows for production of multiple charged ions. Tandem MS helps measure molecular weights of ionized fragments

Prepared by Zainab Arshad

Module 037: Example Case Study III

Tandem MS helps measure the mass of the fragments Those fragments which differ from each other by one amino acid's mass can provide clues on the sequence of proteins

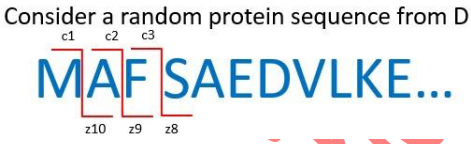


Figure 37 Example peptide sequence tags

Peptide sequence tags help derive clues about the sequence of precursor proteins/peptides. The short peptide sequences help us in shortlisting candidate proteins from the database.

Prepared by Zain Arshad

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Module 038: Example Case Study IV

MS1 helps measure the intact mass of proteins/peptides. A list of candidate proteins/peptides can be formed by comparing MS1 mass to the mass of proteins/peptides in the database. MS2 or Tandem MS was performed after fragmentation of intact proteins. MS2 helped extract peptide sequence tags from MS2 data. Candidate proteins can be further shortlisted by the PSTs

Exhaustive matching of all MS2 peaks with the theoretical fragments of candidate proteins. The set of theoretical fragments contains all possibilities of fragmentation

Theoretical vs experimental fragments comparison helps as the third stage for shortlisting candidate protein list. This shortlisting will help you arrive at a small number of proteins

Prepared by Zahid Arshad

Module 039: Example Case Study V

MS1 and MS2 provide mass of intact molecules and its fragments. This information helps filter proteins from protein database. For a quantitative measure, scoring scheme is required.

MAFSAEDVLKE…

$$M_{Score} = \frac{1}{\sqrt{(M_{Exp} - MT)^2}}$$

Figure 38 Example intact protein mass score

MAFSAEDVLKE…

$$PST_{Score} = \sum_{i=0}^{n} \left(\frac{Length(PSTi)}{RMSE_i}\right)$$

Figure 39 Example peptide sequence tags

Three scoring schemes can be applied to score the match at each stage of protein search. These scoring elements can be integrated to arrive at an overall candidate protein score.

Prepared by Ninad Arshad

–

Module 040: Example Case Study VI

Simply sum the scores up (a linear function)

$$Score \quad Score_{MW} + Score_{PST} + Score_{Exp <> Thr}$$

For comprehensive scoring

A comprehensive scoring scheme can combine all the scores. Several optimizations can be undertaken on the scoring scheme to further improve protein identification

Prepared by Zainab Arshad

Chapter 6 - Protein Structures

Module 001: PROPERTIES OF AMINO ACIDS-I

MCQ: Proteins are made **by polymerization of amino acids** on ribosomes and proteins properties are linked to the properties of **amino acids.**

MCQ: There are **20 amino acids** in nature each has different chemical composition and that's why each protein is **differen** t from other.

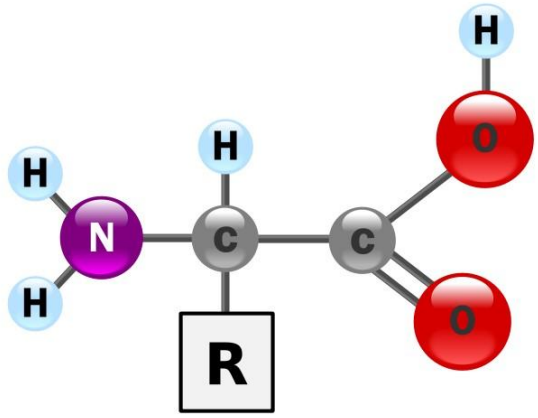


Figure 0.1 chemical structure of amino acid

MCQ: S.Q: Amino acid have **three groups, hydroxyl group, Amine group and R group**. The R group is representing **any group**.

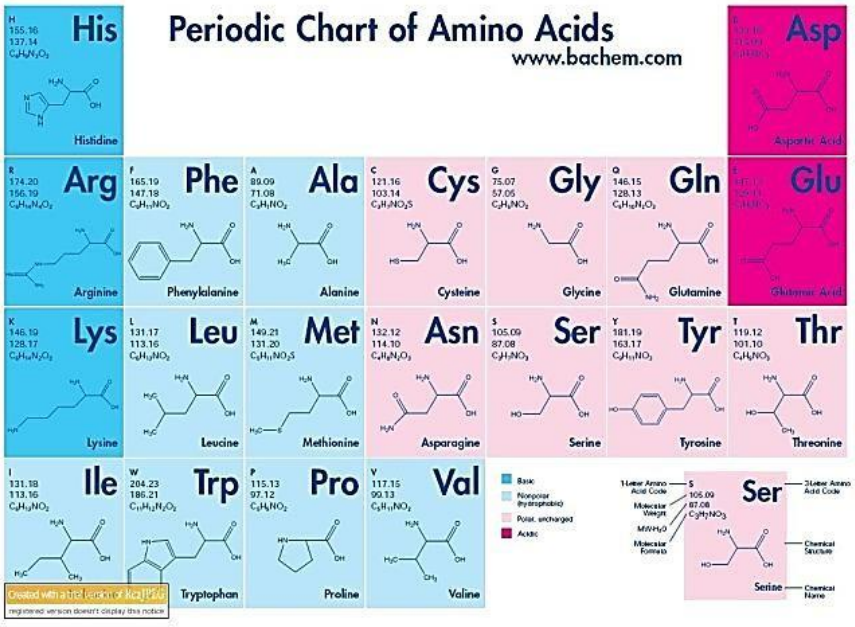


Figure 0.2 periodic chart of amino acid

During polymerization of amino acids the water is formed and amino acids attached with each other.

L.Q & S.Q: MCQ: write the Preoperties of Amino acids

Amino acids have unique properties such as **polarity, charge states and interactions with water**. Each of these properties describes the overall characteristic of an amino acid.

Module 002: PROPERTIES OF AMINO ACIDS-II

Amino acids have characteristics **like polarity, hydrophobicity, and charge states**. These characteristics are governed by the **elemental composition of an amino acid's side chain (R group).**

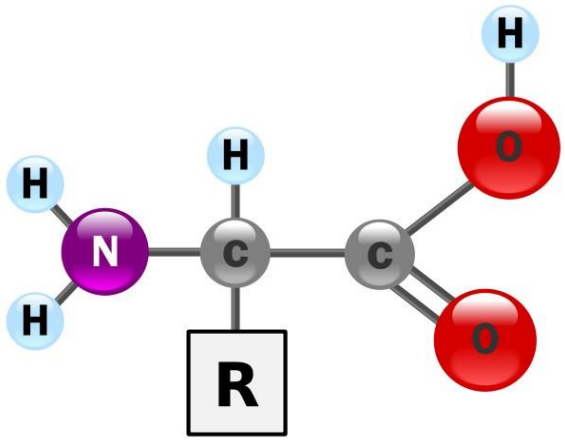


Figure 4 R group in amino acid

s.q: MCQ: What are the HYDROPHILIC AMINO ACIDS

HYDROPHILIC AMINO ACIDS

Since **H and C introduce very little dipole moments in hydrophobic amino acids**, these amino acids are **non-polar**. Hydrophobic amino acids are mostly found at the **inside of folded proteins. Hydrophilic group contain the chain of C and H group in their R group.**

Figure 5 hydrophilic group

Prepared by Arshad

s.q: MCQ: What are the Polar AMINO ACIDS

POLAR AMINO ACID

These amino acids are polar but are **not charged** i.e. no net charge on the amino acid. Prefer to reside / interact with aqueous environments. Mostly found at the **surface of folded proteins.**

Amino acids have unique properties such as polarity, charge states and interactions with water. Each of these properties describes the overall characteristic of an amino acid.

MCQ: Some amino acids are positively charged and some have **negative charge.**

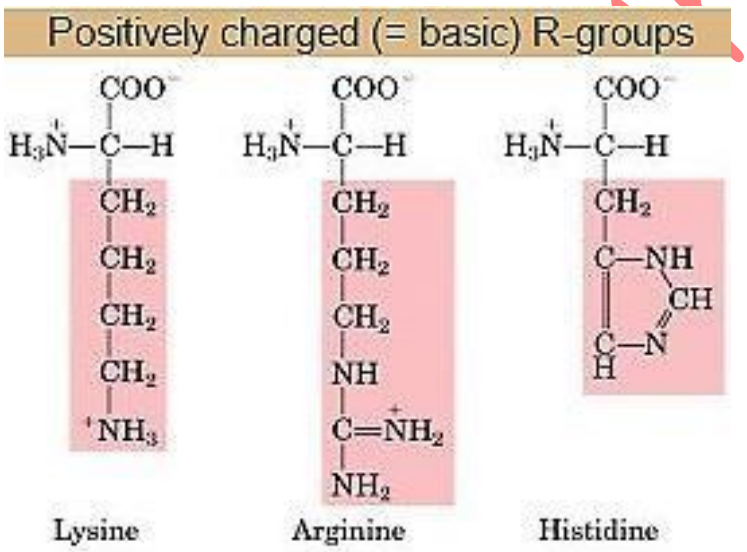


Figure 7 positively charged amino acids

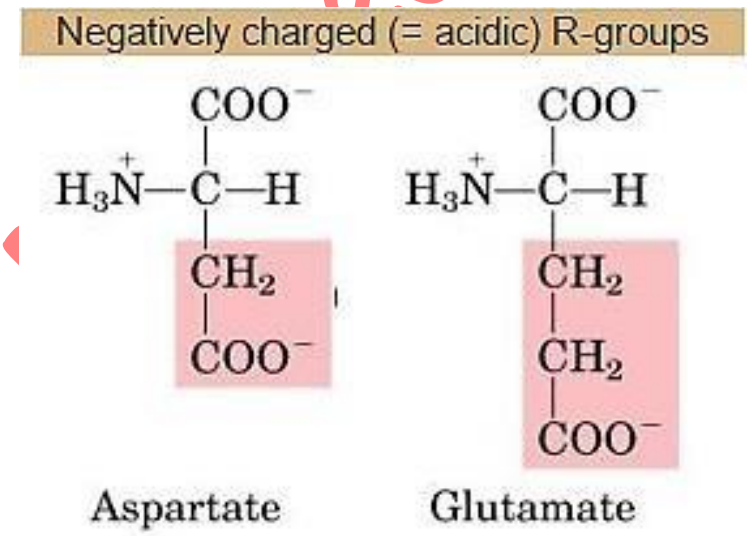


Figure 8 negative charge amino acid

Upon polymerization of amino acids into polypeptide chains**, charged amino acids get neutralized. At pH=7**, five amino acids are charged, **2 negatively and 3 positively.**

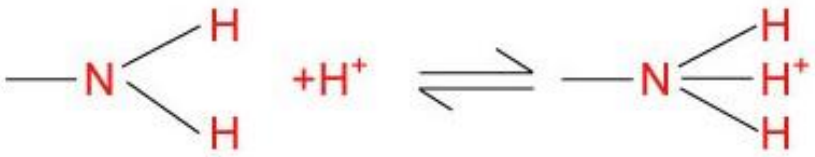
Some amino acids are positively charged and some have negative charge**.**

MCQS: S.Q: Definition

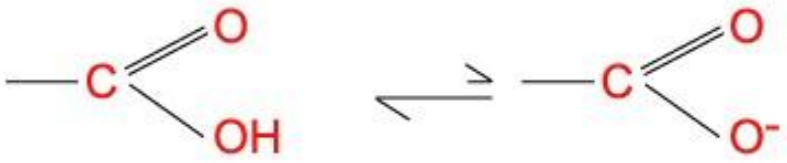
pK is the values for an amino acid is the pH at which exactly half of the chargeable group is charged.

s.q: How ph can effect the polarity of amino acids?

If **pH < pK for an amino acid, the amine side chains gain a proton (H+)** and become positively charged, hence basic.



If **pH > pK for an amino acid, the carboxyl side chains loses a proton (H+) and** become negatively charged, hence acidic.



Prepared by Zoo Arshad

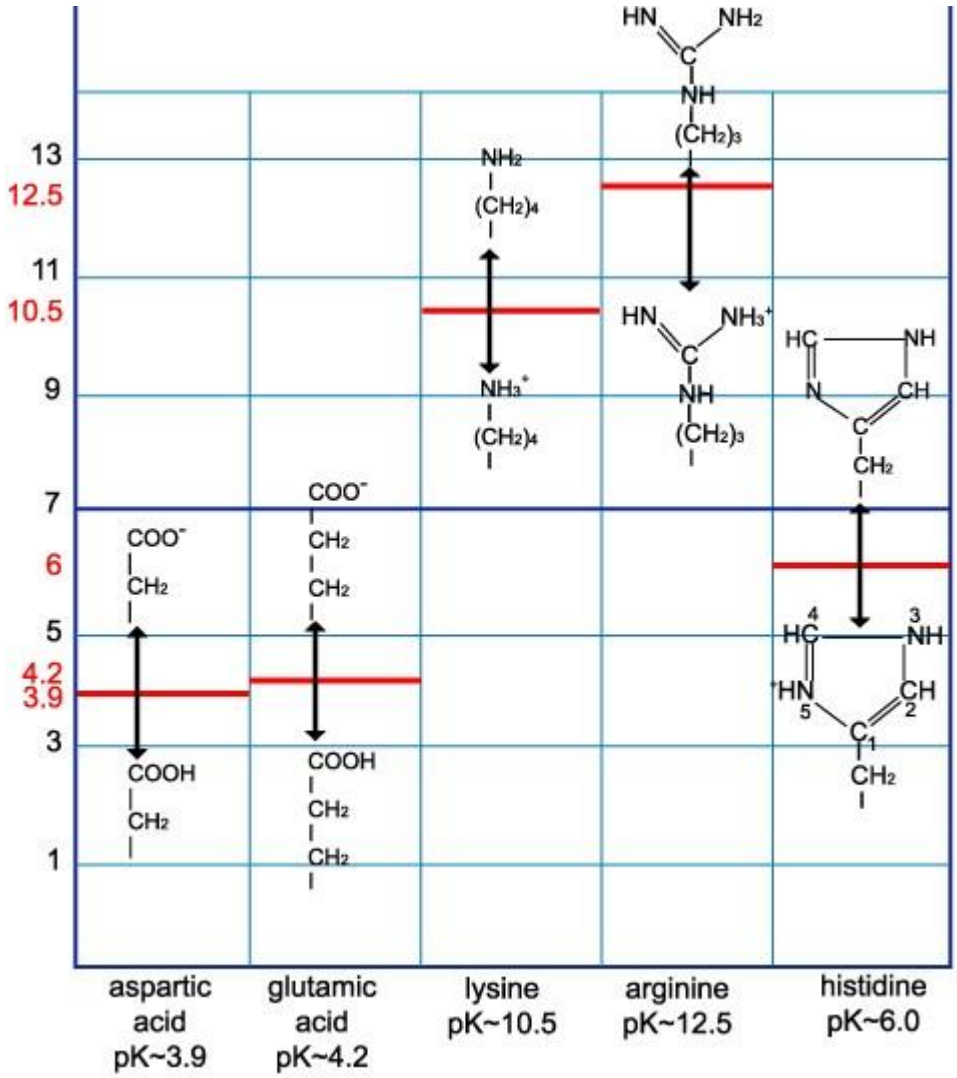


Figure 9 properties of amino acids according to pK and PH.

Depending on the pH, an amino acid may become **charged.** This may be positive or negative depending on the amino acid.

Amino acids may be charged depending on **pH.** This depends on the charge acceptance or donation from within an amino acid. Additionally, amino acids have structures as well.

Prepared

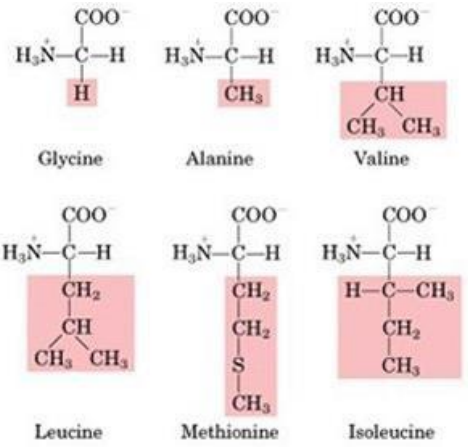


Figure 10 Aliphatic Amino Acids (Non polar C and H chains)

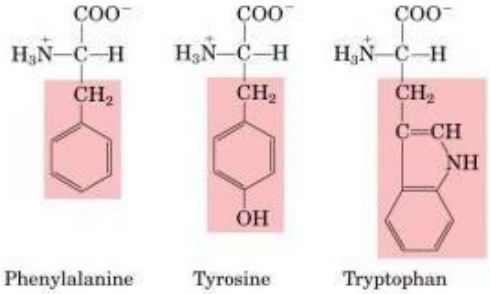


Figure 11 Aromatic R groups

Side chain also impact some properties. Side chains comprising merely of Carbon and Hydrogen are:

- Chemically inert,
- Poorly soluble in water

However, side chains containing organic acids are very different. They are chemically reactive and Soluble in water. Elemental composition plays a very important role in determining properties of amino acids. Solubility and reactivity are key factors participating in protein folding.

Prepared by Zia Arshad

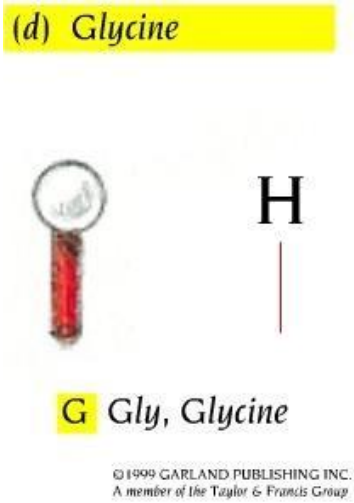
Module 006: STRUCTURAL TRAIT OF AMINO ACID-I

Amino acids have several properties such as charge state, polarity and hydrophobicity.

MCQS: It is important to note that the **physical size of each amino acid also varies**.

EXAMPLE-1: Glycine

MCQS: Glycine residues **increase backbone flexibility because they have no R group (only an H), hence agile.**



EXAMPLE-2: Proline

MCQS: Proline residues **reduce the flexibility of polypeptide chains**. Proline cis-trans isomerization is often a **rate-limiting step in protein folding**.

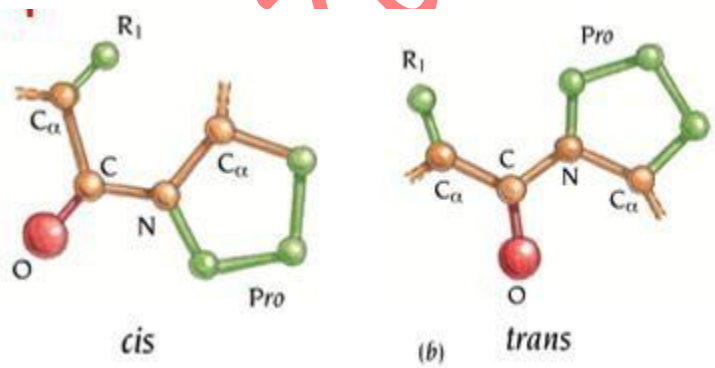


Figure 12 cis and Tran's form of proline

EXAMPLE-3: Cystine

MCQS: Cysteines cement together by **making disulfide bonds to stabilize 3-D protein structures**. In eukaryotes, disulfide bonds can be found in **secreted proteins or extracellular domains**.

Prepared by Qaiad Arshad

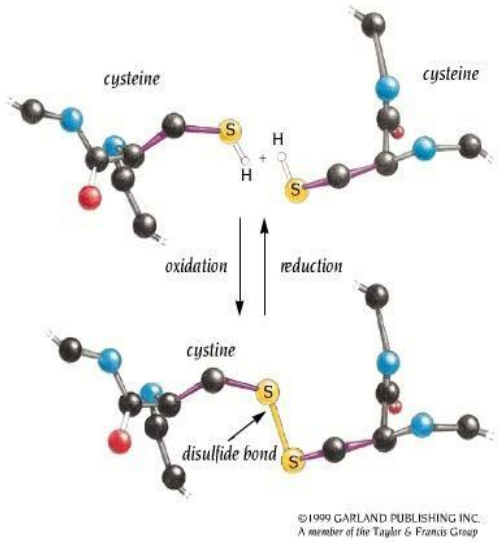


Figure 13 cystine

Amino acids not only have physical and chemical properties, but also structural properties. These structural properties are equally important in giving rise to protein structures.

Module 007: STRUCTURAL TRAIT OF AMINO ACID-II

Each amino acid has a unique set of properties such as charge state, polarity and hydrophobicity. Moreover, it may have unique structural traits as well which can help in protein folding**. Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein. Also, chemically inactive amino acids reduce chances of destabilizing reactions in core.**

There comes a problem in burying hydrophobic amino acids in **protein core Backbone is highly polar (hydrophilic) due to polar -NH and C=O in each peptide unit; these polar groups must be neutralized.**

Form regular secondary structures!

Such as:

- **Alpha Helices**
- **Beta Sheets**

Which are stabilized by H-bonds!

Prepared by Zainab Arshad

Module 008: STRUCTURAL TRAIT OF AMINO ACID-III

The size and structure of each amino acid is unique. Coupled with their chemical properties, each amino acid can uniquely contribute in the protein folding process.

MCQS: Hydrophobic core formed by **packed secondary structural elements provides compact, stable core.** Upon establishment of a stable protein core, unstable or reactive groups can be added.

MCQS: "**Functional groups**" of protein are attached to the hydrophobic core framework. Surface or a **protein or its exterior must have more flexible regions (loops) and polar/charged residues.**

The very few hydrophobic "**patches" on protein surface are involved in protein-protein interactions**. The active regions in a protein are almost all present on the surface.

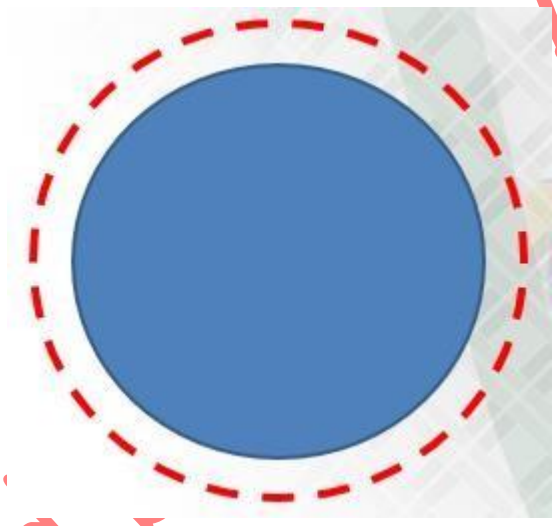


Figure 0.44 Organization of core and surface in a protein

MCQS: Each component of the protein structure has a **unique and precise role in the construction of proteins**. Hydrophobic and hydrophilic components have equally useful roles.

MCQS: The size and structure of each amino acid is unique. Coupled with their chemical properties, each amino acid can uniquely contribute in the **protein folding process**.

Prepared by Arshad

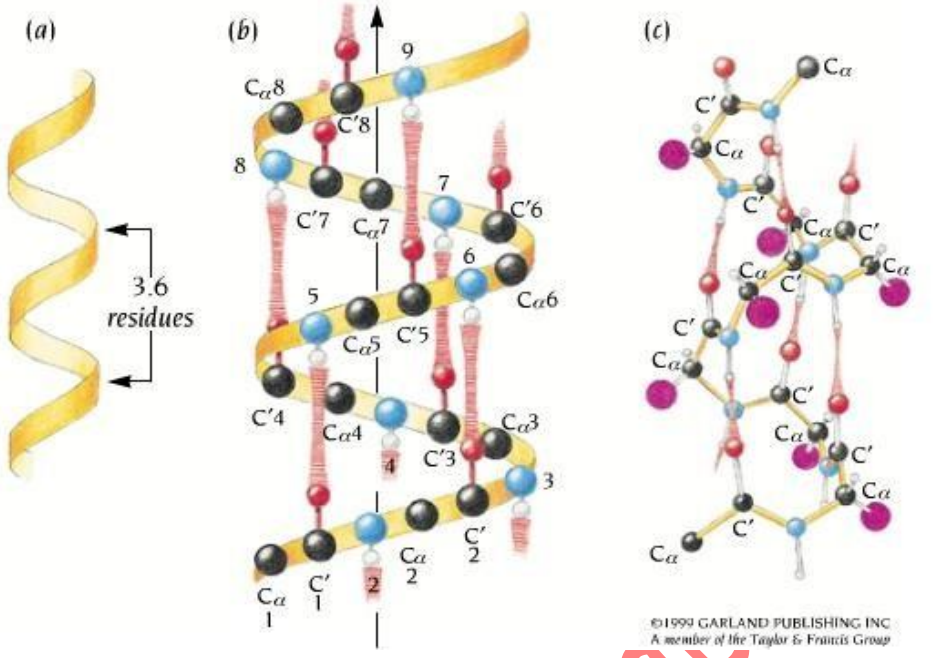


*Figure 0.55 Alpha Helix C = black **O = red N = blue***

MCQS: Alpha Helix is an example of **amino acid folding.** Stabilized by H-bonds between every ~ **4th residues** in backbone. Reactive amino acids are exposed for **external interactions**.

Module 010: INTRODUCTION TO PROTEIN FOLDING

Proteins are made by polymerization of amino acids on ribosomes and proteins properties are linked to the properties of amino acids. There are 20 amino acids in nature each has different chemical composition and that's why each protein is different from other.

But how does a protein actually fold?

The answer is still unknown. Scientists have spent decades in trying to find a definite answer to this question, but to no avail. After polymerization of amino acids, linear chains are formed. When these chains of amino acids are put in water, the proteins fold spontaneously.

MCQS: The folded protein molecule should have **the lowest possible energy**. Anfinsen's dogma (also known as the **thermodynamic hypothesis**) is a postulate in molecular biology that, at least for small globular proteins, the native structure is determined only by the protein's amino acid sequence. Unique, stable and kinetically accessible **minimum free energy**

Prepared by Zainab

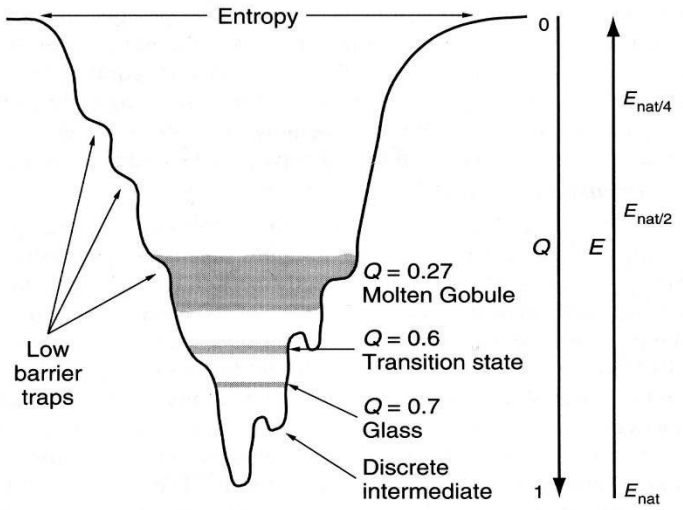


Figure 19.16 Cross section through a folding funnel. *E* corresponds to free energy. [Courtesy of P. G. Wolynes]

Figure 0.66 Overall Energy (stability) of the Protein

S.Q: Why proteins folds?

MCQS: Proteins fold spontaneously **in water**. Proteins fold to **achieve thermodynamic stability.** Proteins fold to organize themselves for performing functions in cells.

S.Q& L.Q: What is the IMPORTANCE OF PROTEIN FOLDING

Module 011: IMPORTANCE OF PROTEIN FOLDING

MCQS: Proteins are like functional **machines in cell**, therefore understanding the folding behavior of proteins can helps us in designing the **suitable drug**. If a protein is **misfolded, then it can lead to a lack of function in the protein.** To study anomalies in structures and to discover newer structural forms, **computational algorithms are used.**

We can study the folding behavior of protein computationally First, we collect clues & evidences from experimentally reported structures. We utilize these observations to analyze unknown structures. The manner in which a newly synthesized chain of amino acids transforms itself into a perfectly folded protein depends both on the intrinsic properties of the amino-acid sequence

Prepared by Zainab Riaz

Module 012: COMPUTING PROTEIN FOLDING POSSIBILITIES

S.Q& L.Q: What are COMPUTING PROTEIN FOLDING POSSIBILITIES

Mcqs. Computing the protein folding can help us study **misfolding, interaction between drugs and proteins etc**. However, first, it is important to know the number of the protein folding possibilities.

Mcqs. Let's assume that each amino acid can fold into **three different conformations**. They are **Alpha Helices, Beta Sheets and Loops**. We know that proteins comprise of **100s of amino acids**

If each amino acid can take 3 different conformations, and its parent protein has 100 amino acids, then $100^3 = 5 \times 10^{47}$ wi

ll be the combination. If it take $1/10^{th}$ of a Nano-second (10^{-10}), then to compute all the folding possibilities will take 1.6×10^{30} years.

In fact, it take a protein less than a second to fold. It's the Amazing speed of folding.

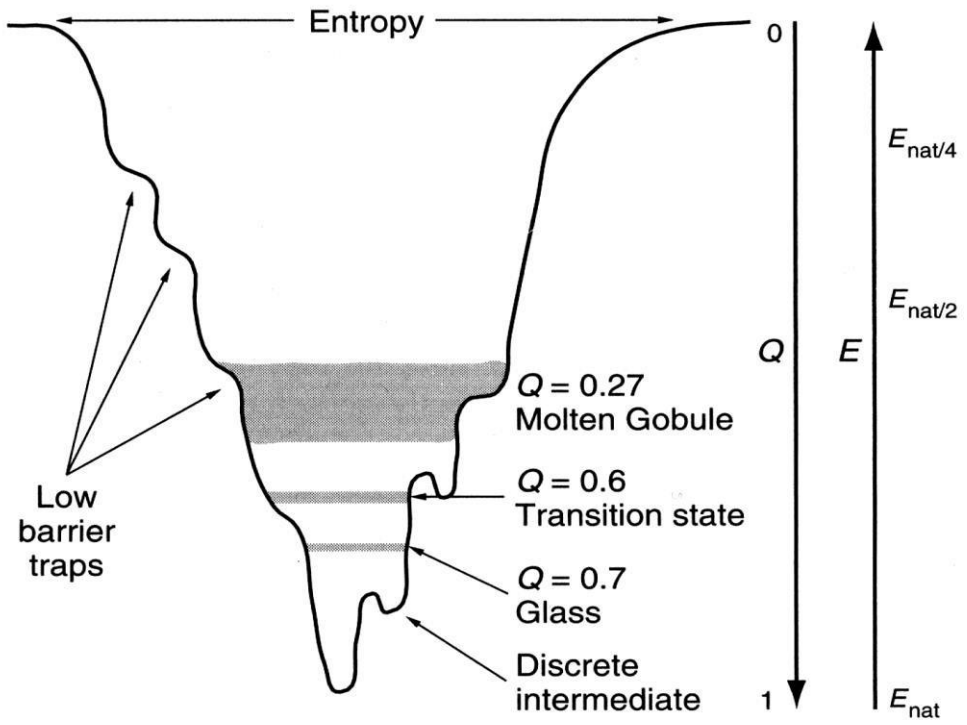


Figure 19.16 Cross section through a folding funnel. *E* corresponds to free energy. [Courtesy of P. G. Wolynes]

Figure 0.77 Overall Energy (stability) of the Protein

This is called "**Levinthal's Paradox".** We will try to understand this folding process using experimental datasets and algorithms. Molecular simulations are also helpful for it.

Module 013: PROCESSING OF PROTEIN FOLDING

Levinthal's Paradox- enormous time required to compute all **folding possibilities**. It's impossible to consider all the **possibilities computationally**. So, we are trying to understand the folding process.

S.Q: What are the forces involves in protein folding

Ans: The forces involved in protein folding include:

- **Electrostatic interactions**
- **van der Waals interactions**
- **Hydrogen bonds**
- **Hydrophobic interactions**

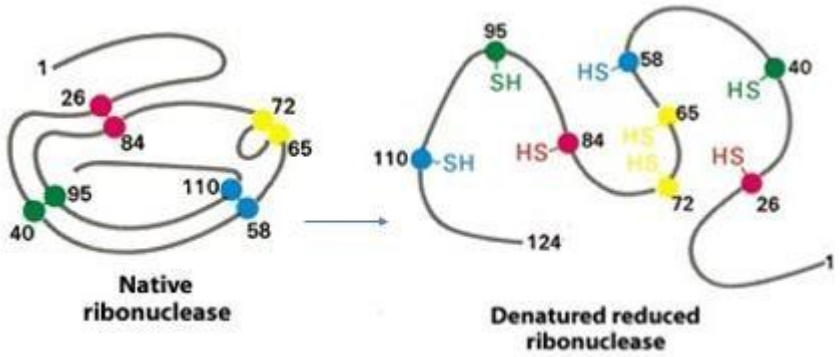


Figure 19 Anfinsen's Experiment

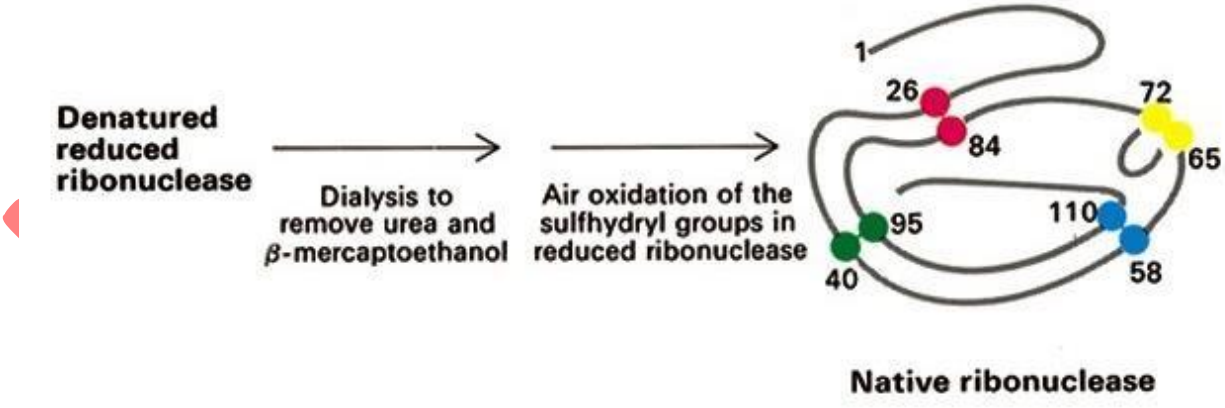


Figure 20 Anfinsen's Experiment

All the information required for folding a protein into its native structure is present within the **MCQ: protein's amino acid sequence**. The native folded form of protein is **thermodynamically most stable as compared to others**

FRAME WORK MODEL

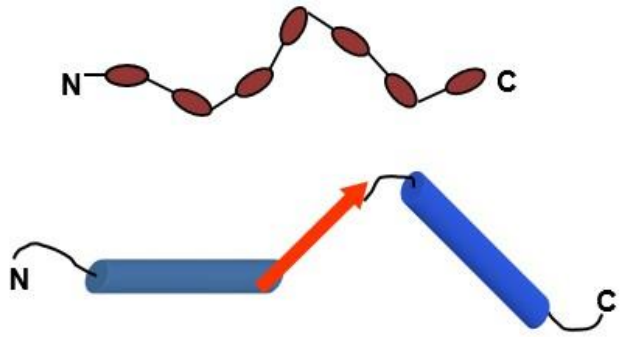


Figure 21 Step 1: Formation of secondary structures

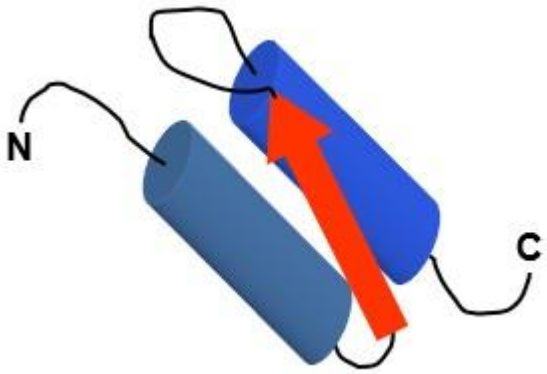


Figure 0.92 Step 2: Arrangement of secondary structures

Prepared by Nauman Rashad

NUCLEAR CONDENSATION MODEL

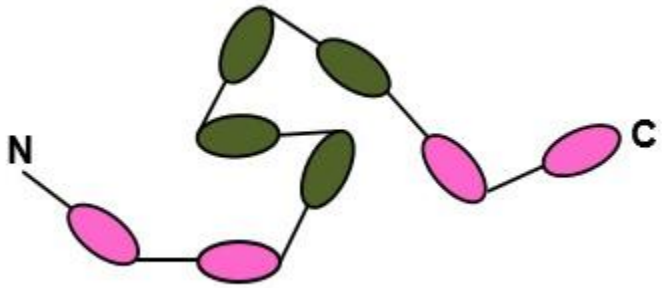


Figure 20.10 Step 1: Formation of a Hydrophobic Core

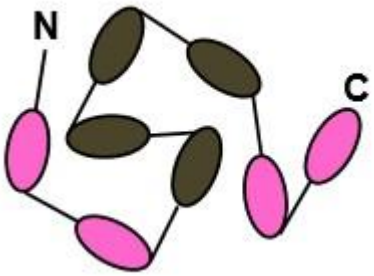


*Figure 20.11 Step 2: Including remaining amino acids and **expanding the nucleus***

MCQS: Several models exist for **folding a protein given its amino acid sequence**. The fundamental requirement is that the folding process remain **spontaneous.** There is still **no definitive folding hypothesis.**

Prepared by Arshad

Module 015: PROTEIN STRUCTURE

MCQS: Proteins spontaneously fold to take **3D forms**. It's a fast yet specific process which leads to a **folded protein.** Several forces act together to fold the protein structure.

Figure 25 Folding funnel

Bond Type	kJ/mol
Covalent Bond	250
Electrostatic	5
van der Waals	5
Hydrogen bond	20

Figure 0.126 Energies of Various Bonds & Interactions

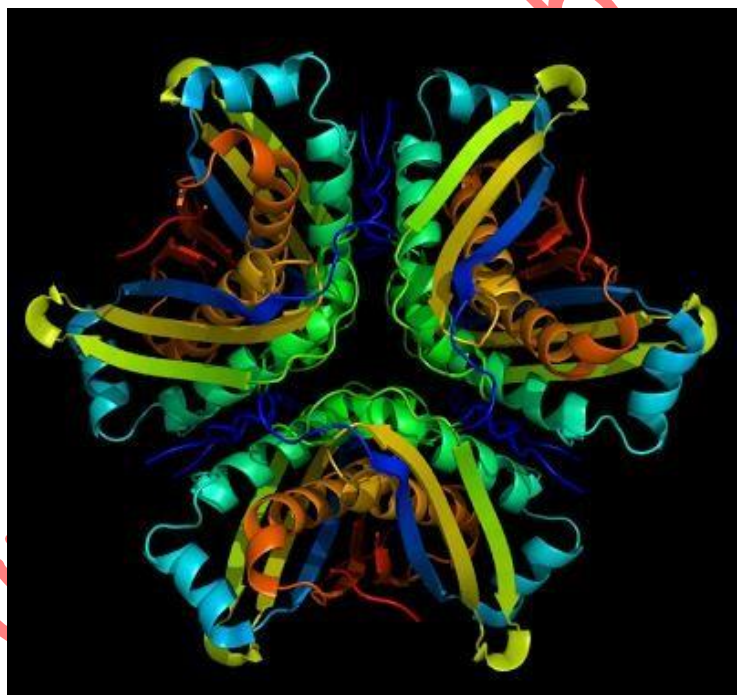


Figure 27 Hydro peroxide resistance protein OsmC (1vla)

Prepared by Arshad

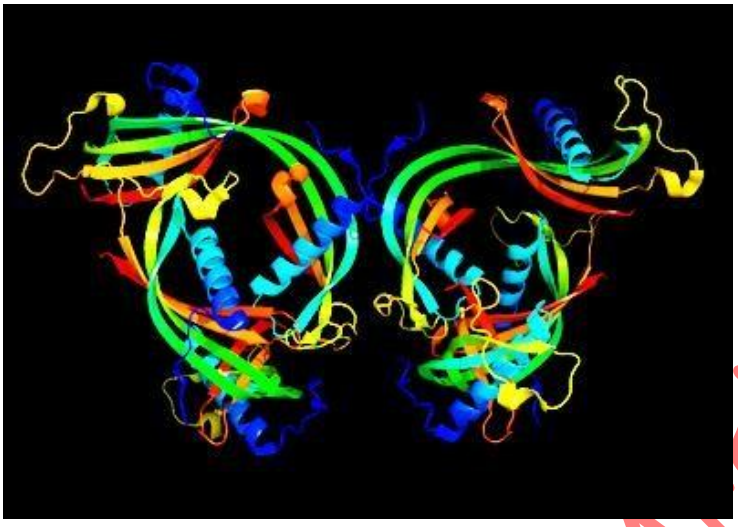


Figure 28 Cystatin – 3 (C) http://beautifulproteins.blogspot.com/

Protein structures are very complex yet they form spontaneously. We will investigate how to develop algorithms to predict such structures.

Module 016: Primary, Secondary, Tertiary and Quaternary Structures

Proteins are made by polymerization of amino acids on ribosomes and proteins properties are linked to the properties of amino acids. There are 20 amino acids in nature each has different chemical composition and that's why each protein is different from other.

Complex protein structures form spontaneously as a protein folds. A huge variety of protein structures exist.

S.Q: Define primary structure of proteins?

MCQS: Each structure is designed to perform a **specific function**. Interestingly, each protein mega structure gets built out of only a few sub-structures. Combinations from the **MCQS: SMALL substructure set are used to construct larger protein structures.**

MCQS: There are many types of structure **Single Alphabet Amino acid tags** can be put together linearly to represent a protein sequence. This sequence is also called the **primary sequence**.
MCQS: Primary sequence can also be referred to **as 1' structure.**

MCQS: Sub-structures are formed as a result of **1' structure's folding**.

S.Q: Define secondary structure of protein?

MCQS: Folded sub structures are called **secondary protein structures** .Secondary structures are also referred to as 2' structures.

S.Q: Define tertiary structures of protein?

Prepared by Zainab Arshad

MCQS: 2’ sub-structures are packed together to form super structures. These protein super structures are called **tertiary structures** .Tertiary structures are also referred to as 3’ structures.

S.Q: Define quaternary structure of protein?

MCQS: 3’ structures represent the complete **monomeric protein structure**.3’ structures can combine with other **polypeptide units to form a quaternary structure**.

Quaternary structures are also called 4’ structures. 4’ structures are exemplified by protein complexes etc.

MCQS: Protein structures are organized into **1’, 2’, 3’ and 4’ modular conformations**. We will investigate how to develop algorithms to predict these structures

S. Q: What is primary structure of protein?

L.Q: Write a note on primary structure of protein

Module 017: Primary Structures of Protein

Protein structures are organized into 1’, 2’, 3’ and 4’ modular conformations. 1’ structures are essentially the amino acid sequence of the proteins.

Prepared by Zainab Arshad

Amino Acid	3-Letter Code	1-Letter Code
Alanine	Ala	A
Cysteine	Cys	C
Aspartic acid or aspartate	Asp	D
Glutamic acid or glutamate	Glu	E
Phenylalanine	Phe	F
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Lysine	Lys	K
Leucine	Leu	L
Methionine	Met	M
Asparagine	Asn	N
Proline	Pro	P
Glutamine	Gln	Q
Arginine	Arg	R
Serine	Ser	S
Threonine	Thr	T
Valine	Val	V
Tryptophan	Trp	W
Tyrosine	Tyr	Y

Figure 30 list of amino acids

***L.Q: what are the methods of obtaining* 1' structure.**

There are two methods for obtaining 1' structure.

- **Edman Degradation (For obtaining primary structure)**
- **Tandem Mass Spectrometry : (For obtaining primary structure)**

S.Q: MCQ:Name the 1' structure protein sequence databases

1' structure databases are essentially protein sequence databases

Examples include Uniprot, Swissprot amongst several others.

Prepared by Arshad

MCQ Protein sequences are the primary structures of **proteins.** The primary or 1' structure of a protein determines its **initial properties**.1' structure lays the **foundation for 2' structures**

S.Q: L.Q: Describe the : Secondary Structures of Proteins?

Module 018: Secondary Structures of Proteins - I

The primary or 1' structure of a protein determines its basic properties and 1' structure lays the foundation for 2' structures. 2' structures are also referred to as secondary structures.

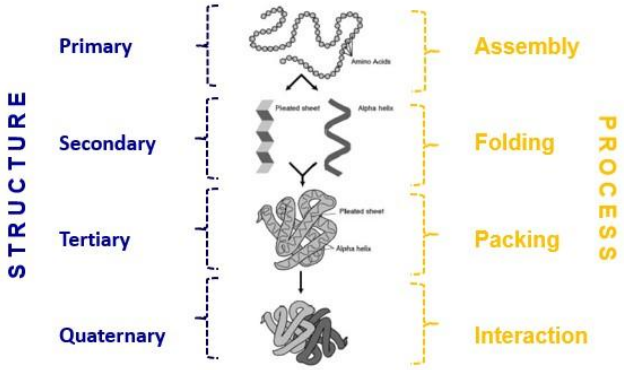
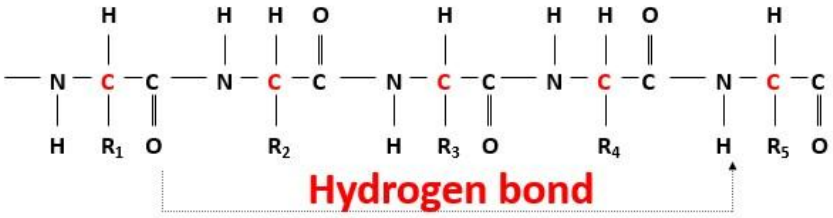


Figure 30.13 Organization of Secondary Structure

Formation of 2' structure

MCQS: C- Terminus is negatively charged .N-terminus is positively charged. C and N termini can therefore make Hydrogen Bonds. Hydrogen Bonds are the reason of 2' structure formation.



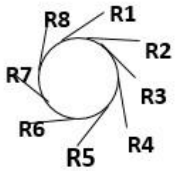


Figure 30.14 Forming Secondary Structure

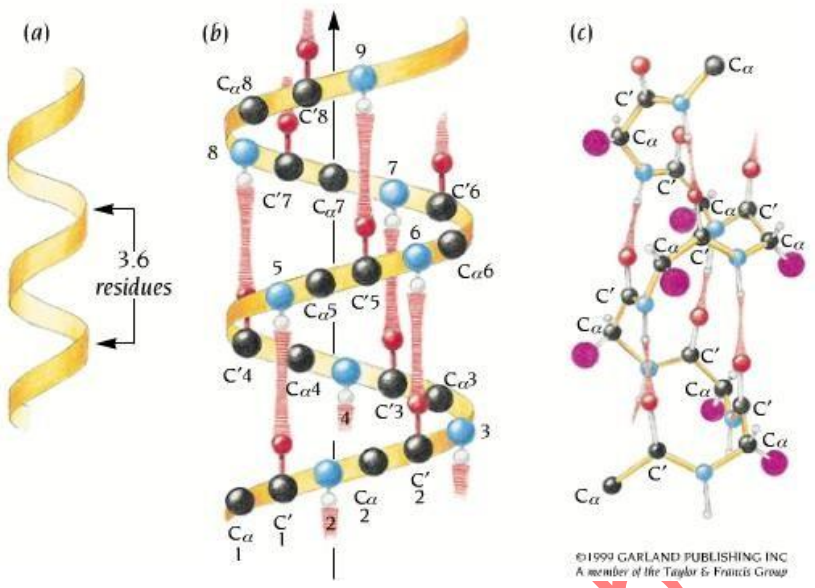


Figure 30.15 Types of Secondary Structures – Alpha Helix

Prepared by Zainab Riaz

MCQS: Protein sequences fold onto themselves and make **H-Bonds to create 2' structures**. Several types of 2' structures exist. These include **Alpha Helices and Beta Sheets.**

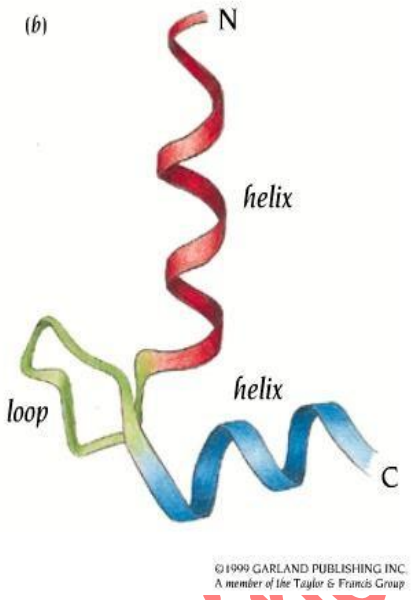


Figure 35 A Special Secondary Structure

S.Q& L.Q: Write the Properties of Loop

Properties of Loop

- **MCQS:** Loops connect **helices and sheets**
- **MCQS:** Loops **vary in length and 3-D configurations**
- **MCQS:** Loops are mostly located **on the surface of proteins**
- **MCQS:** Loops are more "**acceptable"** of mutations
- **MCQS:** Loops are **flexible and can adopt multiple conformations**
- **MCQS:** Loops tend to have **charged and polar amino acids**
- **MCQS:** Loops are frequently components of **active sites**

Prepared by Zaima Arshad

S.Q& L.Q: Write the Properties of Coils

Coils

- **MCQS:** Secondary structure that are **not helices, sheets, or recognizable turns**
- Disordered regions, but also appear to play important functional roles

Loops and Coils are also secondary structure which form the first structures after folding of protein's amino acids. Loops and Coils are very important 2' structures in that they form **active sites of proteins.**

S.Q& L.Q: What is the Tertiary Structures of Proteins?

Write a note on Tertiary Structures of Proteins

Module 020: Tertiary Structures of Proteins

2' structures include **alpha helices, beta sheets, loops and coils**. Upon combination of 2' structures, a tertiary or 3' structure is formed.3' structure is next level of structure organization.



Figure 36 Example of Tertiary Structure

Formation of 3' structure

S.Q: What are force and bonds are present in 3' structure

- **Hydrophobic interactions between nonpolar R-groups**
- **Covalent bonds in the form of Disulphide bridges**

Prepared by Zainab Arshad

Combinations of Alpha helices, Beta sheets, coils and loops help form 3' structures. Covalent bonds, Hydrogen bonds and hydrophobic interactions enforce the 3' structure.

Module 021: Quaternary Structures of Proteins

4' structures or quaternary structures are formed by different peptide chains that make up the protein. Multimeric proteins which comprise of multiple peptides form 4' structures.

Monomeric vs. Multimeric Proteins

Protein comprised of only a single chain (monomeric) do not have a quaternary structure. Proteins with multiple chains can form 4' structures.

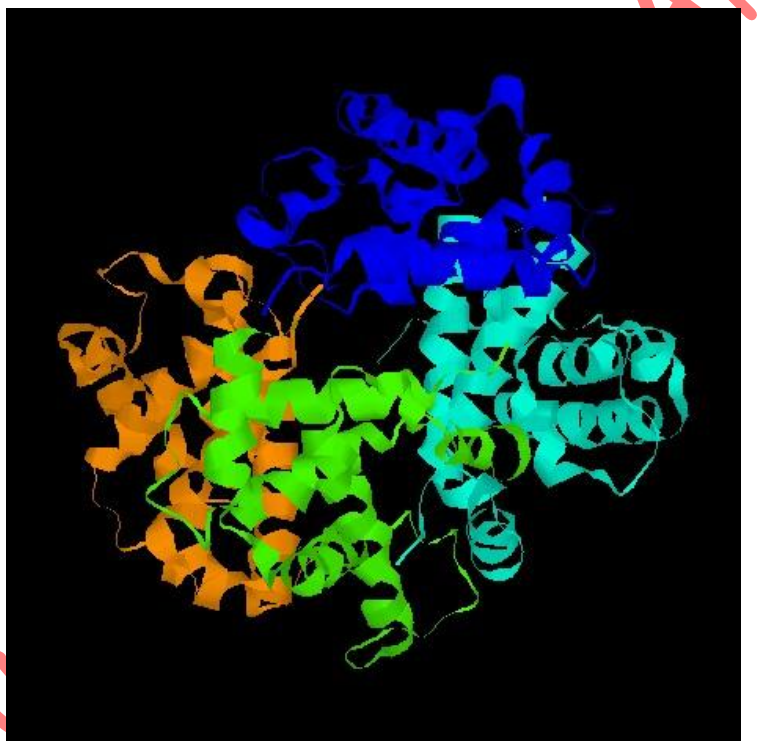


Figure 37 Example of Quaternary Structure See how 2' and 3' structures come together

Which structure of protein is more stable?

4' structures are kept in conformation by Hydrogen Bonds, Covalent Disulphide Bonds, **MCQS:** Hydrophobic Interactions and ionic bonds. **In terms of stability 4' > 3' > 2' > 1'**

Prepared by Arshad

Module 022: Introduction to Bond Angles in Proteins

MCQS: Protein folding results in a linear chain of amino acids getting packed into a compact 3D structure. **<u>This leads to a reduction in bond angles from an initial of 180 degrees (protein's linear form)</u>**

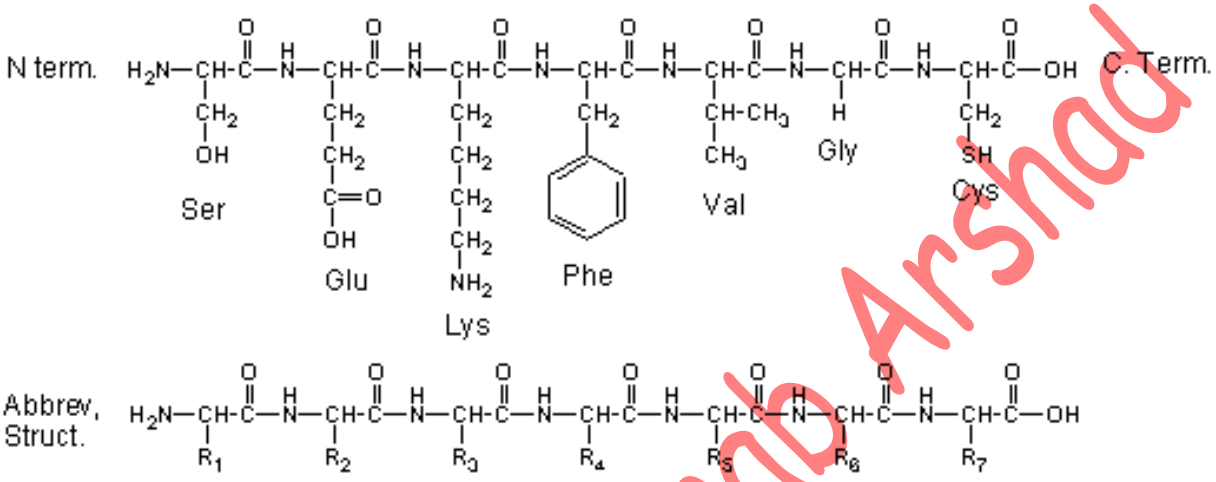


Figure 38 Linear Protein

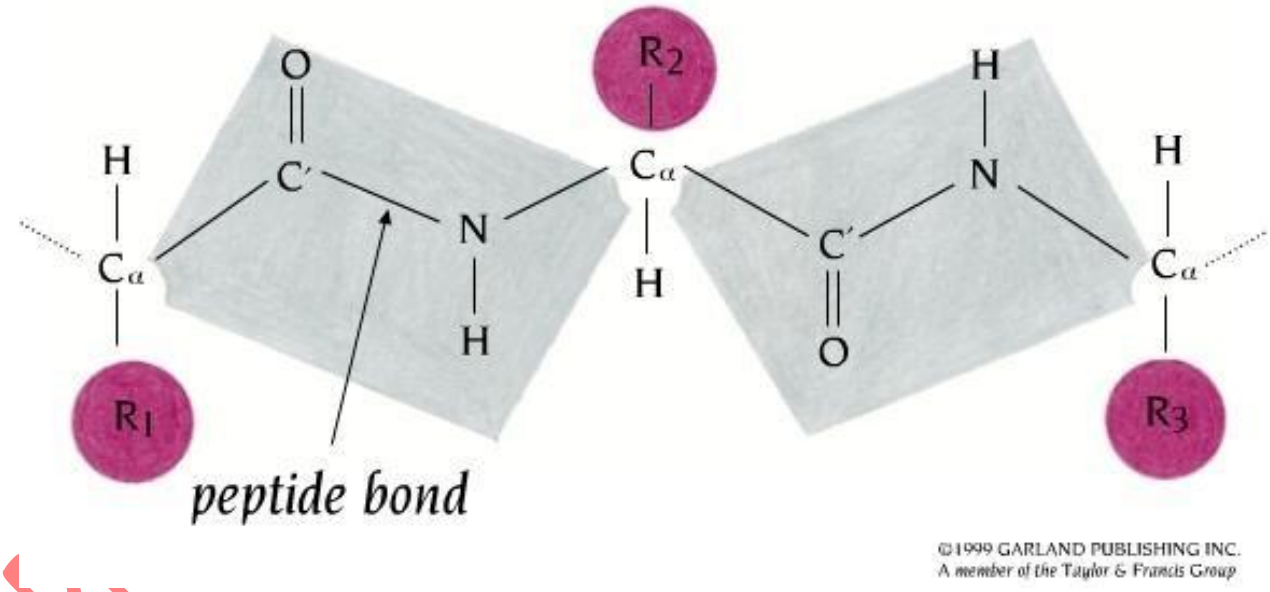


Figure 39 Formation of Planar Peptide Bond

The resultant chain gets its own set of attributes and Peptide bond is planar & rigid.

S.Q: Define <u>Dihedral Angles</u>

- Angle between two planes (i.e. 4 points)!

- Considering the middle two points to be aligned (or overlapped), the angle between the 1st, overlapped and the 4th points forms a dihedral angle.

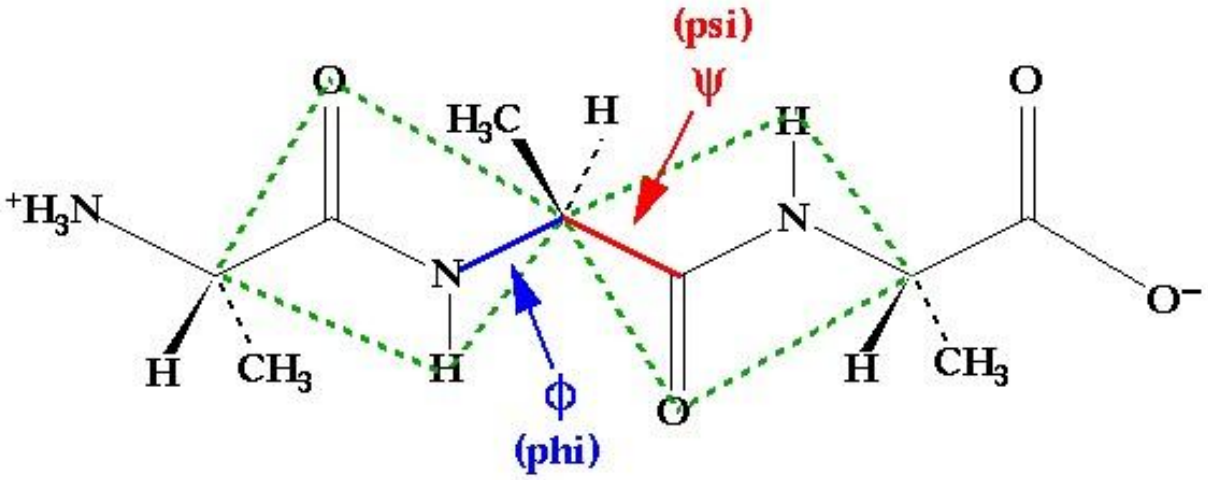


Figure 40 Protein after Folding: Phi and Psi Angles

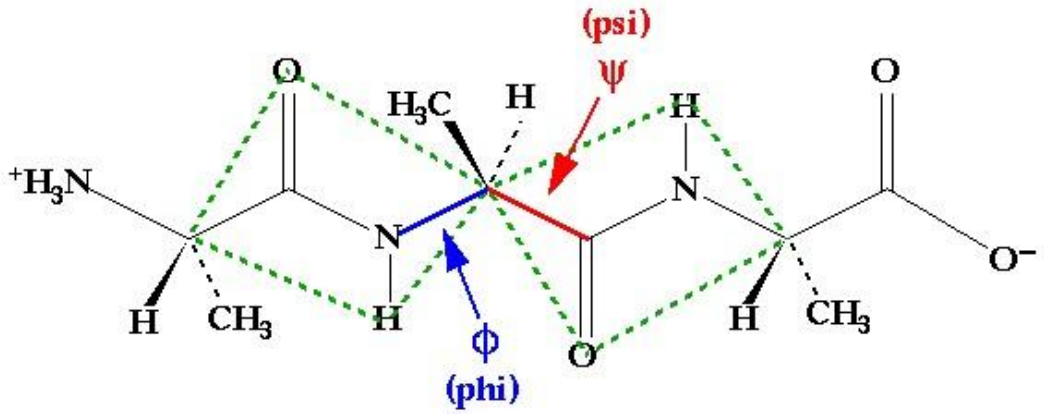


Figure 0.171 Protein after Folding: Phi and Psi Angles

Φ (*phi*, involving C'-N-C^{α}-C') ψ
(*psi*, involving N-C^{α}-C'-N)

Proteins fold into 3D structures. **Phi and psi angles are taken up as a result of folding. These angles can be measured towards understanding the protein structure.**

Module 023: Ramachandran Plot

Phi and Psi angles can be measured with in the folded structures like:

- **ϕ - *phi***
- **Involves C'-N-C^{α}-C'**
- **ψ – *psi***
- **Involves N-C^{α}-C'-N**

Figure 42 Phi and Psi Angles

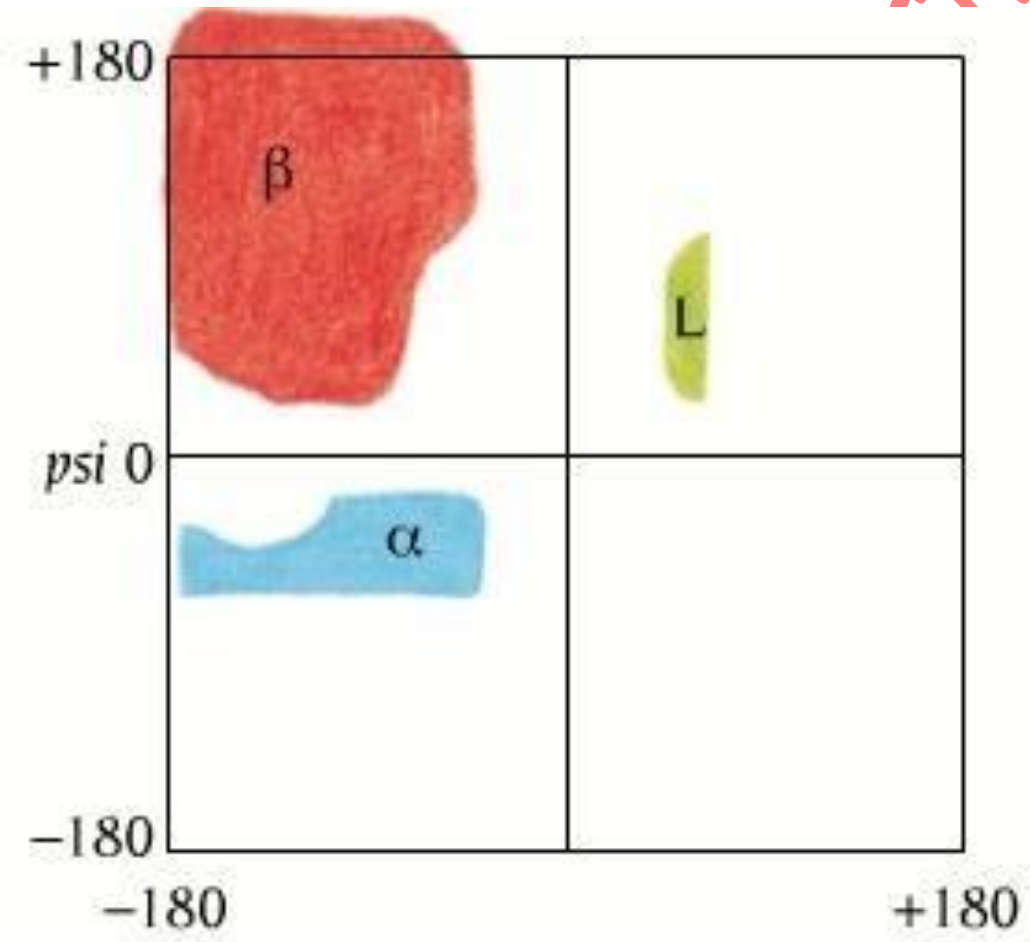


Figure 43 Allowable Phi and Psi Angles

MCQS: Data as in (Lovell et al. 2003) showing about **100,000 data points for several amino-acids**

MCQS: A limited range of Phi and psi angles are taken up as a result of folding. This range of angles constitutes the allowable range **of torsion or rotation angles** that are taken up by the protein.

Module 024: Structure Visualization - I

We know that protein backbone takes up specific rotation angles after folding. A protein consists of multiple amino acids. Each amino acid has **a C-terminus and an N-Terminus**.



Figure 44 Protein Backbone and C atoms

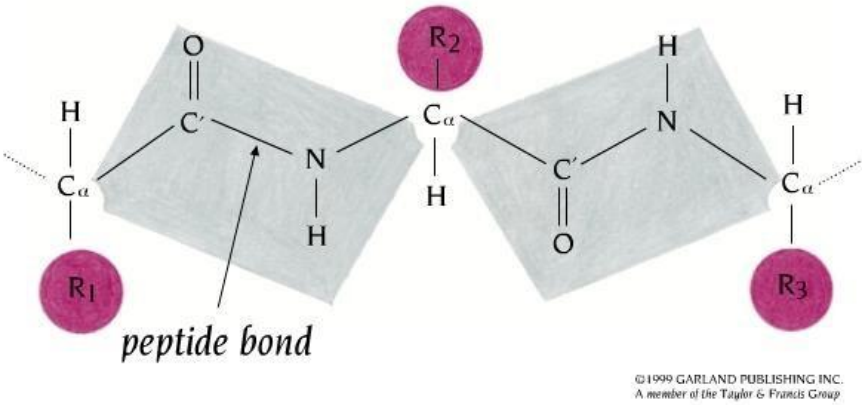


Figure 45 Omitting Planar bonds and Tracing C-Alpha atoms in backbone

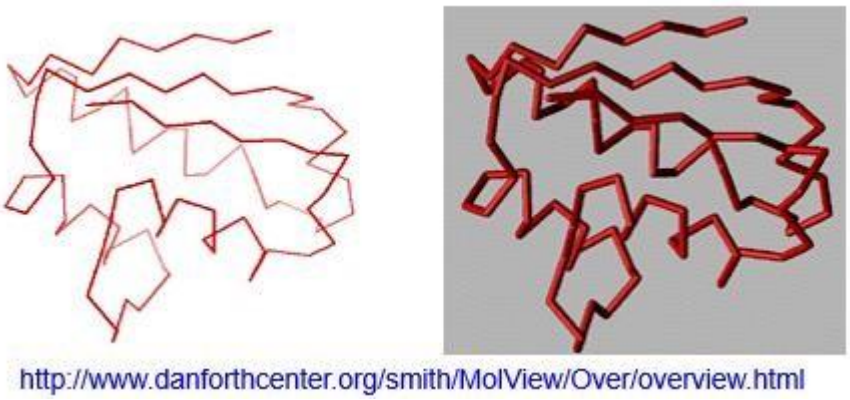


Figure 46 C-Alpha Backbone visualization

MCQS: C-Alpha atoms are traced to **recreate a 3D protein** structure. The choice is made while keeping planar nature of the peptide bond in view. Later we will see how to insert side chains into the visual models as well.

Module 025: Structure Visualization – II

How to visualizalies the structure of protein?

MCQS: C-Alphas can be used to construct the backbone of a protein towards its visualization. We also need a representation of measurements for assigning the atomic distances.

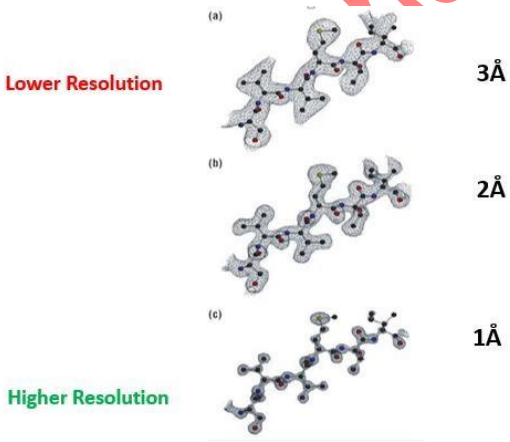
s.q: Define angstrom? What is angstrom? What is the use of angstrom?

MCQS: The **ångström** is used to express the **size of atoms**, molecules and extremely small biological structures, the **lengths of chemical bonds**, the arrangement of atoms in **crystals.**

MCQS: 1 angstrom is a **unit of length equal to 10^{-10} m** (one ten-billionth of a meter) or 0.1 nm

Examples : MCQS: Atoms **of phosphorus**, **sulfur**, and **chlorine are ~1 Å in covalent radius**, while a hydrogen atom is 0.25 Å

MCQ: Ångström is invent by ? Ansedel Anders Ångström (1814–1874)



MCQ: C-Alpha atoms are traced to recreate a 3D protein structure. Each C-Alpha atom is at a distance which can be represented in the **unit "Angstrom".1 A resolution is better than 10 A.**

L.Q: Describe Experimental Determination of Protein Structure

Module 026: Experimental Determination of Protein Structure

C-Alpha atoms are traced to recreate a 3D protein structure. Distances between C-Alphas are measured in the unit "Angstrom".

MCQ: X-Ray Crystallography is used for?

X-Ray Crystallography

Prepared by Mahnoob Arshad

- Crystallography data gives relative positions of atomic coordinates
- Y8u8uyThe data is obtained from **<u>diffractions by the atoms in a protein structure</u>**
- The coordinates of each atom in **<u>x,y and z axis are output</u>**

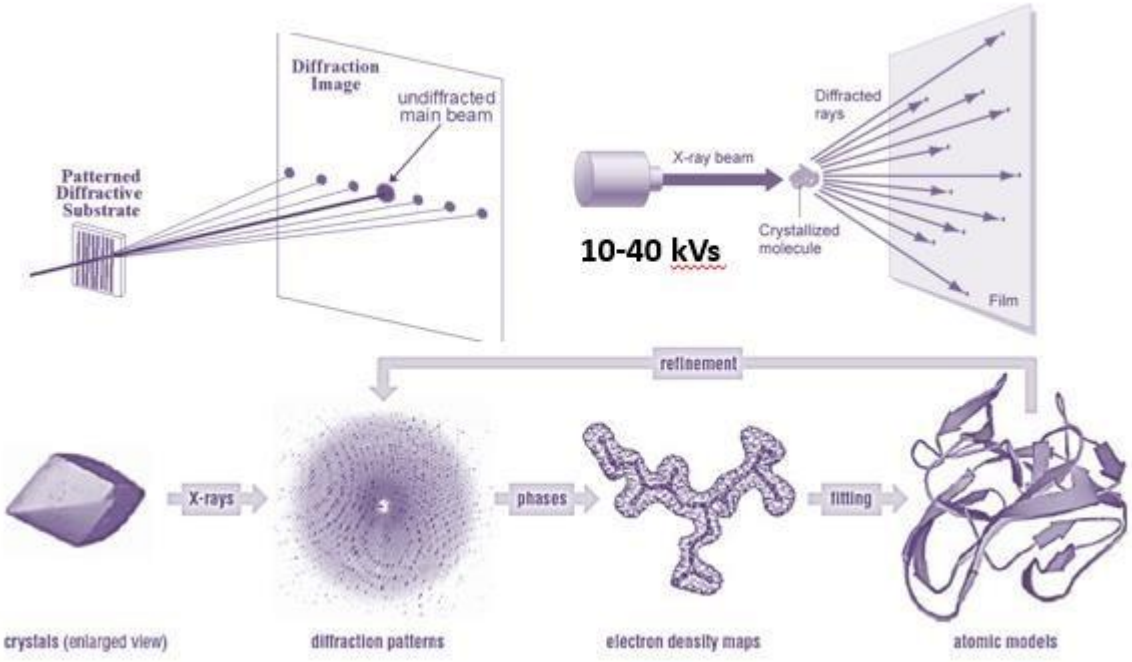


Figure 48 x-ray crystallography

MCQ: Crystallized proteins are used to determine **<u>protein structures</u>**.

MCQ: As X-rays diffract from the atoms in a protein, the atomic distances are noted. These distance in 3D are measured in **<u>Angstroms.</u>**

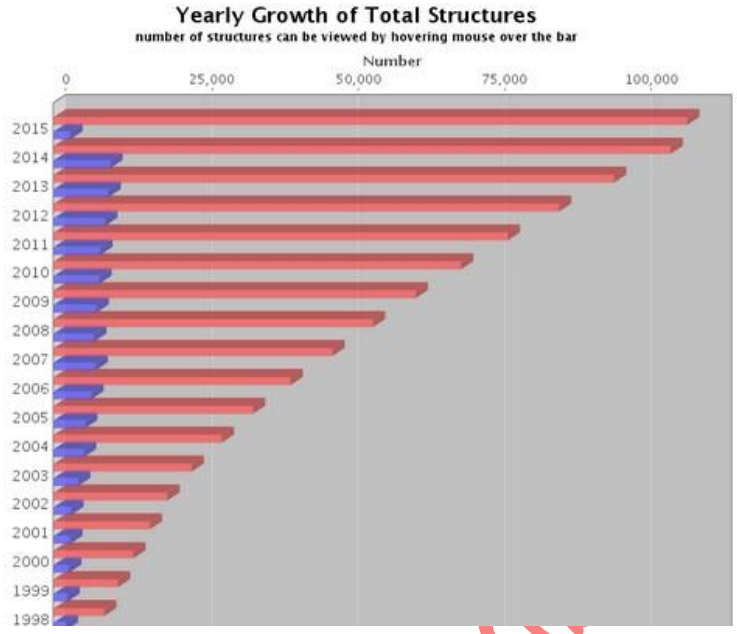
Prepared by Z

Module 027: Protein Databank

S.Q: What are the Protein Databank

MCQ: **Position of <u>C-Alpha atoms are used to construct 3D protein structure</u>**.

MCQ: **X-Ray diffraction data helps <u>measure the atomic positions. X, Y and Z positions of several proteins are available online.</u>**



MCQS:

HEADER – Contains a brief description of the structure, the date and the PDB ID code.

TITLE – The title of the structure.

COMPND – Brief details of the structure.

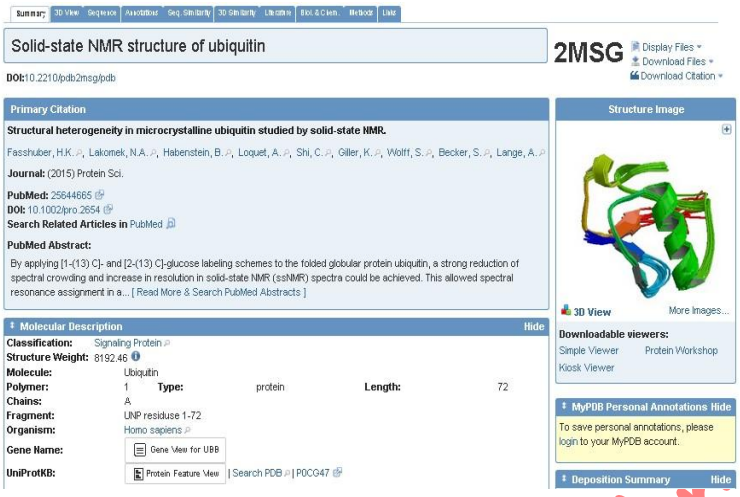
SOURCE – Identifies which organism the structure came from.

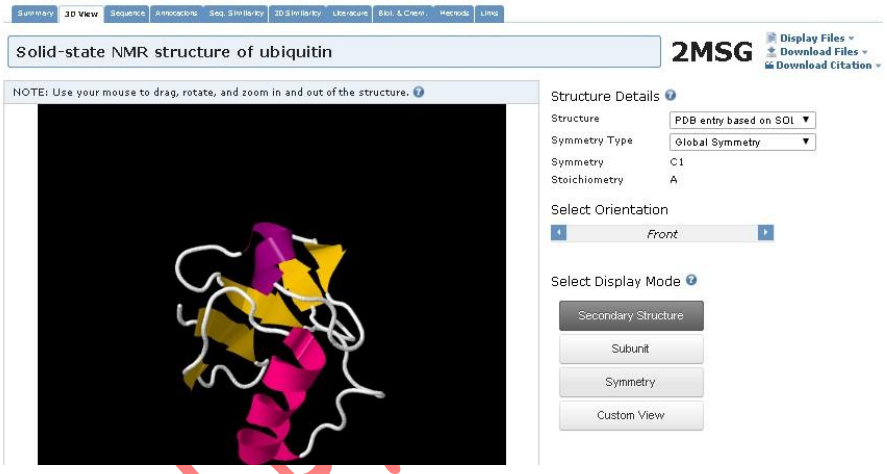
KEYWDS – Lists a set of useful words/phrases that describe the structure.

AUTHOR – The scientists depositing the structure.

REVDAT – The date of the last revision.

Figure 49 PDB File Format





PDB contains protein structure information. It has the coordinates of C-Alphas for **over 50,000 proteins**. Protein structures can be visualized using this information.

Module 028: Visualization Technique

Describe the Proteins **Visualization Technique?**

MCQ: Proteins fold into **3D structures**. **Phi and psi** angles are assumed as **a result of folding**. These angles can be measured and viewed towards understanding the protein structure.

MCQ: To view a protein, we need to **evaluate the physical location of its atoms.**

MCQ: Proteins have **Carbon and Nitrogen in their backbone.**

Prepared By Arshad

CA atomic coordinates

- To trace the backbone of a protein, CA atoms trace can be used
- Note that CA atoms have the side chains attached to them
- A coordinates can be found in the PDB file

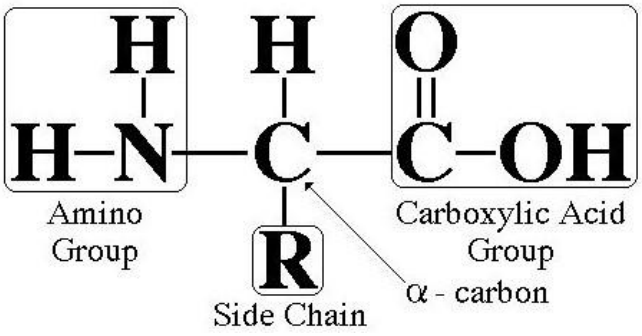
ATOM	1	N	MET	A	1	102.329	111.862	92.452	1.00	78.64	N
ATOM	2	CA	MET	A	1	103.332	112.165	93.516	1.00	77.39	C
ATOM	3	C	MET	A	1	103.877	113.584	93.255	1.00	76.87	C
ATOM	4	O	MET	A	1	103.802	114.075	92.129	1.00	78.54	O
ATOM	5	CB	MET	A	1	104.437	111.099	93.495	1.00	78.51	C
ATOM	6	CG	MET	A	1	105.176	110.881	94.812	1.00	76.25	C
ATOM	7	SD	MET	A	1	106.505	112.076	95.116	1.00	76.95	S
ATOM	8	CE	MET	A	1	107.077	111.551	96.763	1.00	73.31	C
ATOM	9	N	GLU	A	2	104.404	114.235	94.292	1.00	74.31	N
ATOM	10	CA	GLU	A	2	104.917	115.609	94.211	1.00	70.70	C
ATOM	11	C	GLU	A	2	105.963	115.909	93.143	1.00	68.10	C
ATOM	12	O	GLU	A	2	106.048	117.044	92.683	1.00	67.95	O
ATOM	13	CB	GLU	A	2	105.464	116.053	95.574	1.00	75.97	C
ATOM	14	CG	GLU	A	2	106.692	115.246	96.029	1.00	82.57	C
ATOM	15	CD	GLU	A	2	107.263	115.682	97.378	1.00	83.76	C
ATOM	16	OE1	GLU	A	2	106.611	115.396	98.412	1.00	86.90	O
ATOM	17	OE2	GLU	A	2	108.373	116.276	97.401	1.00	81.23	O
ATOM	18	N	ASN	A	3	106.789	114.924	92.784	1.00	64.50	N
ATOM	19	CA	ASN	A	3	107.834	115.134	91.773	1.00	59.93	C
ATOM	20	C	ASN	A	3	107.381	114.767	90.360	1.00	55.41	C
ATOM	21	O	ASN	A	3	108.159	114.856	89.416	1.00	53.33	O
ATOM	22	CB	ASN	A	3	109.086	114.308	92.095	1.00	62.56	C
ATOM	23	CG	ASN	A	3	109.531	114.441	93.535	1.00	62.42	C
ATOM	24	OD1	ASN	A	3	109.045	113.724	94.408	1.00	63.06	O
ATOM	25	ND2	ASN	A	3	110.484	115.326	93.787	1.00	62.31	N

Protein structures can be visualized by tracing the CA atoms. Coordinates of CA atoms can be obtained from the **PDB.** Next, we need a tool to plot these coordinates.

Module 029: Online Resources for Protein Visualization

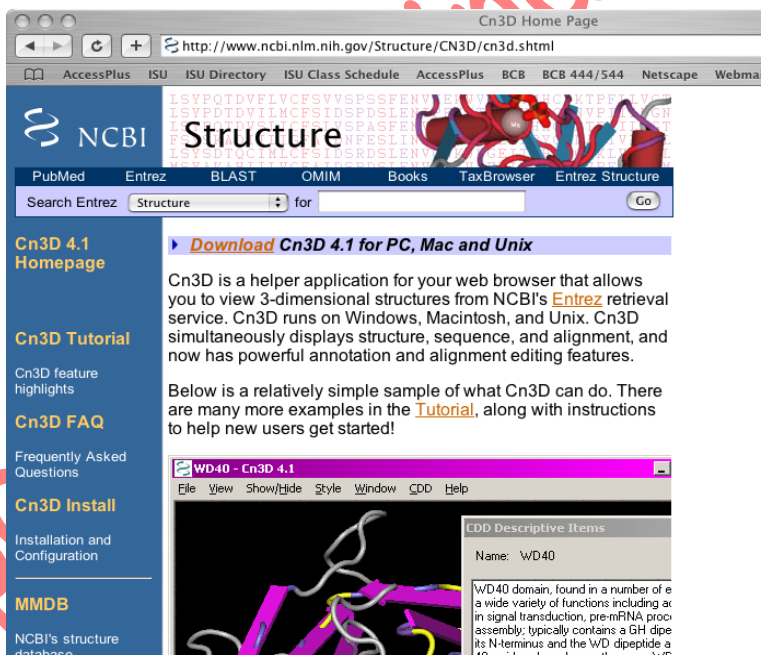
Protein structures can be visualized by tracing the CA atoms. CA Coordinates can be taken from PDB.

Prepared by



Online Tools

- **MCQ**: Rasmol and CHIME are basic tools for **visualizing proteins**
- **MCQ**: Swiss PDB Viewer offers several features such as **protein surface view, alignment of several proteins & modelling secondary structures**
- **MCQ**: PyMOL is a **python-script based tool for visualizing the protein structure**
- Cn3D is another tool which helps us **visualize protein structures**
- It also provides for **annotating protein structures**



Protein structures are visualized using several online tools. These tools include **Rasmol, CHIME, Swiss PDB Viewer and Cn3D.**

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Module 030: Types of Protein Visualizations

To visualize proteins, we use CA coordinates or positions. We can use several online tools to view the resulting model.

S.Q: MCQS: CPK: Corey-Paulin-Koltun Diagrams. In CPK diagrams, each atom is represented by a solid sphere. Spheres are equal to atomic van der Waal radius (the volume of the atom).

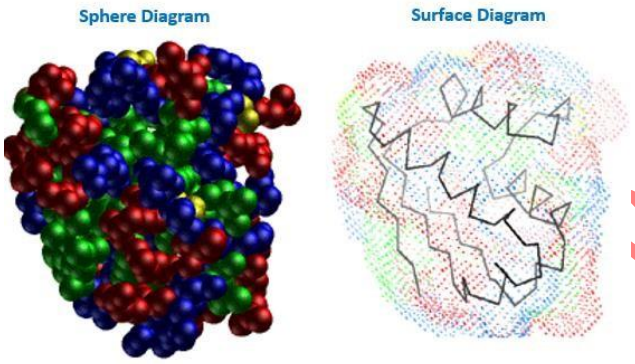


Figure 50 sphere and surface diagrams of protein

http://www.danforthcenter.org/smith/MolView/Over/overview.html

S.Q: What is the use of Ribbon Diagrams ?

Ribbon Diagrams

MCQS: Ribbon diagrams are an **easy and frequently used technique for representing protein structures**. Structure is represented by the secondary structures (fold) using simple **cartoon figures.**

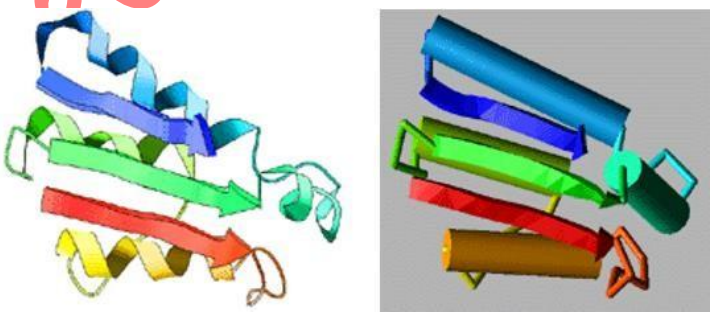


Figure 51 ribbon diagrams

http://www.danforthcenter.org/smith/MolView/Over/overview.html

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Balls & Stick (BS) Models

MCQS: BS model is another popular protein structure representation strategy. **BS Models have atoms as colored balls and intermediate bonds as sticks.**

Figure 52 Balls and sticks model

http://www.danforthcenter.org/smith/MolView/Over/overview.html

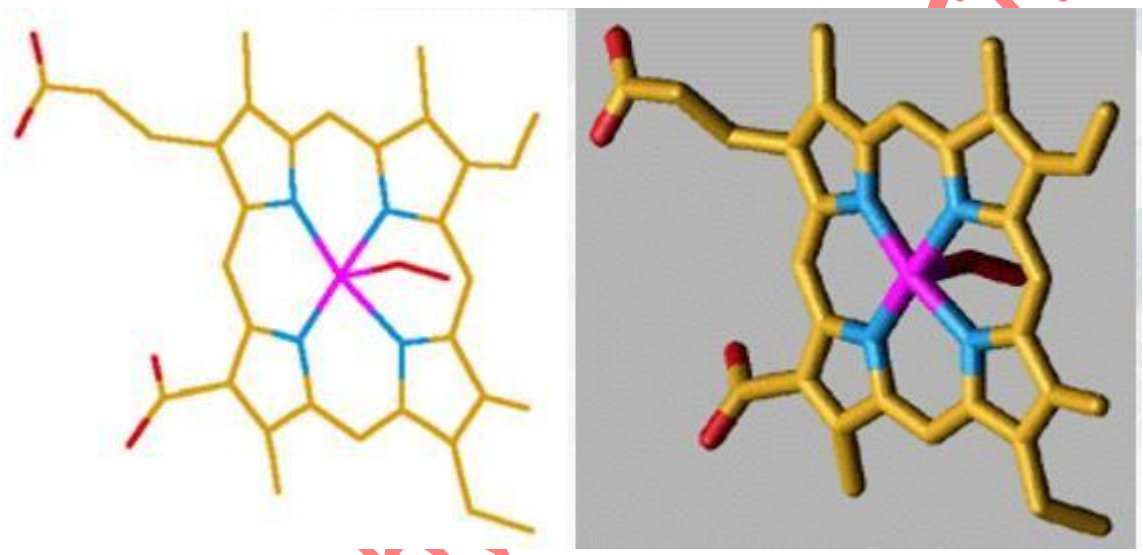


Figure 53 Colored Sticks Models

http://www.danforthcenter.org/smith/MolView/Over/overview.html

S.Q: Name the diagrams of atomic representations?

Protein Structure Visualization can be performed using several atomic representations. These **include CPK, Ribbon and Balls & Stick Diagrams.**

Module 031: Introduction to Energy of Protein Structures

Proteins come together as a result of peptide bond formation between various amino acids. The resulting polymer then goes through the step of folding which leads to the **formation of a 3D structure.**

https://folding.stanford.edu/home/the-science/

L.Q: Write the Role of Amino Acids?

Amino Acids

We know that amino acids can be polar, charged and hydrophobic. Role of polar and charged amino acids in folding. Role of hydrophobic amino acids in folding.

Overall Goal of Folding

MCQS: Anfinsen's thermodynamic hypothesis: Proteins fold for a unique, stable and minimum free kinetic energy structure. What other factors may come into play for **satisfying Anfinsen hypothesis.**

Minimizing Energy

MCQS: We know that if bonds can be formed between two atoms, then energy is **released.** This leads to a situation where there is lesser free energy accessible to each atom for further interactions. So, proteins maximize bonds that can be made between the side chains on each of their constituent amino acids

Such atomic interactions include:

- **Disulphide bonds between Cysteine residues**
- **Hydrogen Bonds**
- **Van der Waals Forces**
- **Electrostatic Interactions between polar/charged amino acids**

MCQS: The greater the number of these bonds, the **more stable a protein becomes**.

MCQS: Hence, the basic idea of thermodynamic stability is to maximize bonding in order to **minimize** the free energy

Prepared by Zainab Arshad

Module

032: Calculating Energy of a Protein Structure

As we know the greater the number of bonds between the amino acids, the more stable a protein becomes.

Figure 54 Energies of Interactions www.ucdavis.edu

Comparison of bond energies

- Hydrophobic interactions >
- Electrostatic interactions>
- Hydrogen bond > van der Waals

Calculating overall energy of a protein structure

Mcq: Given the number of atomic interactions in a protein, you can simply sum the **energy in the protein molecule.**

$$Energy_{TOTAL} = Atoms_{VWF} \times Energy_{VWF} + Atoms_{HB} \times Energy_{HB} + Atoms_{IonicInteraction} \times Energy_{IonicInteraction}$$

Energies of protein structures can be computed by first enumerating the types of interactions between each atom. Then, accumulating the energy of each interaction towards calculating an overall energy of a protein.

S.Q: How to find the number of interactions?

- To determine the number of each type of interaction within a protein, we need to find its inter-atomic distances.
- Based on specific atomic distances, we can guess the type of atomic interaction.
- By looking up at the bond/energy table, we can compute the overall energy.

S.Q: What are the Techniques for structure determination

- **X-Ray Crystallography**
- **Nuclear Magnetic Resonance (NMR) Spectroscopy**

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Module

We need to know the structure of the protein to calculate atomic distances. Atomic distances tell us about atomic interactions with neighboring atoms. To determine the structure, we use **XRay or NMR.**

034: Review of Experimental Structure Determination

The greater the interactions between the amino acids, the more stable a protein becomes. We can calculate energy of a folded protein based on the number and types of atomic interactions.

The structure also dictates which functions a protein can perform via the positioning of hydrophilic & polar amino acids.

For determining stability, structure & function, we need to find the **amino acid interactions**. Several experimental methods exist for structure determination.

- X-Ray Crystallography
- Nuclear Magnetic Resonance (NMR) Spectroscopy

Zainab Arshad



Figure 55 to measure a bond/interaction, we must first see atoms

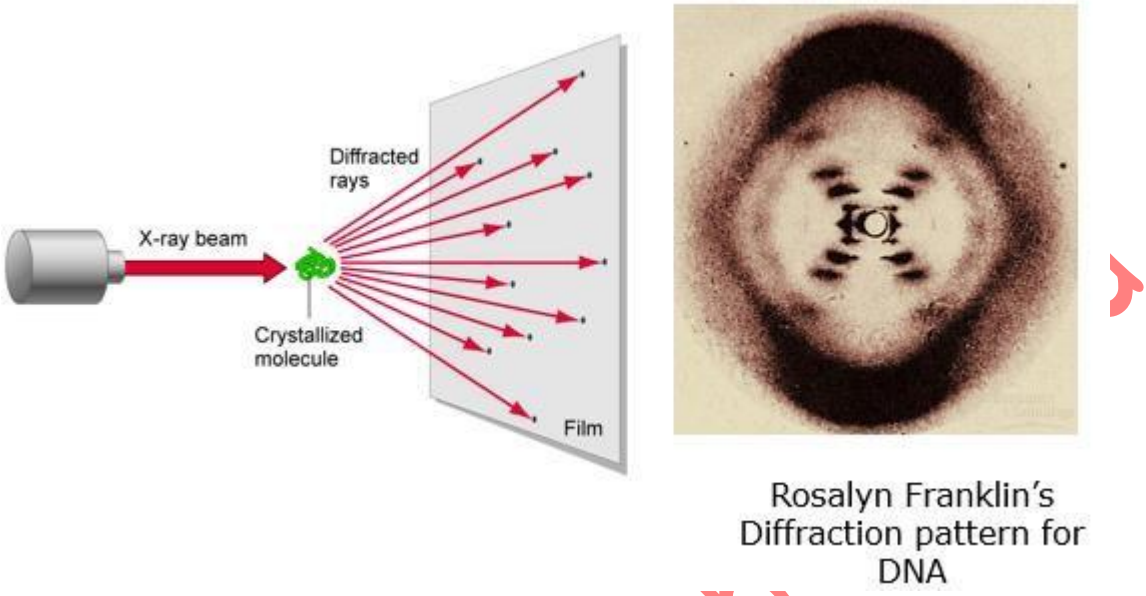


Figure 56 Principle of X-Ray Crystallography

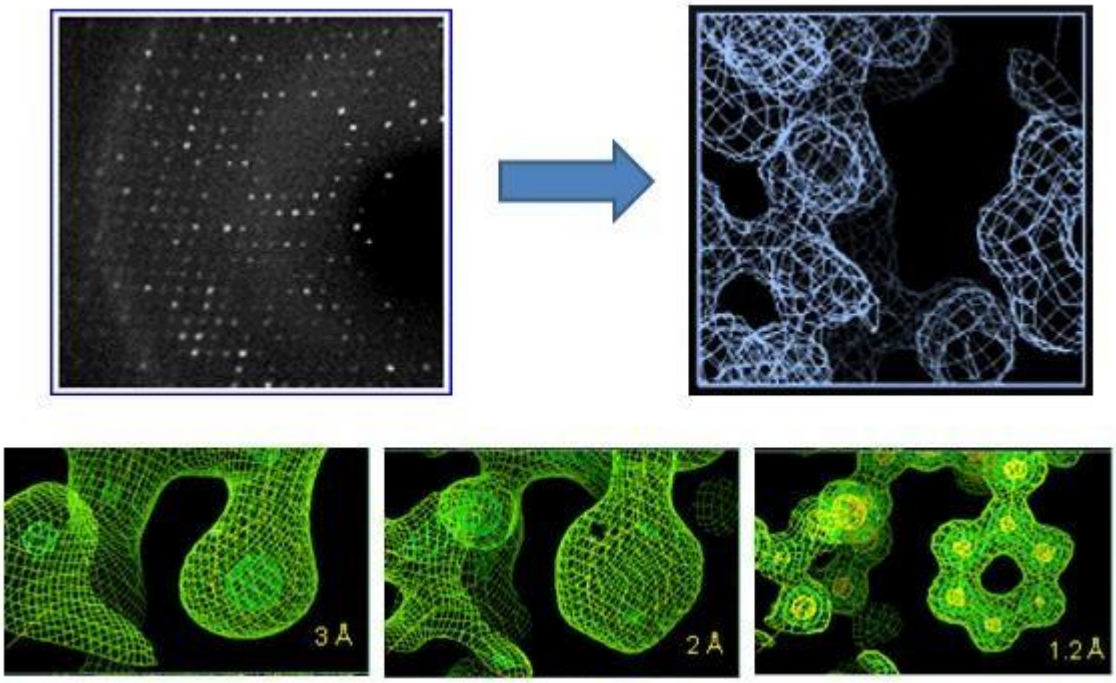


Figure 57 from Diffraction Patterns to Atomic Positions

MCQS: Upon establishing the atomic **positions and distances**, we can then check for possible interaction between the different atoms. Atomic distances can help us classify interaction types e.g. **hydrogen bonds, electrostatic & polar**.

Module 035: Alpha Helices - I

Atomic distances can tell us about their existential interactions. Different types of interactions may occur between atoms. E.g. Hydrogen Bonds, Polar etc. If two atoms are participating in a covalent bond, their distance is ~0.96A. In case of hydrogen bond formation between atoms, the inter-atomic distance is ~1.97A. X-Ray data should have a minimum of 1.97A resolution.

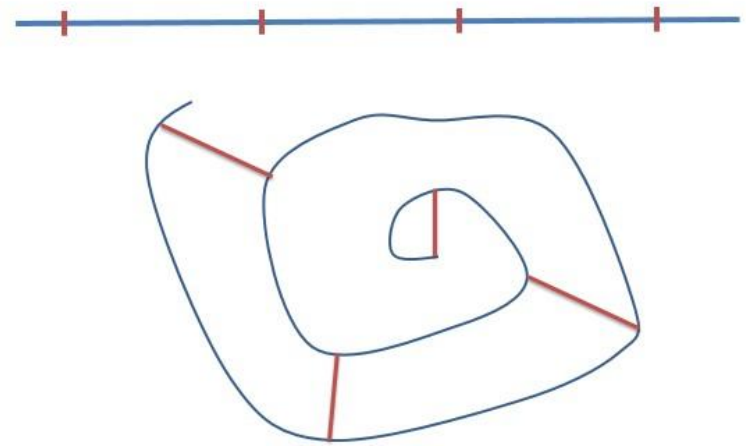
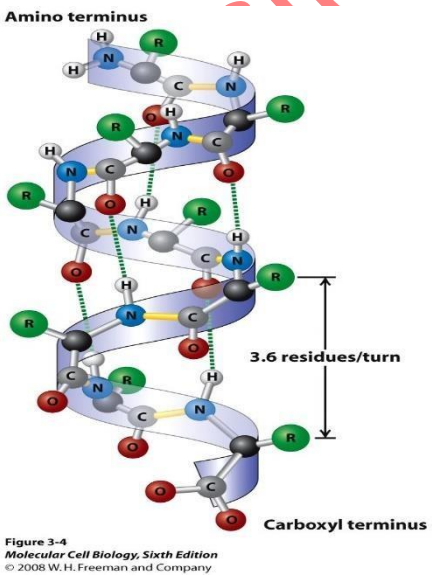


Figure 58 Hydrogen Bonds to Fold an Amino Acid Chain



X-Ray Crystallography data shows that Hydrogen atoms of N-Term may come together with Oxygen atoms of C-term amino acid at 4th neighboring position. Their atomic distance is ~1.9A and hence are considered to be in a hydrogen bonds.

Prepared

Module 036: Alpha Helices – II

MCQS: X-Ray Crystallography of protein shows **that Hydrogen atoms of N-Term come together with Oxygen atoms of C-term amino acid at 4th** neighboring position to make Hydrogen bonds.

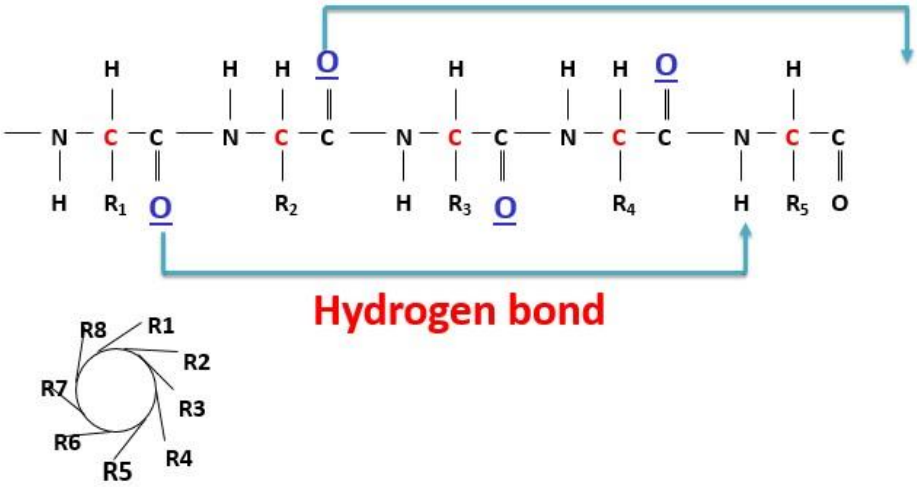
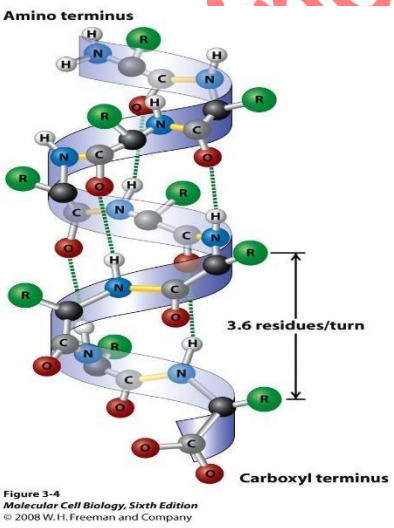


Figure 59 Forming Alpha Helix



Every Oxygen bound to 4th neighboring Amino Group' Hydrogen.

Prepared

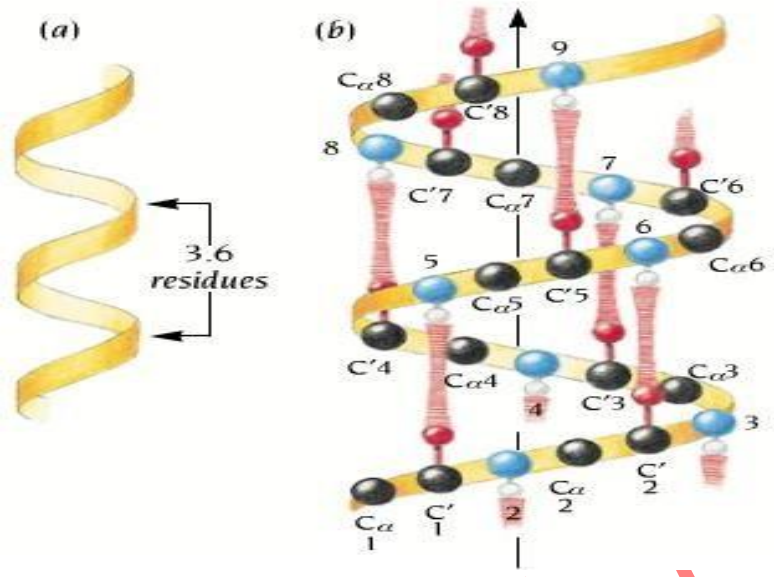


Figure 60 Carbons (Black) & Nitrogen's (Blue): 1-5, 2-6, 3-7…

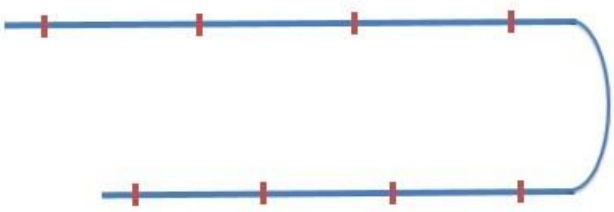
Helix Formers

MCQS: Yes, "Helix Formers" are generally **hydrophobic amino acids (M, A, L…).** Alpha Helices are formed by **hydrogen bonding (O-H) between C_i and N_{i+4} atoms in the protein backbone.**

Module 037: Beta Sheets - I

MCQS: **Beta Sheets are another common secondary structure**. They are constituted by several Beta Strands which come together**. 5 to 10 resides** are needed to make a Beta Strand, typically.

Hydrogen Bonds to make in **Beta Strands**

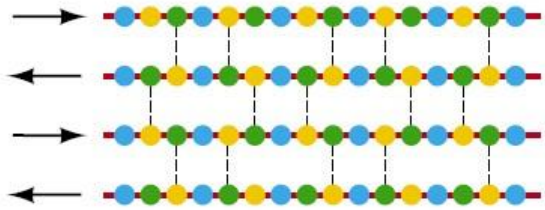


The Beta Sheet is made up of several Beta Strands

C-Alpha atoms and the CO and NH groups are shown in blue, yellow, and green, respectively.

Prepared by Zainab , Shad

This is called a parallel beta sheet.



This is called an anti-parallel beta sheet.

Beta Sheets are another secondary structure that can be formed as a result of hydrogen bonding between the **protein back bones**. Some amino acids have a preference for making Beta Sheet

Beta strands can make hydrogen bonds with each other and organize as beta sheets.

S.Q: & L.Q: What are the properties of beta sheets?

Beta Sheets have different Properties:

- **Beta Strand**
- **Beta Sheet**
- **Beta Barrel**

Beta Sandwiches

Beta Barrels

Beta Barrel is made of a **single beta sheet** that twists and coils upon itself. The first strand in the beta sheet makes a hydrogen bonds with the last strand

S.Q: DEFINATION What is beta barrel

Prepared by Zainab Arshad

MCQ:. A beta barrel is a large beta-sheet that twists and coils to form a closed structure in which the first strand is hydrogen bonded to the last. Beta-strands in beta-barrels are typically arranged in **an antiparallel fashion**.

Prepared by Zainab Arshad

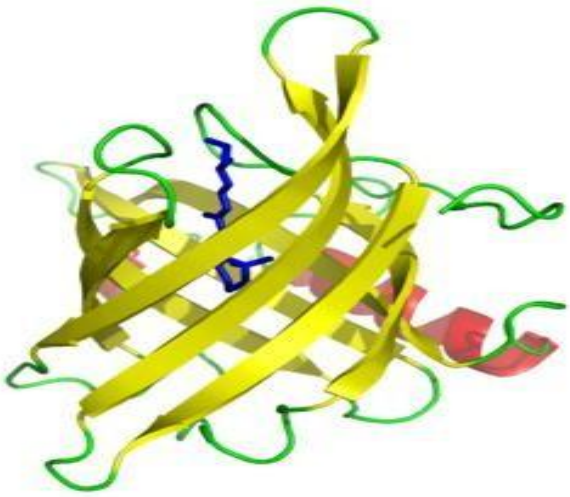


Figure 62 beta barrel

Prepared by Zainab Arshad

s.q: define Beta Sandwiches ?

Beta Sandwiches

MCQ: Beta Sandwiches are made of **two beta sheets** which are usually twisted and packed so their strands are **aligned.**

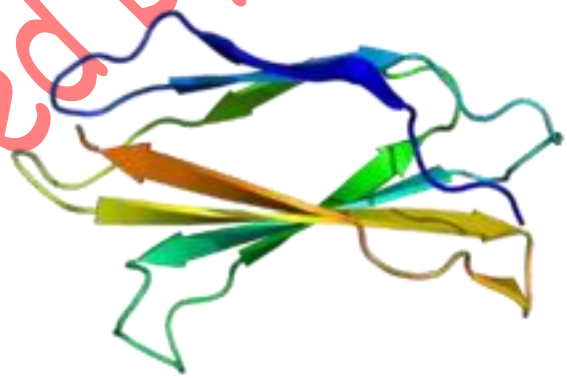


Figure 63 Illustration of the β-sandwich from Tenascin C (PDB entry: 1TEN).

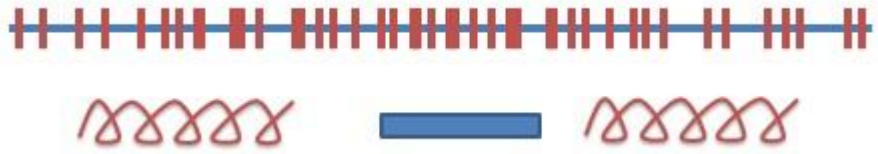
MCQ: Beta Sheets are formed by H bonds between of **5–10 consecutive amino acids** in one portion of the backbone with another **5–10 farther down the backbone**. Beta strands may be adjacent (with a loop in between) or far with other structures in between.

S.Q: What are loops? Define loops?

039: LOOPS-I

Alpha Helices and Beta Sheets are secondary structures formed as a result of hydrogen bonding in between protein backbone atoms.

Protein Backbone and Secondary Structures



MCQ: Loops are formed by amino acids present in the middle of the Alpha Helices and Beta Sheets in a **protein backbone**.

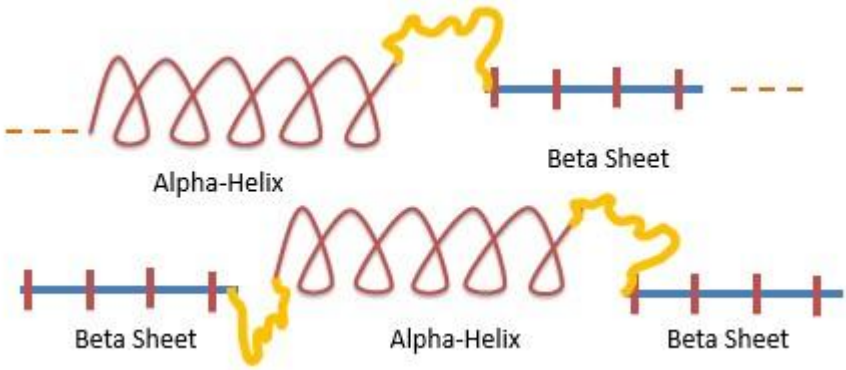
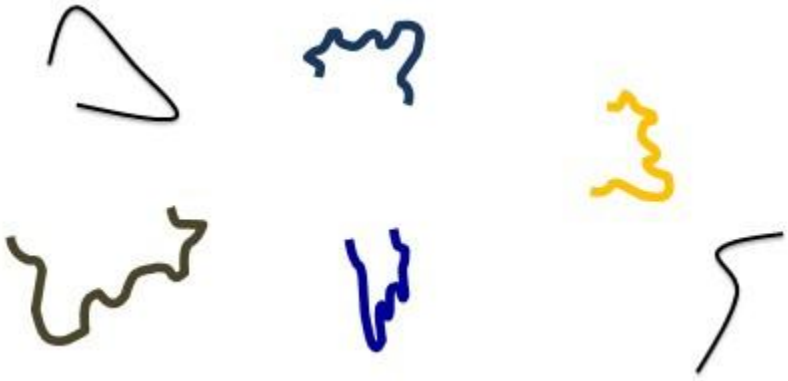


Figure 65 Joining Alpha Helices and Beta Sheets in a Protein Backbone

Variability in length and conformation allows loops to join Alpha Helices and Beta Sheets in a variety of ways. Loops are variable **in length and 3-D conformations**.



Prepared by Shahzad

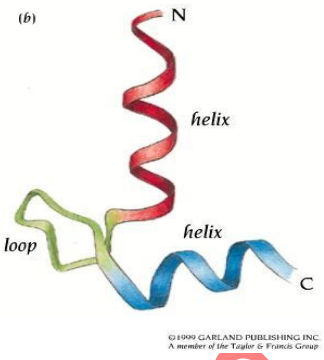
S.Q What are the Characteristics of loops?

Characteristics

- **MCQS:** Loops are mostly located on the **surface of protein structure**
- Mutate in sequence at a much faster rate than Alpha Helices and Beta Sheets
- Loops are **flexible and can adopt multiple conformations**

Loops dictate the overall structure of protein as they couple Alpha helices and beta sheets

Loops dictate the overall structure of protein as they couple Alpha helices and Beta sheets. Loops are flexible and have variable lengths so as to successfully bridge between secondary structures.



Loop Properties

- Loops are mostly comprised of **charged and polar amino acids**
- Loops frequently participate as components of active sites

s.q: write the types of loops:

Types of Loops

- Hairpin loops are two amino acids long and join anti-parallel Beta strands
- Other Loops may be 3 to 4 amino acids long
- Loops fall into various families

MCQS: Loops are the third type of secondary structure after Alpha helices and Beta sheets. Loops are unique in that they are flexible and variable length. Loops constitute active sites.

Prepared by Zainab Arshad

041: COILS

Alpha helices and beta sheets are the regular secondary structures. Loops are flexible secondary structures &connect alpha helices and beta sheets. Coils are another secondary structure. Coils are unstructured and unlike loops**. Essentially, a secondary structure which is not a helix, sheet or loop is a coil.**

S.Q: Write the Functional Aspects of Coils?

Functional Aspects of Coils

- Coils are apparently disordered regions
- They are oriented randomly while being bonded to adjacent amino acids ➢ However, coils also appear to play important functional roles



Figure 68 Coils in Myoglobin

Coils are those secondary structure formed by the protein backbone which are neither **helices, sheets nor loops**. In fact, coils do not have a consistent classifiable structure. Hence, coils are random structure and random length.

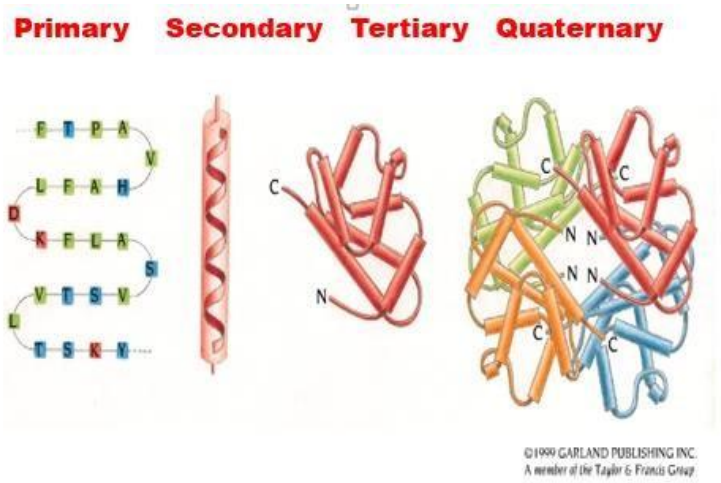
Prepared by Zaheb Arshad

Module 042: Structure Classification - I

Proteins have primary, secondary, tertiary & quaternary structures. Each level of protein structure organization is known to impart specific characteristics to the protein.

Review of the 4 structure levels

- Primary Structures
- Secondary Structures
- Tertiary Structures
- Quaternary Structures



Structural artifacts tend to be more conserved as compared to their sequences. Therefore, it may be useful to look at the secondary/tertiary structures for conservation study.

Classification

- The evolution of protein structures and their hierarchy is not systematized
- Hence, we need to classify the function of protein by examining their secondary and tertiary structures

Motifs (Non-functional Combinations of 2' structures)

Prepared by Zu shad

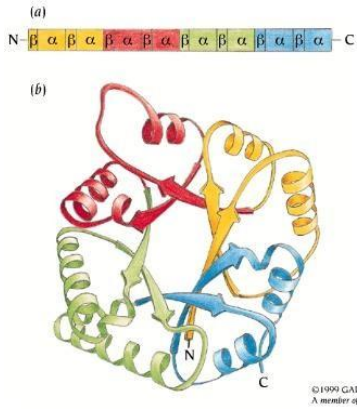


Figure 69 Domain (Functionally Complete)

Domains are semi-independent functional structures in a protein. Have a stable structure. Over ~40 residues. Protein may contain multiple domains.

Prepared by Zainab Arshad

Module 043: Structure Classification - II

Domains are semi-independent functional structures in a protein. Protein may contain multiple domains. Hence, we can try to classify proteins by their domains. Locally Compact – Domains interact (H-bonds) more internally than externally. Domains have a hydrophobic core. Domains are contiguous (min. chain breaks).

Domains have a minimal contact with rest of the peptide. Solvent area in contact with each domain should not vary significantly upon separating two separate domains.

S.Q: Write the Types of Domains ?

Types of Domains

- Alpha Domains
- Beta Domains
- Alpha/Beta Domains
- Alpha + Beta Domains
- Alpha & Beta Multi-Domains
- Membrane & cell-surface proteins

So, by looking at proteins, we can list the domains present in each protein. Once domains in each protein are listed, we can classify whole proteins into various types and classes.

044: Examples of Protein Domains

S.Q: There are many domains for protein structure prediction.

- **Alpha Domains**
- **Beta Domains**
- **Alpha/Beta Domains**
- **Alpha + Beta Domains**
- **Alpha & Beta Multi-Domains**
- **Membrane & cell-surface proteins**

Prepared by Zainab Arshad

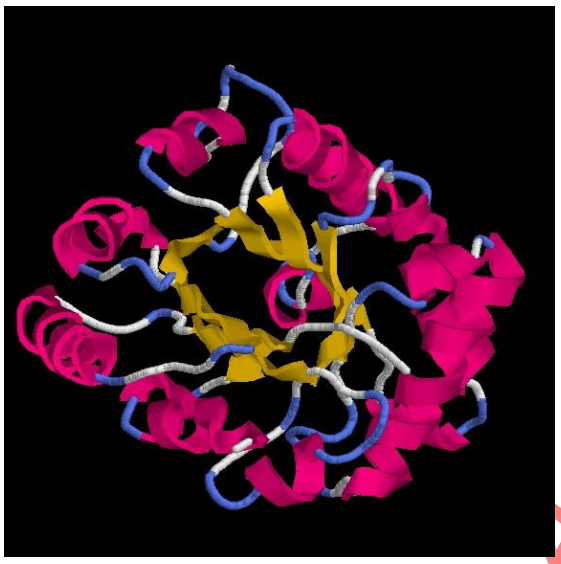


MCQ: ***Figure 70 Alpha Domain: Hemoglobin (1bab)***



Figure 71 Immunoglobulin (8fab)

Prepare



MCQS:Figure 72 Alpha / Beta: Triosephosphate isomerase (1hti)

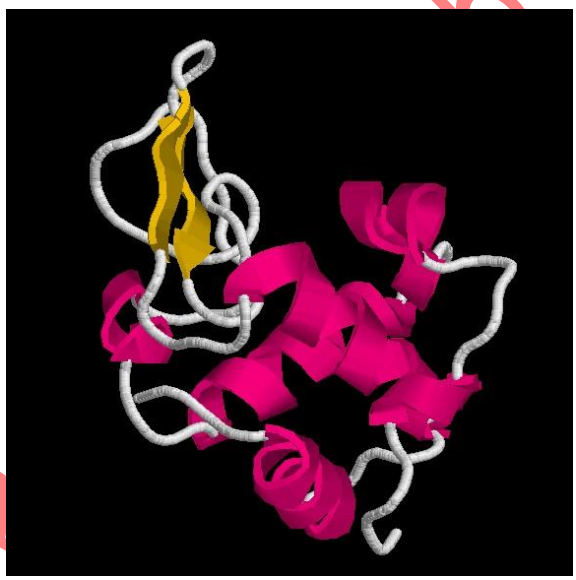


Figure 73 Alpha + Beta: Lysozyme (1jsf)

Various types of domain architectures exist in proteins. Such architectures can be classified into general structural classes. Databases can be made from classes.

045: CATH Classification

Domains can be classified into structural classes. Classes can be further classified into Architecture and Topologies. Let's see how it is done in CATH.

Prepared by Arshad

Class

- Similar secondary structure content
  All α, all β, alternating α/β etc.

S.Q: Architecture

- Also called FOLD
- Major structural similarity
- SSE's in similar arrangement

S.Q: Topology

- Super Family
- Probable common ancestry
- Family membership

S.Q: Homology

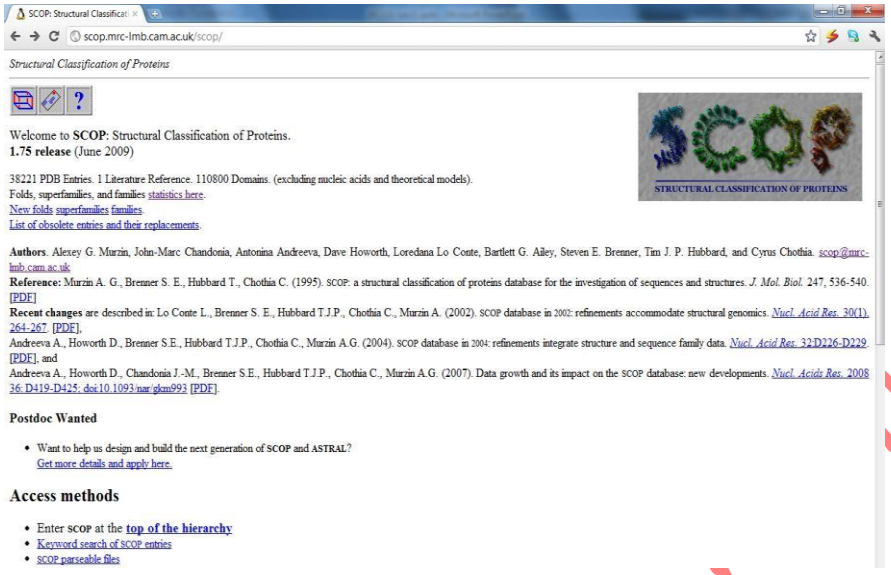
- Same Family
- Clear evolutionary relationship
- Pairwise sequence similarity > 30%

CATH classifies proteins by their structural similarity. It also considers the internal organization of the structural components in proteins.

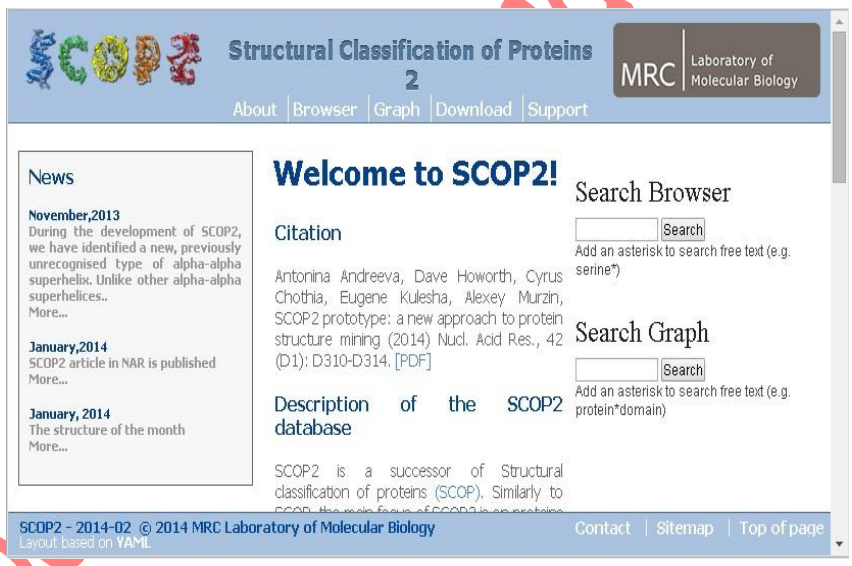
046: Classification Databases

Proteins are classified into various structural classes. CATH is one such system in which proteins are organized into classes, architecture, topology and homology.

Prepared by Zainab Arshad



http://scop.mrc-lmb.cam.ac.uk/scop/



SCOP: Structural Classification of Proteins. **1.75** release
38221 PDB Entries (23 Feb 2009). 110800 Domains. 1 Literature Reference
(excluding nucleic acids and theoretical models)

Class	Number of folds	Number of superfamilies	Number of families
All alpha proteins	284	507	871
All beta proteins	174	354	742
Alpha and beta proteins (a/b)	147	244	803
Alpha and beta proteins (a+b)	376	552	1055
Multi-domain proteins	66	66	89
Membrane and cell surface proteins	58	110	123
Small proteins	90	129	219
Total	1195	1962	3902

Figure 75 SCOP Classification Statistics

http://scop.mrc-lmb.cam.ac.uk/scop/count.html

FSSP - Family of Structurally Similar Proteins, based on the DALI algorithm. Pclass - Protein Classification, based on the LOCK and 3Dsearch algorithms.

Prepared by Zainab Ahmad

047: Algorithms for Structure Classification

Several algorithms exist for classifying protein structures.

S.Q: &L.Q: What are the Algorithms for Structure Classification?

Intra-Molecular Distance Algorithms.

- **McqS:**Proteins are considered as **rigid bodies.**
- They are placed in a **3D Cartesian coordinate system**.
- Structural alignment in 3D.
- **E.g. VAST, LOCK**

Inter-Molecular Distance Algorithms

- Proteins are considered as rigid bodies.
- They are **placed in 2D.**
- Structural alignment using **internal distances and angles**.

The basic idea is to capture internal geometry of protein structures. E.g. DALI, and SSAP.

Such algorithms are also very useful to compare whole protein structures. They can help determine evolutionary relationship. Also, functional similarity can be estimated.

Prepared by Zainab Arshad

Module

048: Protein Structure Comparison

Proteins are assembled into primary (1'), secondary (2'), tertiary (3') and quaternary (4') structures. Protein sequence is less conserved than its structure. Protein structure determines function. Since protein structure dictates function, comparing two structures can help us evaluate if the proteins do the same or similar function.

S.Q: Write a note on comparison of Protein Structure?

Comparing Whole Protein Structures

Proteins contain multiple structural subunits e.g. secondary structures, motifs and domains. Structures of all such subunits are to be considered as one and compared. We know that domains are functionally independent components of the protein structure. Proteins may have **multiple domains.** So for two different proteins, sharing the same domain, we may want to compare only a **portion of the overall structure** i.e. a domain. For comparing the complete or partial protein structures, the position of Alpha Carbon atoms can be used. **The (x, y, z) positions of Alpha Carbon atoms can be obtained from the PDB.**

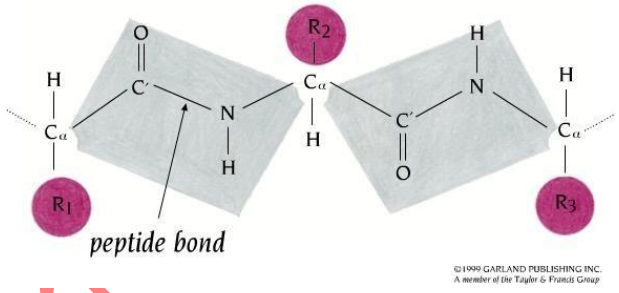


Figure 74 C-Alpha atoms in backbone

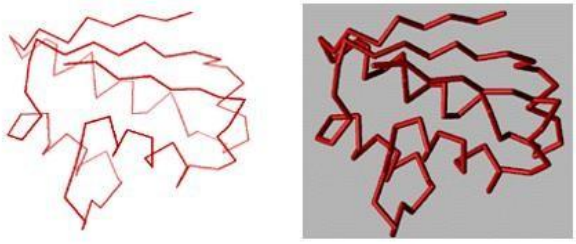


Figure 76 Tracing and Visualizing C-Alpha Backbone

http://www.danforthcenter.org/smith/MolView/Over/overview.html

PDB coordinates of Alpha Carbons in the protein back bone can be used for comparison. In this way, whole protein structure or domains etc. can be compared.

Prepared by Sajjad Ahmad Arshad

049: Strategies for Structure Comparison - I

PDB coordinates of Alpha Carbons in the protein back bone can be used for comparison. Thus, two whole protein structures or domains within each structure can be compared.

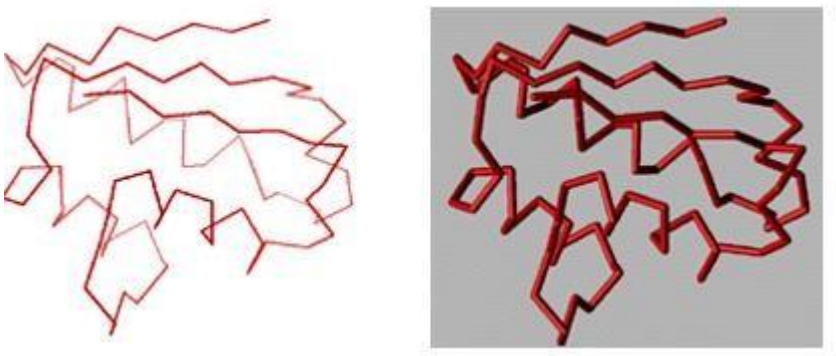


Figure 77 Tracing and Visualizing C-Alpha Backbone

http://www.danforthcenter.org/smith/MolView/Over/overview.html

S.Q: WRITE a note on Strategy # 1 – Whole Protein Structure Comparison by Intermolecular distances

- Two protein sequences are pair-wise aligned with each other ➢ Corresponding Alpha Carbons are identified
- Coordinates of corresponding Alpha Carbons are retrieved from PDB
- Their individual differences calculated
- Root Mean Square Distance is computed to assess the similarity

MCQs: Whole protein structures can be compared by calculating the **root mean squared difference (RMSD)** between their Alpha Carbons positions. The lower the RMSD, the similar are the proteins.

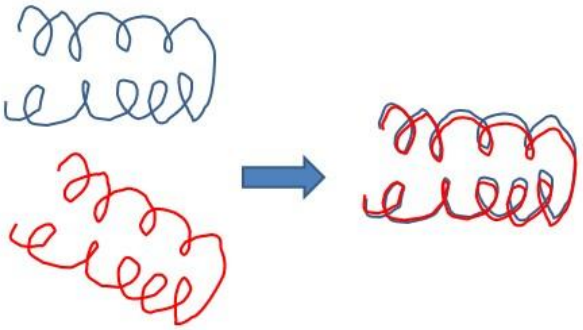
Prepared by Zunaira Ahmad

MCQs: Module 050: Strategies for Structure Comparison - II

Full protein structures can be compared and ranked by the overall differences in positions between their **Alpha Carbons**. But proteins are 3D and in various conformations.

Full Protein Comparison

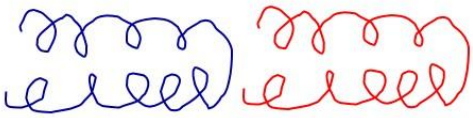
(Translation -> Rotation -> RMSD)



Calculating RMSD – An Example

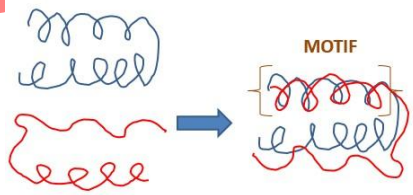
$$RMS(A,B) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d(a_i, b_i)^2}$$

where $d(a_i - b_i)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$



Domain or Motif Comparison

(Region Selection ->Translation -> Rotation -> RMSD)

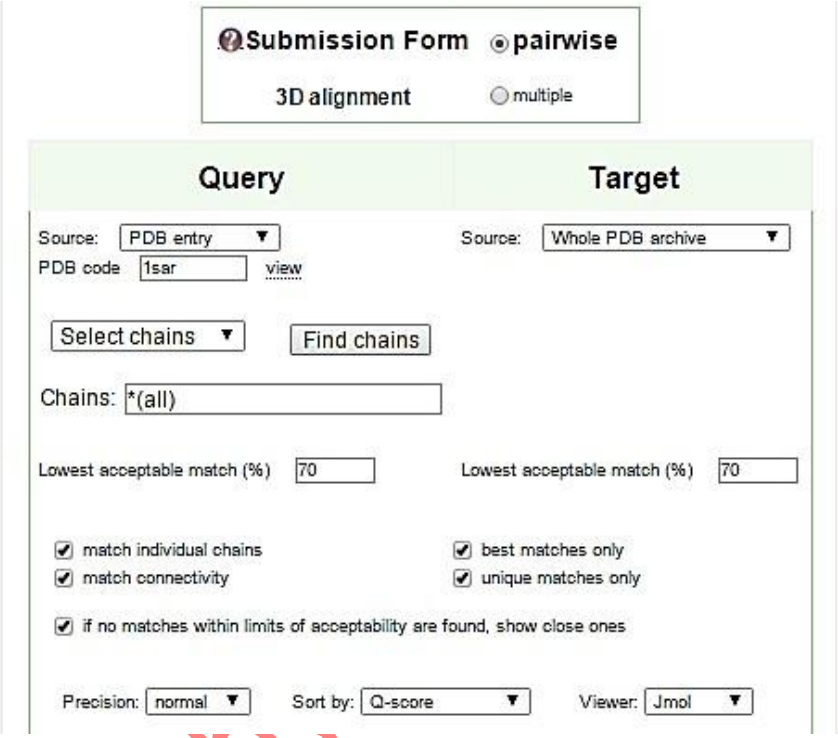


Motifs, Domains and Full Proteins can be compared by using the rigid body super-positioning. Depending on the RMSD, proteins, their motifs and domains can be selectively compared.

Prepared by Zainab Arshad

S.Q: What are the Online Resources for Structure Comparison ?

Multiple types of comparison can be performed between Proteins, Motifs, and Domains by rigid body super-positioning. RMSD tells us about the quality of the matches.



Protein structures can be compared in multiple ways. Till now, we can compare proteins by their motifs, domains and full structures. There are several advanced techniques for this as well.

Prepared by

Module 052: Protein Structure Prediction

MCQs:Complex protein structures enable proteins to perform complex functions. We know over a **million protein sequences but only about 100,000 protein structures**. Estimating exact protein structures is **very difficult.** It's difficult to **crystallize proteins**. Even if we manage to get **protein's X-Ray,** to reconstruct the structure is extremely complex.

Since we know so many sequences, they can be used for predicting protein structures. This indeed is possible and helpful.

The Basic Idea

- Amino acids determine the protein structure
- We have a large protein sequence dataset (uniprot) Hence, we

can fold protein sequences and predict their structures

- **S.Q: Why predict and why not exact solutions?**

A deterministic solution of protein folding is a major unsolved problem in molecular biology. Proteins fold spontaneously or with the help of enzymes or chaperones. To computationally predict protein structures, we need to copy or mimic the natural folding.

S.Q.To fold we must learn the steps

Step 1: "Collapse"- leading to burial of hydrophobic AA's

Step 2: Fluid globule - helices & sheets form, but are unorganized

Step 3: Compaction, and rearrangement of 2'structures

Protein structure prediction involves learning how the amino acids in primary sequence fold. Using this information, upon getting a protein sequence, we can try to predict how it folds

053: Predicting Secondary Structures

By looking at the structures in PDB, we know that Alanine mostly found in Alpha Helices. So if we have several **Alanines** in the sequence, then we can **anticipate** that a helix may be formed by them. What if we survey the entire PDB and check the presence of each amino in each type of secondary structure. If we know which amino acid is found in which specific secondary structure, then we can use it for prediction.

Prepared by Zaindb Arshad

Module

Amino Acid	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Figure 78 Chou & Fasman (1974 & 1978)

Several algorithms have been designed to predict 2' given an amino acid sequence. The first such algorithm was the Chou-Fasman Algorithm. We will see it in the upcoming modules.

S.Q: L.Q: What is Chou Fasman Algorithm and its uses?

3D Structure of proteins is determined by their Amino Acid sequence. Note that we only know **100,000 3D protein** structures, **but 10 times more sequences**. For those proteins whose structure is already known, can we evaluate their amino acid sequence?

Amino Acid	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Chou & Fasman (1974 & 1978)

Figure 79 Propensity Table

Prepare by Zainab Arshad

Module

Predicting the 2' structures

Now, let's consider that if we are given an amino acid sequence, we can simply look up the propensity table and assign the tentative secondary structure.

Sequence: E M A V I Y P G

2' Structure: α α α β β β T T

1.51 * 1.45 * 1.42 * 1.7 *
1.6 * 1.47 * 1.52 * 1.56

= 29.47

	P_α	P_β	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Chou & Fasman (1974 & 1978)

The propensity for forming "ααα β β β T T" is = 29.47

Given an amino acid sequence, look up the propensity table for each amino acid's propensity for various 2' structures. Product of these propensity values will give you the overall propensity for formation of each 2' structure.

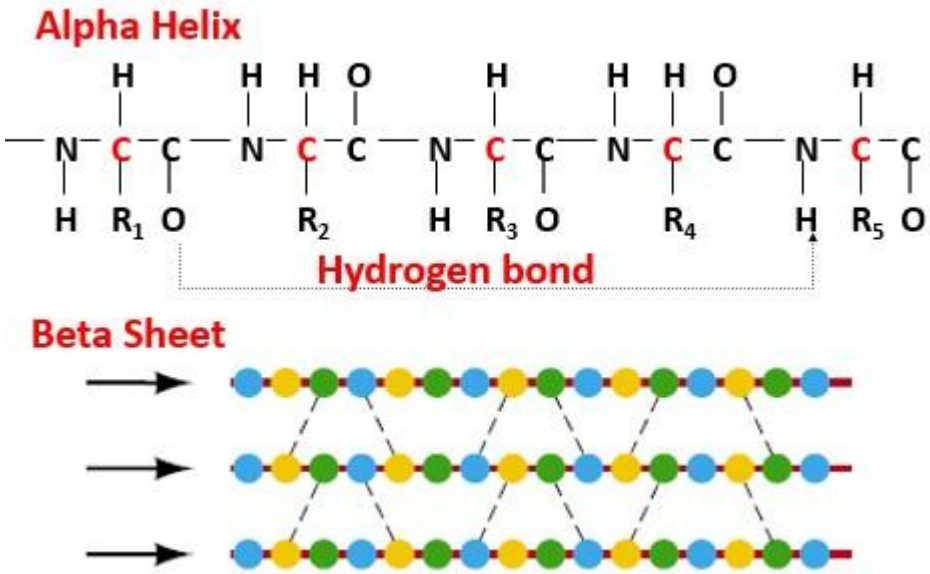
055: 2' Structures in Chou Fasman Algorithm

For a primary sequence, and a tentative 2' structure, propensity table can help us compute the overall propensity. Product of propensity values is computed for overall propensity for each 2' **MCQS:** structure. An important point to note here is that **2' structures are formed due to hydrogen bonding between amino acids.**

So, we need to consider the neighboring amino acids as well.

Prepared by Zainab Irshad

Module



You only need to compute propensities for a small number 2' structures. The highest **net propensity** will be the most probably secondary structure that will be formed.

Prepared by Zam

Module : Chou Fasman Algorithm -

Only a small number of combinations of secondary structures are possible due to their **individual properties.** Such as 4 amino acids are needed to start an Alpha Helix and 5 amino acids for **Beta Sheet**. Note that besides the alpha helix and beta sheets, LOOPS are another secondary structure. Loops are small ~ 3-4 amino acids.

1. Scan through the sequence : E M A V I Y P G
2. Identify sequence regions where:
    - 4 out of 6 contiguous residues give a P(α) > 1.0
    - That region is declared as alpha-helix
    - Extend helix to both sides until 4 out of 6 contiguous residues give a P(α) < 1.0

That is declared end of the helix. For Alpha Helices, 4 contiguous amino acids are required. Their Alpha-Helix propensity should be more than 1.0. Once this propensity falls below 1.0, Alpha-Helix stops.

057 II

MSQ: Alpha Helices are formed from 4 contiguous amino acids having an Alpha-Helix propensity over 1.0. The Alpha-Helix stops if this propensity falls below 1.0. Once Alpha Helices are constructed, and concluded, the remaining amino acids can be evaluated for Beta sheets and turns etc. Let's see how Beta sheets are evaluated using Chou Fasman Algorithm.

Prepared by Zainab Arshad

Module : Chou Fasman Algorithm -

Name	P(a)	P(b)	P(turn)	f(i)	f(i+1)	f(i+2)	f(i+3)
Alanine	1.42	0.83	0.66	0.06	0.076	0.035	0.058
Arginine	0.98	0.93	0.95	0.070	0.106	0.099	0.085
Aspartic Acid	1.01	0.54	1.46	0.147	0.110	0.179	0.081
Asparagine	0.67	0.89	1.56	0.161	0.083	0.191	0.091
Cysteine	0.70	1.19	1.19	0.149	0.050	0.117	0.128
Glutamic Acid	1.39	1.17	0.74	0.056	0.060	0.077	0.064
Glutamine	1.11	1.10	0.98	0.074	0.098	0.037	0.098
Glycine	0.57	0.75	1.56	0.102	0.085	0.190	0.152
Histidine	1.00	0.87	0.95	0.140	0.047	0.093	0.054
Isoleucine	1.08	1.60	0.47	0.043	0.034	0.013	0.056
Leucine	1.41	1.30	0.59	0.061	0.025	0.036	0.070
Lysine	1.14	0.74	1.01	0.055	0.115	0.072	0.095
Methionine	1.45	1.05	0.60	0.068	0.082	0.014	0.055
Phenylalanine	1.13	1.38	0.60	0.059	0.041	0.065	0.065
Proline	0.57	0.55	1.52	0.102	0.301	0.034	0.068
Serine	0.77	0.75	1.43	0.120	0.139	0.125	0.106
Threonine	0.83	1.19	0.96	0.086	0.108	0.065	0.079
Tryptophan	1.08	1.37	0.96	0.077	0.013	0.064	0.167
Tyrosine	0.69	1.47	1.14	0.082	0.065	0.114	0.125
Valine	1.06	1.70	0.50	0.062	0.048	0.028	0.053

1. Compute P(β) for contiguous regions of 5 Amino Acids
2. From these regions, identify regions where:
3. 5 contiguous residues have P(α) > P(β)

That region is finalized as alpha-helix. Repeat this step for the full amino acid sequence to finalize all possible alpha helical regions in the sequence.

MCQ: Alpha Helices can be finalized if their **propensity** is higher than the propensity for **Beta Sheets** in regions of **5 amino acids**. For those regions where that is not the case, further evaluation is required.

Alpha Helices are formed from **4 contiguous amino acids** having an Alpha-Helix propensity over **1.0.** The Alpha-Helix stops if this propensity falls below **1.0.** Alpha Helices were **finalized** if their propensity was higher than the propensity for Beta Sheets in regions of **5 amino acids**.

We can evaluate such regions **for Beta Sheets.** Let us see step by stop how to find a beta sheet and how to differentiate them from alpha helices.

Scan the sequence to identify regions where:

- 3 out of 5 amino acids have P(β) > 1.0

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- That region is declared as beta sheet
- Extend beta sheet to both sides until
  4 contiguous residues **average P(β) < 1.0**
- That is declared end of the beta sheet
- Those regions are finalized as beta-sheets which have average **P(β) > 1.05 and the average P(β) > P(α) for that region.**

MCQ: Regions where overlapping alpha-helices and beta-sheets occur are declared helices if

- the average **P(a-helix) > P(b-sheet)** for that region

Else, a beta sheet is declared if

- average **P(b-sheet) > P(a-helix)** for that region

Using the strategy of higher propensity, alpha helices and beta sheets can be completely resolved. Assignments for each beta sheet and alpha helix can be **finalized.**

After computing the propensity of alpha helices and beta sheets, we need to settle for loops. Let's see how we can find out the loops using **Chou Fasman Algorithm.**

Name	P(a)	P(b)	P(turn)	f(i)	f(i+1)	f(i+2)	f(i+3)
Alanine	1.42	0.83	0.66	0.06	0.076	0.035	0.058
Arginine	0.98	0.93	0.95	0.070	0.106	0.099	0.085
Aspartic Acid	1.01	0.54	1.46	0.147	0.110	0.179	0.081
Asparagine	0.67	0.89	1.56	0.161	0.083	0.191	0.091
Cysteine	0.70	1.19	1.19	0.149	0.050	0.117	0.128
Glutamic Acid	1.39	1.17	0.74	0.056	0.060	0.077	0.064
Glutamine	1.11	1.10	0.98	0.074	0.098	0.037	0.098
Glycine	0.57	0.75	1.56	0.102	0.085	0.190	0.152
Histidine	1.00	0.87	0.95	0.140	0.047	0.093	0.054
Isoleucine	1.08	1.60	0.47	0.043	0.034	0.013	0.056
Leucine	1.41	1.30	0.59	0.061	0.025	0.036	0.070
Lysine	1.14	0.74	1.01	0.055	0.115	0.072	0.095
Methionine	1.45	1.05	0.60	0.068	0.082	0.014	0.055
Phenylalanine	1.13	1.38	0.60	0.059	0.041	0.065	0.065
Proline	0.57	0.55	1.52	0.102	0.301	0.034	0.068
Serine	0.77	0.75	1.43	0.120	0.139	0.125	0.106
Threonine	0.83	1.19	0.96	0.086	0.108	0.065	0.079
Tryptophan	1.08	1.37	0.96	0.077	0.013	0.064	0.167
Tyrosine	0.69	1.47	1.14	0.082	0.065	0.114	0.125
Valine	1.06	1.70	0.50	0.062	0.048	0.028	0.053

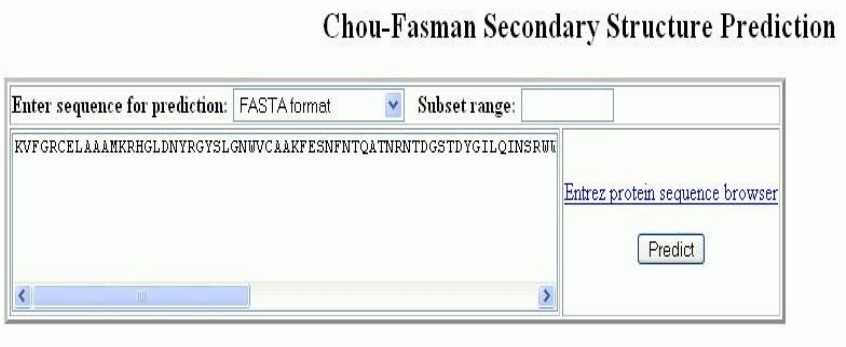
For any *jth* residue in sequence, we calculate f
(Total) = f(j) f(j+1) f(j+2) f(j+3) (tetrapeptide)

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If

- f(Total) > 0.000075
- the average value for P(turn) > 1.00 in the tetra peptide
- the averages for the tetra peptide are such P(a-helix) < P(turn) > P(b-sheet)



http://fasta.bioch.virginia.edu/fasta_www/chofas.htm

MCQ: **Chou Fasman Algorithm helps predict Alpha Helices, Beta Sheets and Turns.** The algorithm is based on **statistical occurrence** of Amino Acids in known structures.

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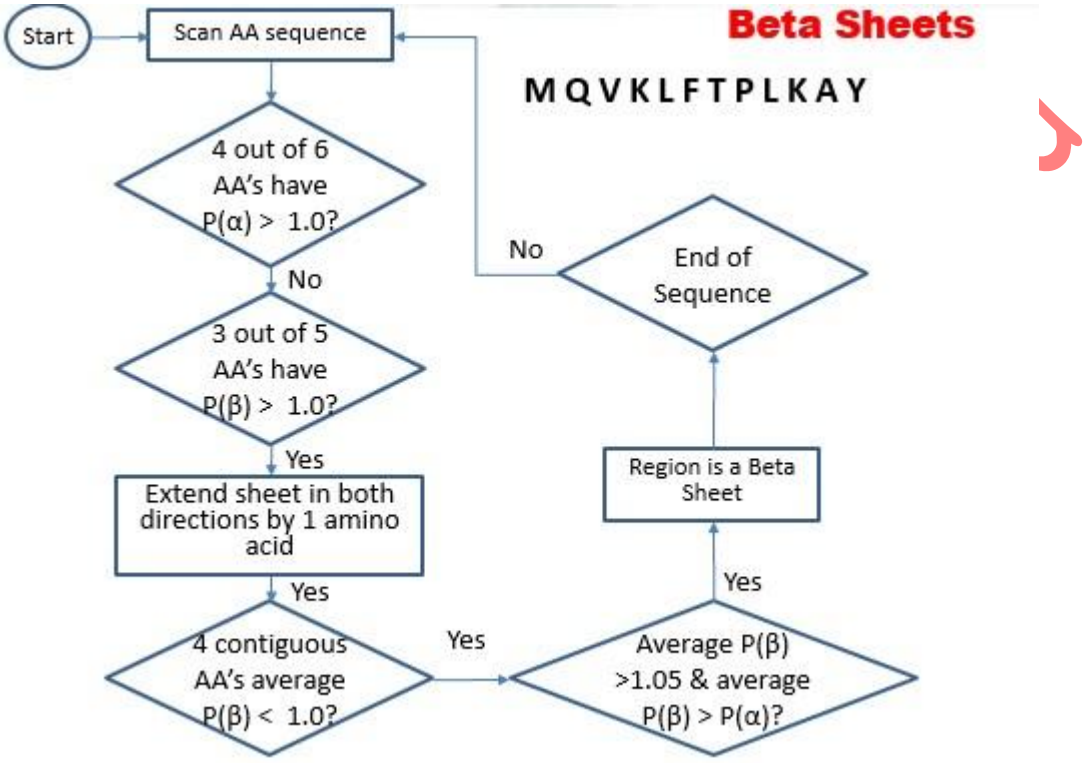
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Module : Chou Fasman Algorithm -

Module 060: Chou Fasman Algorithm – Flowchart I

Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets and Turns. Step by step flowchart of the entire algorithm.



Beta sheets can be predicted from primary amino acid sequences. Next, we will see the flowchart of Alpha Helices and Beta Turns.

Module 061: Chou Fasman Algorithm – Flowchart II

Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets and Turns. Step by step flowchart of the entire algorithm.

Prepared by

Module : Chou Fasman Algorithm -

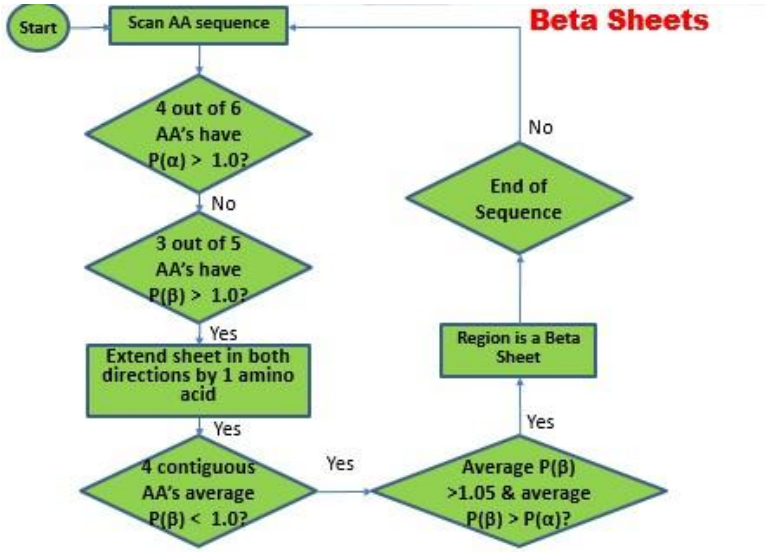


Figure 80 Beta sheet flowchart

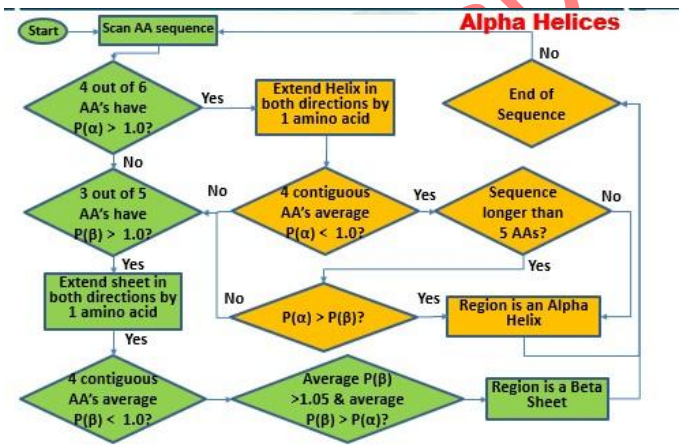


Figure 81 Alpha helices flowchart

Now we have reviewed flowcharts for Alpha Helices and Beta Sheets. Next up is the flow chart for Beta Turns.

Module 062: Chou Fasman Algorithm – Flowchart III

Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets and Turns. Step by step flowchart of the entire algorithm.

Prepared by: shad

Module : Chou Fasman Algorithm -

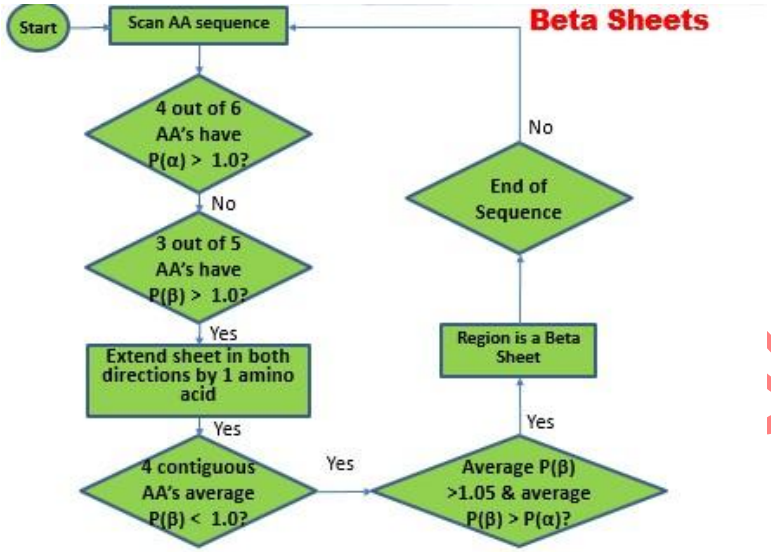


Figure 82 Beta sheet flowchart

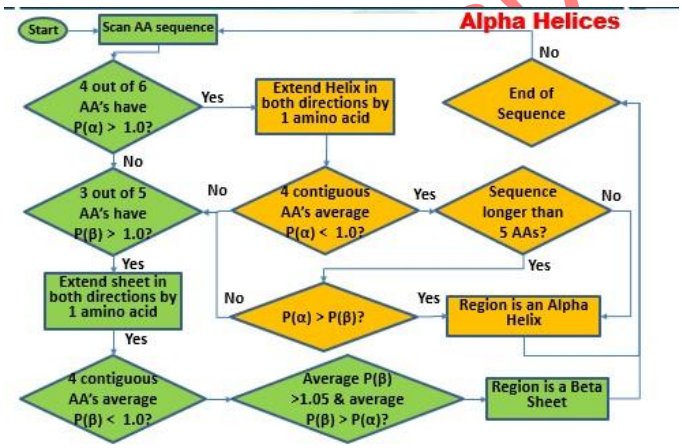


Figure 83 Alpha helices flowchart

Prepared by : Shad

Module : Chou Fasman Algorithm -

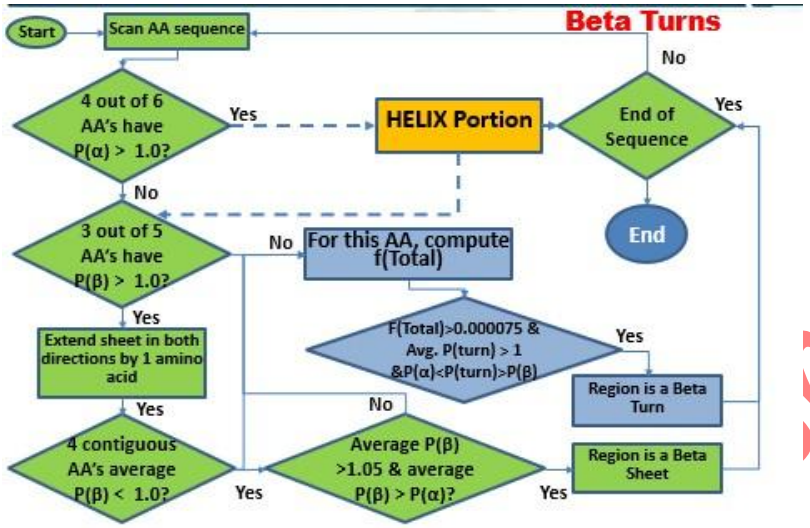


Figure 84 Beta turn

Alpha helices, beta sheets and turns can be predicted using Chou Fasman Algorithm. This algorithm is based on statistical analysis of amino acid occurrences in proteins.

Write a note on Improvements Chou Fasman Algorithm?

Module 063: Chou Fasman Algorithm – Improvements

Alpha helices, beta sheets and turns can be predicted using Chou-Fasman Algorithm. The algorithm is based on statistical analysis of amino acid occurrences in proteins.

Secondary structure propensity values of alpha helix, beta sheet and turns should be recalculated with the latest protein data sets.

IMPROVEMENTS

Special consideration for:

- **Nucleation regions**
- **Membrane proteins**
- **Hydrophobic domains**
- **Consider variable coil and loop sizes besides the from tetra peptide turns**
- **Consider local protein folding environments**
- **Solvent accessibility of residues**

Prepared by Zainb Nawaz

Module : Chou Fasman Algorithm -

- **Protein structural class**
- **Protein's organism**

Chou Fasman can be improved to better predict secondary structures by incorporating biochemical factors and updated statistics!

Module 064: Summary of Visualization, Classification & Prediction

Structure Classification

- relationship between protein structure and function
- There is need to classify proteins
- Hierarchy of classification

Structure visualization, classification and prediction equip us to perform functional evaluation of proteins. This is important for understanding disease and designing drugs for treating them.

Prepared by Zainab Arshad